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Chapter 32

GELFAND-P001425

[source number: 1]

8/II–88 Gelfand Seminar.

I. M. G. Problems. 1. Evaluate

$$\det \left\| \frac{1}{x_i - y_k} \right\|_{i,k=1}^{n}. \tag{1}$$

2. Prove that the Legendre polynomial

$$\left((x^2-1)^n\right)^{(n)}$$

has only real zeros.

3. Can a function that is the limit of continuous functions at every point of a closed interval

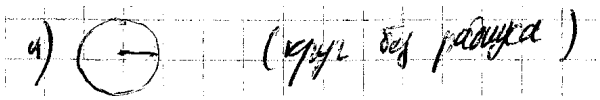
a) have discontinuities?

b) have an everywhere dense set of points of discontinuity?

c) be discontinuous everywhere (i.e., at every point)?

4. **Map conformally onto the disc**

a) (disc with a radius removed!)



Original source details: GELFAND-P001425

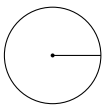
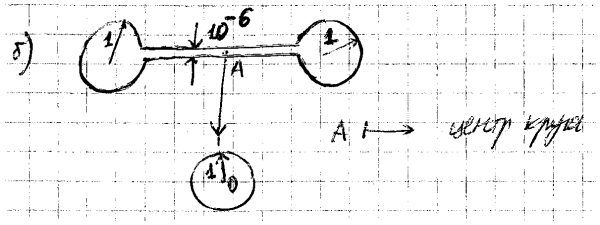


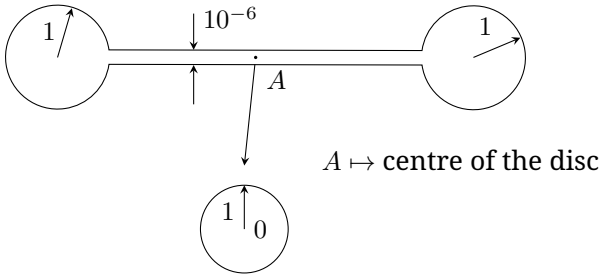
Diagram and typed description.

b)



Original source details: GELFAND-P001425

Diagram and typed description.



A. Beilinson. What does one need to know about string theory?

Conformal field theory. (2-dimensional)

Operator expansion.

Connection with representations of natural infinite-dimensional Lie algebras.

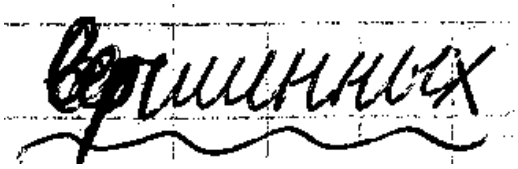
W-algebras, parafermions.

[Crossed out: Be... / Bep...?] Algebra of vertex operators.



Original source details: GELFAND-P001425

Uncertain reading. The crossed-out beginning reads “Be...”, possibly “Bep...” (the start of “vertex”).



Original source details: GELFAND-P001425

Uncertain reading. The beginning of the word “vertex” is written over earlier strokes.

Various reductions of representations of $\widehat{SL}_2$ to the Virasoro algebra.

Connection with moduli spaces of algebro-geometric structures.

Explicit formulas for partition functions and [Crossed out: correlation functions; inserted above the line: correlators].

Branching property of characters.

GELFAND-P001426

D. Alekseevsky. G-structures

Scalar field quantity.

[crossed out: Field] — a function of the point $x \in \mathbb{R}^3$ with vector values

(e.g., with values in $\mathbb{R}$)

Field quantity: depends on the coordinate system

G-structure.

Def. 1 $G \subset GL(V), V = \mathbb{R}^n$.

[Under G: linear group.]

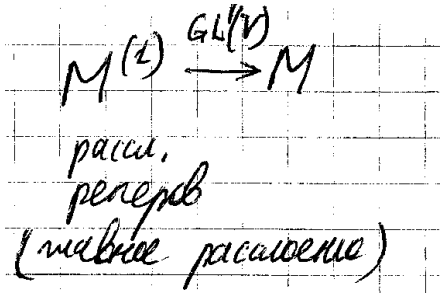
G-structure [above the line: on M^n] — a principal subbundle of the manifold of all [crossed out: co?]frames.

[On the right, under «frames»: (coframes) — is better to say.]



Original source details: GELFAND-P001426

Uncertain reading. The crossed-out prefix before “frames” tentatively reads “co.”



Original source details: GELFAND-P001426

Diagram and typed description.

$$M^{(1)} \xrightarrow{GL(V)} M$$

frame bundle
(principal bundle)

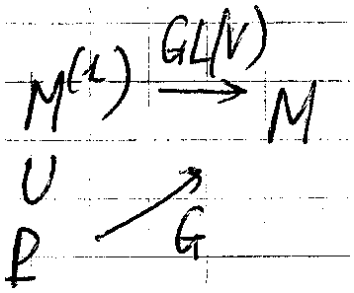
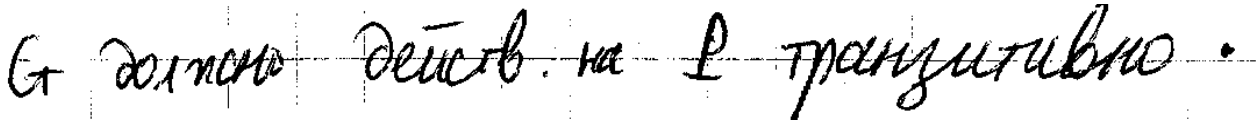


Diagram and typed description.

$$\begin{array}{ccc} M^{(1)} & \xrightarrow{GL(V)} & M \\ \cup & \nearrow_{G} & \\ P & & \end{array}$$

Original source details: GELFAND-P001426

G must act on P transitively.



Original source details: GELFAND-P001426

Uncertain reading. The manuscript says "on P transitively."

Then this is precisely a G-structure.

Examples: 1) a Riemannian metric is an $O(V)$-structure

2) Let S be a tensor field. When does it define a G-structure?

If they are all of the same canonical form

(at $\forall$ point $\exists$ a coframe [crossed out: θ / φ?], bringing S to a fixed form)



Uncertain reading. The crossed-out sign after "coframe" resembles θ or φ.

Original source details: GELFAND-P001426

Then if G — the group of automorphisms of the canonical form, then S defines a G-structure.

3) An affinor defines a G-structure, if at $\forall$ point the Jordan form is the same

4) Almost complex structure J: $G = GL\left(\frac{n}{2}, \mathbb{C}\right)$

5) Almost symplectic structure: 2-form σ of maximal rank,

(symplectic structure: $d\sigma = 0$)

Def. 2 G-structure — a principal G-bundle $\pi\colon P \xrightarrow{G} M$

$G \subset GL(V)$

equipped with a vector-valued 1-form $\theta\colon TP \to V$,

satisfying the following conditions:

GELFAND-P001427

[source number: 2]

a) G-equivariance

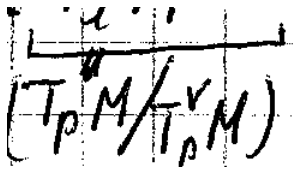
b) Ker $\theta = T^v P$ (vertical space), i.e., θ is strictly horizontal.

Equivalence of definitions 1 and 2.

$p \in P$, $x = \pi p$;

$$\theta_p \colon \underbrace{T_x M}_{\downarrow (T_p M / T_p^v M)} \xrightarrow{\simeq} V.$$

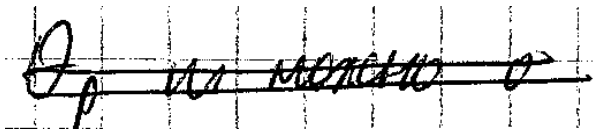
[On the right:] this is precisely the coframe with which we identify the point p.



Original source details: GELFAND-P001427

Uncertain reading. Below $T_x M$ is $T_p M / T_p^v M$, with M written in both terms.

[Crossed out: θ_p on the vertical [space]? 0.]



Original source details: GELFAND-P001427

Uncertain reading. The word between "on" and 0 tentatively begins "вертик..." ("vertical...").

Coframe $\simeq$ an isomorphism $\theta_p \colon T_x M \xrightarrow{\simeq} V$

θ is called the solder form

Suppose there is a G-structure

$$(\pi \colon P \xrightarrow{G} M; \theta)$$

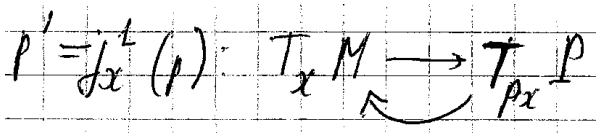
Consider the bundle of 1-jets of the bundle π

$$\pi' \colon P' \to P$$

$x \to p_x$ is a section

$$p' \colon T_x M \to T_{p_x} P$$

Def. A 1-jet is a map p' — inverse to the projection, $p' = j^1_x(p)$



Original source details: GELFAND-P001427

Diagram and typed description.

$p' = j^1_x(p) \colon \quad T_x M \longrightarrow T_{p_x} P$

i.e., $\pi_* \circ p' = \mathrm{id}$

Def. A horizontal subspace $H \subset T_pP$, [Crossed out: illegible] — is a subspace such that



Original source details: GELFAND-P001427

Undeciphered passage. The crossed-out mark after T_pP is illegible.

$$T_pP = T_p^v \oplus H$$

Second definition of a 1-jet

$$\underbrace{\text{Elements of } P'}_{(=\text{1-jets})} = \text{horizontal subspaces}$$

$$p' \in P' \quad \Longleftrightarrow \quad H(p') \text{ — a horizontal subspace}$$

Relation:

$$p' \colon T_xM \to H(p') \subset T_pP$$

Def. [Crossed out: P'?] π' is called the bundle of connections.



Original source details: GELFAND-P001427

Uncertain reading. The crossed-out mark before π' may read P'.

GELFAND-P001428

Space of connections: $\Gamma(\pi')^G$

(i.e., a connection is a G-equivariant section of the bundle $\pi' \colon P' \to P$)

Connection:

$$h \colon P \to P'$$
$$p \mapsto H(p)$$

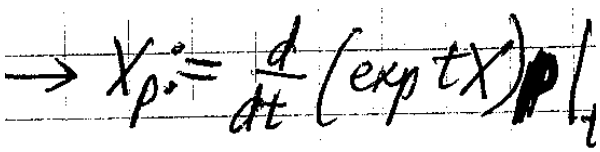
(G-invariant field of horizontal subspaces)

This gives parallel transport (restrict to a curve)

Torsion function (analogue of the structure constants of a Lie group)

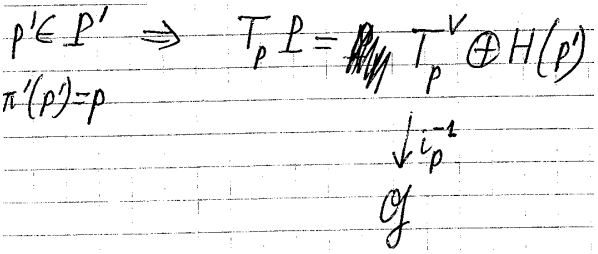
There is a vertical parallelism:

$$i_p \colon \mathfrak{g} \xrightarrow[\simeq]{} T_p^v P$$
$$X \mapsto X_p := \frac{d}{dt}(\exp tX)p\Big|_{t=0}$$



Original source details: GELFAND-P001428

Uncertain reading. The p following $(\exp tX)$ is retraced. It is written to the right of the exponential.

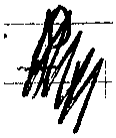


Original source details: GELFAND-P001428

Diagram and typed description.

$$p' \in P' \Longrightarrow T_pP = T_p^v \oplus H(p')$$
$$\pi'(p') = p \qquad \downarrow i_p^{-1}$$
$$\mathfrak{g}$$

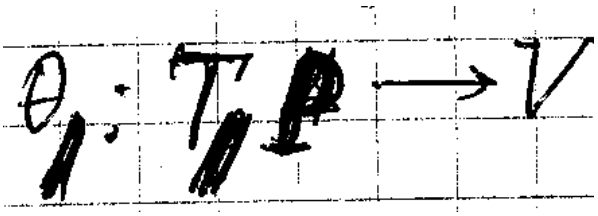
[After the $=$ sign, crossed out: illegible]



Original source details: GELFAND-P001428

Undeciphered passage. The mark between $=$ and T_p^v is crossed out and illegible.

Take the form $\theta_p \colon T_pP \to V$



Original source details: GELFAND-P001428

Uncertain reading. Both subscripts p and the letter P are retraced or overwritten.

and set

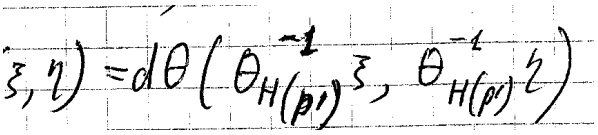
$$\theta_{H(p')} = \theta|_{H(p')} \colon H(p') \xrightarrow{\simeq} V$$

Def. The torsion function — is the map

$$t \colon P' \longrightarrow \mathrm{Hom}(V \wedge V, V)$$

(tensors of type $1, 2$)

$$p' \mapsto t_{p'}, \qquad t_{p'}(\xi, \eta) = d\theta\big(\theta^{-1}_{H(p')}\xi, \theta^{-1}_{H(p')}\eta\big)$$



Original source details: GELFAND-P001428

Uncertain reading. The last symbol has a differently shaped initial stroke and reads as η.

Properties of the torsion function.

1) G-equivariance and $G' = \mathrm{Hom}(V, \mathfrak{g})$-equivariance

GELFAND-P001429

[source number: 3]

15/II–88 Gelfand Seminar (from the notes of A. K. Zvonkin)

In I. M. G.'s lectures on quasilinear equations (UMN, 1959, vol. 14, no. 2) three pages present all of thermodynamics.

A concrete problem in geometry: what is the angle between two 2-dimensional subspaces in 4-dimensional space?

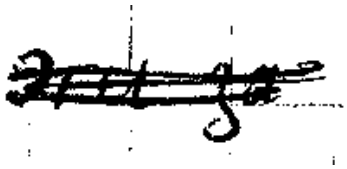
The angle is an invariant of relative motion (for lines).

There are 2 versions, for oriented lines and for unoriented lines.

For planes, how many such invariants are there and what are they?

Question: find all the invariants of two subspaces of a Hilbert space.

[Crossed out: Zam...?]



Original source details: GELFAND-P001429

Uncertain reading. The crossed-out writing above the name tentatively begins "Zam...".

Zamolodchikov, Sasha. Conformal (2-dimensional) field theory.

$$\langle A_1(x_1)A_2(x_2)\dots A_N(x_N)\rangle, \qquad x_i \in \mathbb{R}^2, \tag{1}$$

— «correlation functions»

$\mathcal{A} \ni A$ — fields (in general infinite-dimensional)

(1) — N-linear function on the space of fields $\mathcal{A}$ (also a function of $(x_1, \dots, x_N)$)

Let us specify the meaning of the average in (1):

$$(1) = Z^{-1} \int \mathcal{D}\varphi\, e^{-S[\varphi]} A_1(x_1)\dots A_N(x_N), \tag{2}$$

where $\varphi = \varphi(x)$,

$$S = \int \sigma\big(\varphi(x), \partial\varphi(x)\big)\, dx, \tag{3}$$

$$Z = \int \mathcal{D}\varphi\, e^{-S[\varphi]}. \tag{4}$$



Original source details: GELFAND-P001429

Uncertain reading. The number (4) is written over a digit resembling 3.

Properties of the correlation functions.

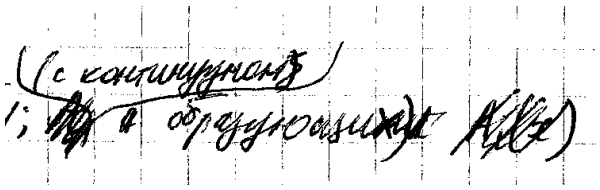
1) «Algebra of fields»

Let the fields form an associative algebra; [Crossed out: illegible] with [inserted above the line: (with a continuum)] generators $A_i(x)$.



Original source details: GELFAND-P001429

Undeciphered passage. The writing immediately after the semicolon is crossed out and illegible.



Original source details: GELFAND-P001429

Uncertain reading. The ending of the Russian word for “generators” is overwritten, possibly over a genitive plural. The strokes in $A_i(x)$ are also retraced.

Let $\{A_i\}$ — be a basis in $\mathcal{A}$. Then

$$A_i(x)A_j(y) = \sum_k \underbrace{c_{ij}^k(x,y)}_{\text{numerical functions of } x,y} A_k(y)$$

Set

$$\langle A_k(x)\rangle = a_k(x)$$

GELFAND-P001430

(Knowing the functions c_{ij}^k and a_k is sufficient to compute the correlation functions: we replace the pair A_1A_2 by $\sum c^k A_k$ and so on)

2) «Invariance»

Consider a field T of the form

$$T^{\mu\nu}, \quad \mu,\nu = 1,2; \quad T^{\mu\nu} \in \mathcal{A}.$$

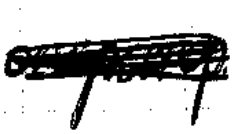
Incidentally

$$\frac{\partial}{\partial x^\mu} A_i(x)A_j(y) = \sum \frac{\partial}{\partial x^\mu} c_{ij}^k(x,y)A_k. \tag{5}$$

$$\frac{\partial T^{\mu\nu}}{\partial x^\mu} = 0 \tag{6}$$

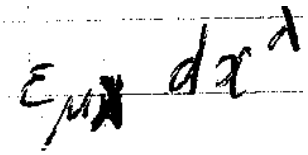
Consider [crossed out: operator?] the integral

$$\oint T^{\mu\nu}(x)A(y)\,\varepsilon_{\mu\lambda}\,dx^\lambda \tag{7}$$



Original source details: GELFAND-P001430

Uncertain reading. The crossed-out word tentatively reads “operator.”



Uncertain reading. The second index of ε is overwritten. It reads λ, possibly over ν.

Original source details: GELFAND-P001430

[To the right of the integral]



Diagram and typed description. 

Original source details: GELFAND-P001430

[Callout to ε in (7):]

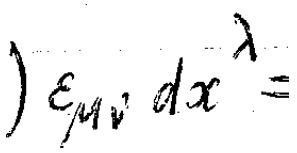
$$\varepsilon_{\mu\lambda} = -\varepsilon_{\lambda\mu}, \qquad \varepsilon_{12} = 1$$

It defines a linear operator in $\mathcal{A}$.

Required: $\exists$ a field T such that this operator equals $\dfrac{\partial}{\partial y}$.

$T^{\mu\nu}$:= «energy–momentum tensor»

$$\oint T^{\mu\nu}(x) A_i(y)\, \varepsilon_{\mu\nu}\, dx^\lambda = \frac{\partial}{\partial y^\nu} A_i(y)$$



Uncertain reading. In the repeated formula the second index of ε reads ν, alongside dx^λ.

Original source details: GELFAND-P001430

$1')$ $c_{ij}^k(x,y) = c_{ij}^k(x-y)$ — functions of one variable in $\mathbb{R}^2$.

3) «Positivity condition»

$A \mapsto A^+ \in \mathcal{A}$ — involution

$$A_i^+(x) A_j^+(y) = \sum \bar{c}_{ij}^k(\widetilde{x} - \widetilde{y}) A_k^+(y) \tag{8}$$

$$x = (x^1, x^2) \quad \Rightarrow \quad \widetilde{x} = (-x^1, x^2)$$

Requirement: the matrix $\langle A_i^+(\widetilde{x}) A_j(x) \rangle$ is positive definite for all x.

The problem is to construct substantive examples of all this.

GELFAND-P001431

[source number: 4]

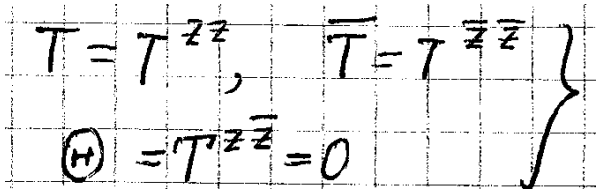
Let us introduce in $\mathbb{R}^2$ complex coordinates

$$z = x^1 + ix^2 \qquad ds^2 = dz\, d\bar{z}$$

$$T^{\nu\mu} = T^{\mu\nu} \tag{9}$$

$$\left.\begin{array}{l} T = T^{zz}, \quad \bar{T} = T^{\bar{z}\bar{z}} \\ \Theta = T^{z\bar{z}} = 0 \end{array}\right\} \quad \begin{array}{c} \text{under these additional conditions} \\ \text{the theory is called} \\ \text{«conformal field theory»} \end{array} \tag{10}$$

[Arrow from below to $\Theta = 0$:] Important special case: massless theory



Original source details: GELFAND-P001431

Uncertain reading. In (10) both upper indices in the definition of $\bar{T}$ have a bar: $\bar{T} = T^{\bar{z}\bar{z}}$.

Check question: write T^{zz}, $T^{z\bar{z}}$, $T^{\bar{z}\bar{z}}$ in terms of $T^{\mu\nu}$
(this is a linear algebra problem)

$$\frac{\partial T}{\partial \bar{z}} = 0, \quad \frac{\partial \bar{T}}{\partial z} = 0 \qquad \text{— Cauchy–Riemann equations} \tag{11}$$

$$\Updownarrow$$

T — an analytic function of z, $T = T(z)$, $\bar{T} = \bar{T}(\bar{z})$.

$$T(z)A_i(z_1, \bar{z}_1) = \sum f_{ij}(z - z_1)A_j(z_1, \bar{z}_1) \tag{12}$$

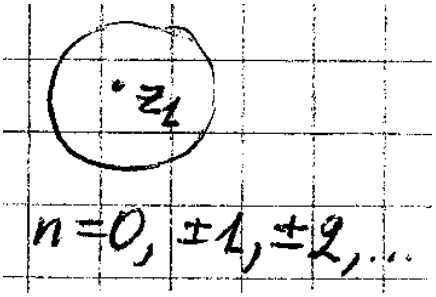
In electrodynamics.
ψ — electron
A — photon

$$\{A_i\} = \{\psi, \psi^2, \psi\partial\psi, \ldots, A, A^2, A\psi, \ldots\}$$

ψ^4 still exists, but $\psi^5 = \psi^6 = 0$ (since the spinor has only 4 components)
for A all powers exist
But, in general, A^2 does not exist.
Do not take the fields at the same point. Take them at different points.

$$\oint \frac{dz}{2\pi i} z^{n+1} T(z)A_i(z_1, \bar{z}_1) = L_n A_i(z_1, \bar{z}_1) \tag{13}$$

[To the right of (13):]



Original source details: GELFAND-P001431

Diagram and typed description.



$n = 0, \pm 1, \pm 2, \ldots$

$$L_{-1} = \frac{\partial}{\partial z_1}$$

$$\bar{T}(\bar{z}) \to \bar{L}_n; \quad \bar{L}_{-1} = \frac{\partial}{\partial \bar{z}_1} \tag{14}$$

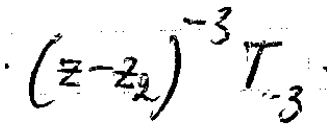
$T(z_1)T(z_2)$ — is a function of z_1, analytic everywhere except at z_2; let us expand it in a Laurent series.

$$T(z_1)T(z_2) = \sum_{m=-\infty}^{\infty} (z_1 - z_2)^m T_m(z_2), \qquad T_m \in \mathcal{A} \tag{15}$$

Let us make an assumption:

$$(15) = (z_1 - z_2)^{-4} T_{-4} + (z - z_2)^{-3} T_{-3} + \dots \tag{16}$$

(i.e., the series terminates at its lower end at term -4)



Uncertain reading. The second power in (16) is written $(z - z_2)^{-3}$, without a subscript on its first z.

Original source details: GELFAND-P001432

$$T_{-1} = \frac{dT}{dz_2}(z_2) \tag{17}$$

$$T_{-2} = 2T(z_2) \quad \leftarrow \quad \text{follows from the fact that } T \text{ is a rank-2 tensor} \tag{18}$$

The most important thing is to determine the most singular terms T_{-3} and T_{-4}.

It follows from the positivity condition that

$$T_{-4} \sim \text{identity operator}: \quad T_{-4} = \frac{c}{2}, \qquad T_{-3} = 0 \tag{19}$$

$$T(z)T(z') = \frac{c}{2(z - z')^4} + \frac{2T(z')}{(z - z')^2} + \frac{1}{z - z'} T'(z') + \dots \tag{20}$$

[Arrow to the ellipsis in (20):] the further terms are of no interest

From this expansion we obtain

$$\boxed{[L_n, L_m] = (n - m)L_{n+m} + \frac{c}{12}(n^3 - n)\delta_{n+m,0}} \tag{21}$$

These are the relations of the Virasoro algebra (central extension of the algebra of vector fields on the circle)

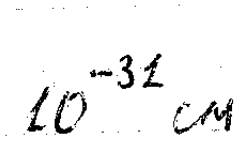
(In ordinary quantum mechanics, the Lie algebra of the group of translations on the line: $\frac{d}{dx}$ — small translation — momentum)

In general the energy–momentum tensor comes from coordinate transformations

In this sense Sophus Lie invented quantum mechanics. He was the first to replace coordinate transformations with infinitesimal transformations.

(The group for the Virasoro algebra is the group of diffeomorphisms of the circle.)

Lengths smaller than 10^{-31} cm have no meaning. This statement is already independent [crossed out: от / не?] of any theory — from the relations between universal constants.



Uncertain reading. The manuscript exponent reads as -31.

Original source details: GELFAND-P001432



Uncertain reading. The word crossed out after "independent" tentatively reads «от» ("of"), possibly «не» ("not").

Original source details: GELFAND-P001432

GELFAND-P001433

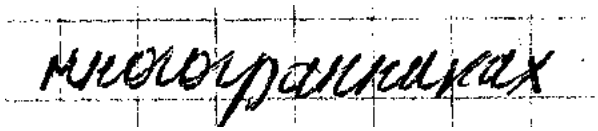
[source number: 5]

p-adic theory has computational advantages. Some determinants are best computed adelically

But there is another point. Physics can say nothing about distant points, or about nearby ones.

[Two vertical lines in the left margin] Everything is known about two-dimensional polyhedra, but as little is known about three- and four-dimensional ones as 2000 years ago.

This is new combinatorics.



Uncertain reading. The manuscript reads "polyhedra." The two rules at the left also cover the line "This is new combinatorics."

Original source details: GELFAND-P001433

How are we to think of space if small distances are ruled out? We lack imagination. (Thickness, discreteness, curvatures, ...)

Conformal invariance. The theory is unchanged under the substitution

$$\begin{cases} z \mapsto z + \varepsilon(z) \\ \bar{z} \mapsto \bar{z} + \bar{\varepsilon}(\bar{z}) \end{cases} \tag{22}$$

Taking $\varepsilon(z) \sim z^{n+1}$, we obtain the Virasoro algebra

Let us see how the average changes

$$\langle A_1(x_1) \dots A_N(x_N) \rangle = Z^{-1} \int A_1(x_1) \dots A_N(x_N) e^{-S[\varphi]} \, \mathcal{D}\varphi \tag{23}$$

under the substitutions

$$x^\mu \to x^\mu + \varepsilon^\mu(x) \tag{24}$$

Variation

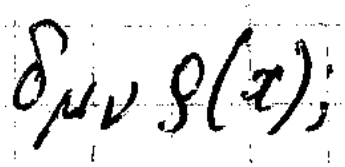
$$\begin{aligned} \delta \langle A_1(x_1) \dots A_N(x_N) \rangle = &\sum_{k=1}^{N} \langle A_1 \dots \delta_\varepsilon A_k \dots A_N \rangle \\ &+ \int \partial_\mu \varepsilon_\nu \langle T^{\mu\nu}(x) A_1 \dots A_N \rangle \, d^2x = 0 \end{aligned} \tag{25}$$

transfer (by parts) ∂_μ onto $T^{\mu\nu}$ and use $\partial_\mu T^{\mu\nu} = 0$;
integrating along small contours around $x_1, \dots, x_N$, we obtain

$$\delta_\varepsilon A_k(y) = \oint_{c_y} \varepsilon_\nu(x) T^{\lambda\nu}(x) \varepsilon_{\lambda\mu}\, dx^\mu A_k(y),$$

but

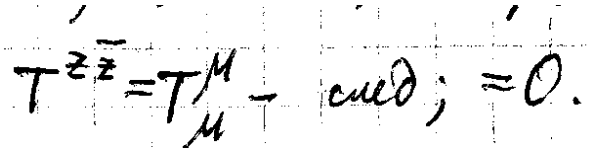
$$\partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu = \delta_{\mu\nu} g(x);$$



Uncertain reading. The factor after $\delta_{\mu\nu}$ tentatively reads $g(x)$, possibly $\rho(x)$.

Original source details: GELFAND-P001433

$T^{z\bar{z}} = T^\mu_\mu$ — the trace; $= 0$.



Uncertain reading. The manuscript has an equals sign between $T^{z\bar{z}}$ and T^μ_μ.

Original source details: GELFAND-P001433

[At the lower right]

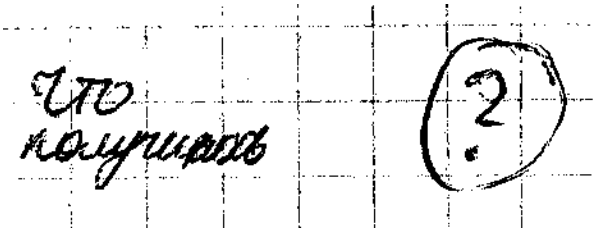


Diagram and typed description.
What did we get ?

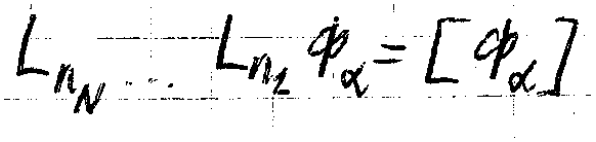
Original source details: GELFAND-P001433

GELFAND-P001434

Fields are classified by representations of the Virasoro algebra.

$$\begin{cases} L_n \phi_\alpha = 0, \quad n > 0; & \bar{L}_n \phi_\alpha = 0; \\ L_0 \phi_\alpha = \Delta_\alpha \phi_\alpha; & \bar{L}_0 \phi_\alpha = \bar{\Delta}_\alpha \phi_\alpha; \end{cases} \tag{26}$$

$$L_{n_N} \dots L_{n_1} \phi_\alpha = [\phi_\alpha]$$



Uncertain reading. The descendant line has $[\phi_\alpha]$ to the right of the equals sign, with a lowercase ϕ.

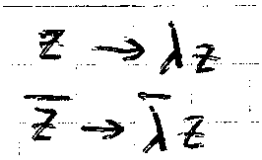
Original source details: GELFAND-P001434

$$n_i < 0; \quad \Delta_\alpha, \bar{\Delta}_\alpha \text{ — two numbers } \geq 0$$

$$\mathcal{A} = \bigoplus_\alpha [\Phi_\alpha]$$

$$\mathrm{I} \in \mathcal{A}; \qquad \mathrm{I} \cdot A_i(x) = A_i(x), \qquad \text{for it } \Delta, \bar{\Delta} = 0$$

$$\begin{matrix} z \to \lambda z \\ \bar{z} \to \bar{\lambda} z \end{matrix} \qquad \phi_\alpha \to \lambda^{\Delta_\alpha} \bar{\lambda}^{\bar{\Delta}_\alpha} \phi_\alpha,$$



Uncertain reading. The right-hand side of the second line is $\bar{\lambda} z$, without a bar over z.

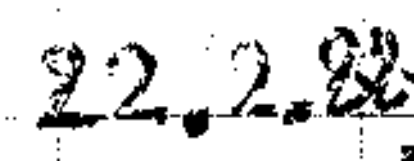
Original source details: GELFAND-P001434

$$\Delta_\alpha - \bar{\Delta}_\alpha = S_\alpha \text{ — the spin of } \phi_\alpha.$$

GELFAND-P001435

[source number: 10]

22.2.8? Angles between subspaces.



Uncertain reading. The final digit of the date is overwritten. Possible readings are 22.2.88 and 22.2.82. The MCCME catalogue dates Gelfand's talk to 22 February 1988.

Original source details: GELFAND-P001435

A series of problems on angles between subspaces

(I. M. Gelfand; from the notes of A. K. Zvonkin)

Some are accessible to an eighth-grader (problems in ($\mathbb{R}^2$ and $\mathbb{R}^3$), others are for a doctor of science (problems in Hilbert space).

What is the «angle» between two lines in the plane? It is an orthogonal invariant of this pair of lines, i.e., a numerical characteristic that is unchanged by every orthogonal transformation of the plane.

Moreover, the angle is a complete system of invariants, i.e., an orthogonal transformation takes one pair of lines to another if and only if the two pairs have the same angle.

Hereafter "angle" means a complete system of invariants of some object under the action of some group. Sometimes this is a number, sometimes a finite or countable set of numbers, and sometimes even a continuum of numbers.

(In cases where we are speaking, say, of four subspaces $\Rightarrow$ of a linear space, we will avoid the expression «angle between...», since it is unclear, between what?, and will say «angle for...».)

In every problem two things are of interest, what the angle is, and also what values it can take.

I. Problems for school pupils.

1. What is the angle between a line and a plane in three-dimensional space.

(Two versions, with the line oriented and unoriented).

2. What is the angle between two planes in three-dimensional space?

II. Problems for university students.

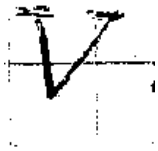
3. Given a 4-dimensional linear space with an inner product. (distinct from $\mathbb{R}^4$, since no axes are singled out). What is the angle between two of its two-dimensional subspaces?

4. Given $\{L, (\cdot,\cdot)\}$, $\dim L = n$, $v \in L$, $V \subset L$, $\dim V = k$.

Determine the angle between v and V.

GELFAND-P001436

5. Given $\{L, (\cdot,\cdot)\}$, $\dim L = n$, $V, W \subset L$, $\dim V = k$, $\dim W = m$. Determine the angle between V and W.



Original source details: GELFAND-P001436

Uncertain reading. The V at the end of problem 5 has two short strokes at its upper tips. A separate overbar cannot be distinguished.

6. Let L — just a linear space (without an inner product). Determine the angle for

(a) a subspace of L

(b) a pair of subspaces

(c) a triple of subspaces.

(Here the group is understood to be $GL(n)$)

7. Given the Minkowski plane, i.e., L, $\dim L = 2$ with the indefinite scalar product $-x_1^2 + x_2^2$. Determine the angle between two lines (i.e., an invariant under the group of transformations preserving this scalar product).

8. Further variations:

— for a Minkowski space of higher dimension and two of its subspaces

— for other spaces with an indefinite metric (i.e., when the scalar product has more than one minus sign)

— for spaces with a degenerate form ($\mathbb{R}^3$ with scalar product $x_1^2 + x_2^2$)

For all the listed problems consider the cases of the groups $O(n)$ and $SO(n)$.

III. Problems for graduate students.

9. Given a linear space L and four of its subspaces. Find the angle of this quadruple (this is the subject of the work of Gelfand and Ponomarev)

10. (A. A. Kirillov Jr.) Given $\{L, (\cdot,\cdot)\}$; then the tensor powers of L also carry a scalar product. Let $a, b \in L \otimes L$. Find the angle between the subspaces $a \otimes L$ and $L \otimes b$, embedded in $L \otimes L \otimes L$.

(This is needed for Hecke algebras and the Yang–Baxter equation)

GELFAND-P001437

[source number: 11]

IV. Problems for Candidates and Doctors of Science.

11. Determine the angle for two subspaces of a Hilbert space.

Give examples in which the value of the angle is a countable set of numbers and examples in which it is a continuum of numbers.

12. The same problem for a Hilbert space with an indefinite metric.

Comments.

1. The second part of these problems is particularly difficult. For example, given a subspace and an angle, construct another subspace that makes the given angle with the given subspace.

2. A linear operator can be specified by four subspaces.

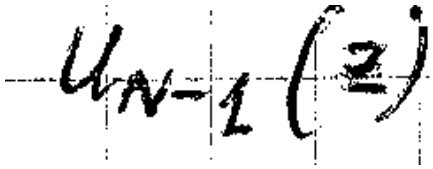
GELFAND-P001438

21/III-88 Gelfand Seminar

Radul, A.

Quantum Gelfand–Dickey brackets (following S. Lukyanov) Consider on S^1 the operator

$$L = \partial_z^N + \sum_{i=0}^{N-1} u_{N-1}(z)\partial_z^i. \tag{1}$$



Original source details: GELFAND-P001438

Uncertain reading. The coefficient subscript in (1) reads $N-1$.

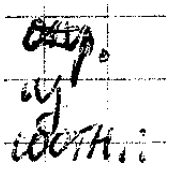
and the space of functionals of the form

$$f[u] = \oint P\big[z, u_i^{(k)}(z)\big]dz \tag{2}$$

and their finite products. P is a polynomial in the $u_i^{(k)}$.

Two Gelfand–Dickey brackets are defined on such functionals.

$$\frac{\delta f}{\delta u_i} \quad \text{defined by the relation:} \quad \frac{d}{d\varepsilon} f[u + \varepsilon\delta u]\Big|_{\varepsilon=0} = \oint dz\, \frac{\delta f}{\delta u_i}\delta u_i. \tag{3}$$



Original source details: GELFAND-P001438

Uncertain reading. The abbreviation «опр.» (“defined”) is written over earlier strokes.

I. M. Gelfand

A digression on functional analysis All the intuition of functional analysis proceeded from the idea that functionals are also functions, but on an infinite-dimensional space.

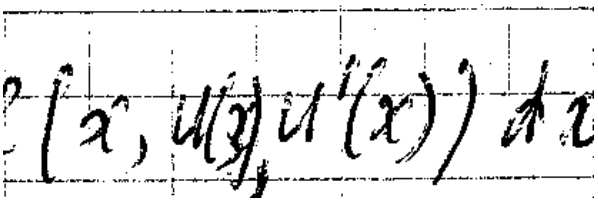
I. M. G. changed his point of view, and two consequences follow.

1) There are different types of functionals, and they must be classified.

2) Functionals are not functions ⇒ this affects geometry.

1) In the calculus of variations there are functionals

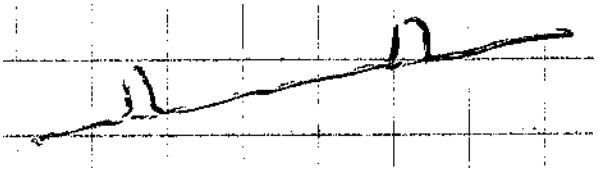
$$\int_0^L f\big(x, u(x), u'(x)\big)\, dx \qquad (*)$$



Original source details: GELFAND-P001438

Uncertain reading. The argument of the middle u tentatively reads x.

This is a functional of u, but it has the following property:



Original source details: GELFAND-P001438

Diagram and typed description.



A slanting line with two raised bumps. There are breaks in the strokes at the tops of the bumps.

if there are increments [inserted above the line with an insertion mark: of u] at two places, the increment of the functional (*) will be the sum of [crossed out: illegible] the increments

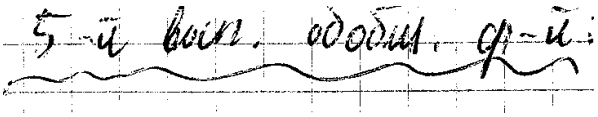


Original source details: GELFAND-P001438

Uncertain reading. The writing after “sum” is crossed out. The strokes may conceal a 2 or the beginning of «двух» (“two”).

(locality!)

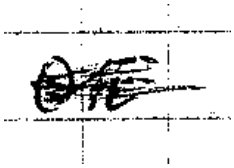
5th possible generalization of functions:



Original source details: GELFAND-P001438

Uncertain reading. The first sign tentatively reads 5. The manuscript abbreviates «обобщ.».

it follows from this (and from continuity) that the functional then [crossed out: «от» / «он»?] has this form



Original source details: GELFAND-P001438

Uncertain reading. The crossed-out writing after “then” tentatively reads «от» or «он».

[Crossed-out mark: illegible.] This is type 0.



Uncertain reading. The crossed-out mark before “This” resembles 1 or the Cyrillic letter И.

Original source details: GELFAND-P001438

GELFAND-P001439

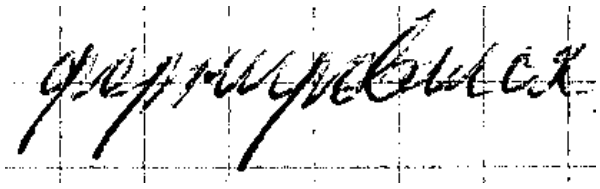
[source number: 27]

Type 1: if

$$\frac{\delta}{\delta u}F(u) \quad \text{is local.}$$

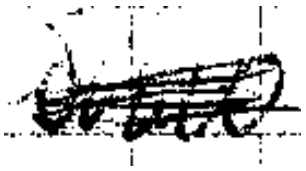
Such are the functionals that occur in Atiyah–Singer theory

The trouble with functional analysis is that it took shape when everyone [crossed out: «это» (“this”)?] was trying to rationalize.



Uncertain reading. The verb reads «формировался» (“took shape”).

Original source details: GELFAND-P001439



Uncertain reading. The crossed-out word after “everyone” may read «это» (“this”).

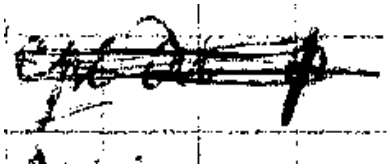
Original source details: GELFAND-P001439



Uncertain reading. The verb before “to rationalize” reads «пытались» (“were trying”).

Original source details: GELFAND-P001439

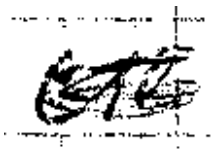
And [crossed out: spectrum?] in the theory of functions even continuity belongs to the domain of death.



Uncertain reading. The crossed-out passage after “And” tentatively reads «спектр» (“spectrum”).

Original source details: GELFAND-P001439

Functionals: a) length of a curve
b) area of a surface (local)
Transcendental functionals: (as in analysis — integrals that cannot be evaluated — elliptic functions)
passing from the coefficients to the eigenvalues!!
λ_1 — is such a functional
Functional of transcendence degree 1 — if the variational derivative is a local functional.
and so on.
Functional [crossed out: «ст.» / «от»?] of type 2: if $\frac{\delta F}{\delta u}$ is of type 1.



Uncertain reading. The crossed-out abbreviation before “type 2” resembles «ст.», possibly «от».

Original source details: GELFAND-P001439

Gelfand, Gabrielov and Loshik thought it through in their work, but it got stuck because it is difficult.



Uncertain reading. The handwritten surname resembles «Лошик» (Loshik). The paper on Atiyah–Patodi–Singer functionals lists M. V. Losik as the third author.

Original source details: GELFAND-P001439

Functionals of all types will appear in the theory of [inserted above the line: characteristic] classes.

Functionals must be considered on infinite-dimensional sets. But one must do more than simply replace an n-dimensional space with an infinite-dimensional one!

[Crossed out: «Род» / «Тут»?] The main thing is that locality plays an exceptional role!



Uncertain reading. The crossed-out writing before “The main thing” tentatively reads «Род» or «Тут».

Original source details: GELFAND-P001439

Then one must proceed differently.

Functions form an algebra.

Local functionals do not form an algebra!!

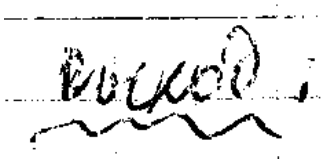
Problem: a calculus of variations is needed for sums and products of such functionals.

$$\int f\left(x, y; u_i(x), u_i'(x), v_i(y), v_i'(y)\right) dx\, dy$$

So one can no longer construct differential geometry.

Adding products formally is a bad idea!

A way out; structures introduced by G. and Dorfman and then change the connection, de Rham cohomology, Yang–Baxter brackets...



Original source details: GELFAND-P001440

Uncertain reading. The first word tentatively reads «Выход» (“a way out”).

Radul
(continued)

Paper: IPM preprint. Translation in vol. 1 of the collected works.

Digression: symbols of pseudodifferential operators

$$A = \sum a_i(z)\partial_z^i, \qquad B = \sum b_j(z)\partial_z^j. \tag{4}$$

$$\partial_z \longrightarrow \xi$$

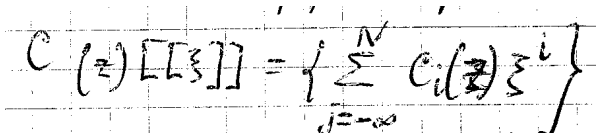
$$AB = A(z,\xi) \circ B(z,\xi)|_{\xi=\partial_z}, \tag{5}$$

where

$$A(z,\xi) \circ B(z,\xi) := \sum_{n\geq 0} \partial_\xi^n A(z,\xi)\, \partial_z^n B(z,\xi)/n!.$$

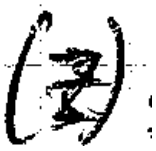
Let us switch to formal Laurent series:

$$C(z)[[\xi]] = \left\{ \sum_{j=-\infty}^{N} c_i(z)\xi^i \right\}. \tag{6}$$



Original source details: GELFAND-P001440

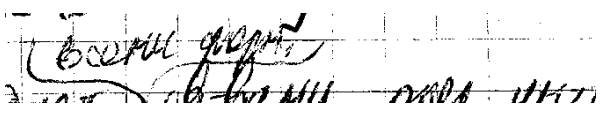
Uncertain reading. The lower summation limit reads $j = -\infty$, while i is written twice in the summand $c_i(z)\xi^i$.



Original source details: GELFAND-P001440

Uncertain reading. The argument of c_i shows z written over earlier strokes.

$\xi^{-1} = \partial_z^{-1}$ has [inserted above the line: all the formal] properties of the integration operator



Original source details: GELFAND-P001440

Uncertain reading. «всеми форм.» (“all the formal”) is written above the line with an insertion mark.

Operations on symbols

$$\left(\sum_{-\infty}^{N} a_i \xi^i\right)_+ = \sum_{0}^{N} a_i \xi^i \tag{7}$$

$$\operatorname{res}\left(\sum_{-\infty}^{N} a_i \xi^i\right) = a_{-1} \tag{8}$$

Gradient of f.

$$\nabla f := \sum_{j=0}^{N-1} \xi^{-j-1} \circ \frac{\delta f}{\delta u_j}. \tag{9}$$

GELFAND-P001441

[source number: 28]

Then

$$\delta f = \oint \operatorname{res} \delta L \circ \nabla f \tag{10}$$

here L is operator (1)

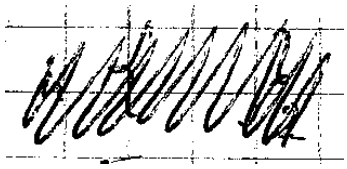
1st bracket (Hamiltonian structure)

$$1 \text{ HS:} \quad \{f, g\}_1 := \oint \operatorname{res}[L, \nabla f]\nabla g \tag{11}$$

2nd bracket:

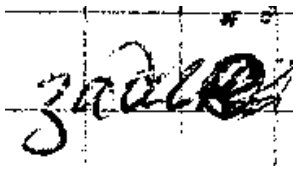
$$2 \text{ HS:} \quad \{f, g\}_2 = \oint \operatorname{res}\big(L(\nabla f L)_+ - (L\nabla f)_+ L\big)\nabla g \tag{12}$$

(13) Claim. $\{\,\}_i$ [crossed out: illegible] gives the space of functionals a Lie algebra structure.



Original source details: GELFAND-P001441

Undeciphered passage. The writing between $\{\,\}_i$ and “gives” is crossed out and illegible.



Original source details: GELFAND-P001441

Uncertain reading. The end of «задаёт» (“gives”) is written over.

(14) Example. $N = 2$, $u_1 = 0$, $\partial_z^2 + u_2$.

Let

$$u_2(n) = \frac{1}{2}\frac{1}{2\pi i}\oint u_2(z) z^{n+1}\, dz \tag{15}$$

Then

$$\{u_2(n), u_2(m)\}_2 = \frac{n^3 - n}{12}\delta_{n+m,0} + (n-m)u_2(n+m) \tag{16}$$

(this gives the commutation relations of the Virasoro algebra!)

(17) Example. $N = 3$, $u_1 = 0$, $\partial_z^3 + u_2\partial_z + u_3$.

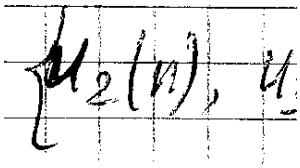


Uncertain reading. The second digit of (17) is overwritten and reads as 7.

Original source details: GELFAND-P001441

$$\{u_2(n), u_2(m)\}_2 = \frac{n^3 - n}{6}\delta_{n+m,0} + (n-m)u_2(n+m) \tag{18}_{\text{а}}$$

$$\{u_2(n), u_3(m)\}_2 = 3(n+1)u_3(m+n) - (m+n+3)u_3(m+n) \tag{18}_{\text{б}}$$



Uncertain reading. In (18)б the u of the first $u_2(n)$ is retraced, and the subscript reads 2.

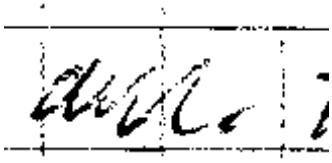
Original source details: GELFAND-P001441

$$\{u_3(n), u_3(m)\}_2 = \ldots + \sum_{k=-\infty}^{\infty}(m+n-k)u_2(m+n-k)u_2(k) \tag{18}_{\text{в}}$$

(square of the energy–momentum tensor)

Henceforth $\{\ \}$ always denotes $\{\ \}_2$.

Together this is the classical limit of some quantum current algebra.



Uncertain reading. The abbreviation before «токов» (“currents”) tentatively reads «алг.» (“algebra”).

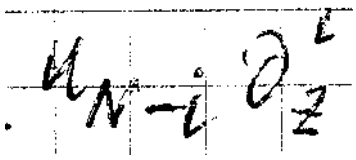
Original source details: GELFAND-P001441

GELFAND-P001442

Moral. one obtains an algebra with quadratic relations in place of a classical Lie algebra.
(Khovanova, T.)
Dynkin also introduced quadratic relations.
S. Lukyanov was working on the quantization of these brackets.
[On the left below the name: (Seryozha)] (S. Lukyanov is a graduate student of V. Fateev)

Digression The Miura transformation. Let

$$L(z) = \partial_z^N + \sum u_{N-i}\partial_z^i = (\partial_z + v_1)\dots(\partial_z + v_N) \tag{19}$$



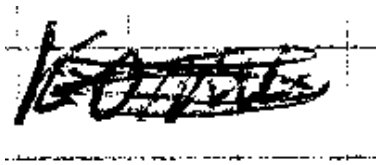
Original source details: GELFAND-P001442

Uncertain reading. In (19) the subscript on u reads $N - i$, with a dot over i.

— · —

remark
I. M. G. If $(\partial_z + v_1)(\partial_z + v_2)u = f$, then u is obtained by quadratures.

Algebraic equation in radicals — Galois theory
For differential equations the corresponding theory is Picard–Vessiot theory.
Kolchin [crossed out before the name: Kol...?] is working on this now.



Original source details: GELFAND-P001442

Uncertain reading. The crossed-out writing before «Колчин» (Kolchin) tentatively begins «Кол...».

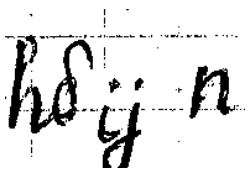
— · —

R.
Factorization (19): Miura
Kupershmidt — Wilson
Drinfeld — Sokolov

(20) Theorem. Let us give v_j the commutation relations

$$\{v_i(n), v_j(m)\} = h\delta_{ij}\, n\delta_{m+n,0}. \tag{21}$$

[On the right: (Lie algebra — [crossed mark] Heisenberg algebra)]



Original source details: GELFAND-P001442

Uncertain reading. The factor before δ_{ij} reads h.



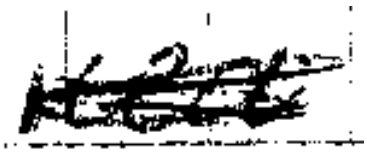
Uncertain reading. Before «Гейзенберга» (“Heisenberg”) is a crossed mark resembling H or #.

Original source details: GELFAND-P001442

Then u_i satisfy the algebraic G.–D. relations (Gelfand–Dickey)

(2nd G.–D. structure, here and below)

It turned out that the Lie algebra is isomorphic to a quadratic algebra [crossed out before “quadratic”: «квад...»?] which is not a Lie algebra.



Uncertain reading. The crossed-out writing before «квадр.» (“quadratic”) tentatively begins «ква...» or «квад...».

Original source details: GELFAND-P001442

Now let

$$L(z) = :(\partial_z + v_1)\dots(\partial_z + v_N): \tag{22}$$

What are the relations?

Introduce

GELFAND-P001443

[source number: 29]

$$M_0 = \mathrm{Hol}\big((\mathbb{C}\backslash 0)_z \times (\mathbb{C}\backslash 0)_w \longrightarrow \mathbb{C}\big)$$

[Below Hol: holomorphic maps.]

$$M = \mathrm{Hol}\big(((\mathbb{C}\backslash 0)_z \times (\mathbb{C}\backslash 0)_w \backslash \Delta) \longrightarrow U(v_1, \dots, v_N)\big)$$

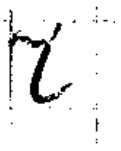
[A line connects U to the explanation: universal enveloping algebra of the Heisenberg algebra.]

An element of M is an operator-valued function.

Consider also the operator

$$\tau = \frac{\nu}{2}\ln(z-w)^2\,\sigma, \qquad \sigma f(z,w) = f(w,z).$$

on functions of two variables



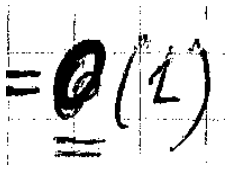
Uncertain reading. The sign with a top bar tentatively reads τ, possibly r. It recurs in (23).

Original source details: GELFAND-P001443

Then, if $\nu(\nu-1)=h$,

$$\exp(-\tau)L(z)L(w)\exp\tau[f]=\underline{\underline{O}}(1)\quad \text{as } z\longrightarrow w \tag{23}$$

$$\forall f\in M_0$$



Uncertain reading. The sign before (1) is retraced and double-underlined. It tentatively reads O, possibly 0.

Original source details: GELFAND-P001443

This is a set of relations on the u_i (on the coefficients)

Result:

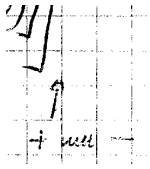
$$[u_i(z),u_j(w)]=\left[\sum_{k\geq 2}(z-w)^{-k}\sum_{p+q=i+j-k}c_{ij}^{pq}(k)[u_p(z),u_q(w)]\right] \tag{24}$$

[Below the outer right bracket is an arrow ↑ from the label "+ or −".]



Uncertain reading. The lower limit of the first sum tentatively reads $k\geq 2$. The sign between k and 2 is indistinct and may read $=$.

Original source details: GELFAND-P001443



Uncertain reading. The arrow from the label "+ or −" points to the outer right bracket of (24).

Original source details: GELFAND-P001443

$c_{ij}^{pq}(k)$ are expressed in terms of hypergeometric functions, but it is complicated

The form [below the word: «соотн.» ("relation")?] of (23) resembles a relation in the quantum inverse problem method.



Uncertain reading. The abbreviation below "The form" tentatively reads «соотн.» ("relation").

Original source details: GELFAND-P001443

$$L(z)L(w)=R^{-1}L(w)L(z)R. \tag{25}$$

M. Kapranov. Quiver representations and the derived category.

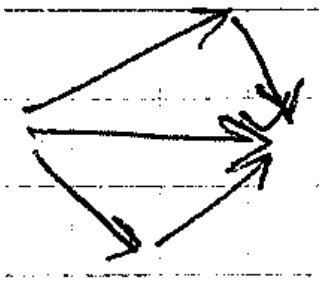
(On Parthasarathy's work on quiver representations.)



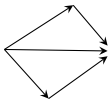
Original source details: GELFAND-P001444

Uncertain reading. The name tentatively reads «Партасарати» (Parthasarathy). See R. Parthasarathy's paper on t-structures in the derived category of quiver representations.

A quiver — a directed graph without cycles



Original source details: GELFAND-P001444

Diagram and typed description. 

Λ — graph

Ω — orientation

$M(\Lambda, \Omega)$ — [crossed out: «пр...»?] category of representations of the quiver



Original source details: GELFAND-P001444

Uncertain reading. The crossed-out writing before "category" tentatively begins «пр...», possibly «кат...».

$\mathcal{D}(\Lambda, \Omega)$ — derived category of this category.

(1) Lemma.

Then $\mathcal{D}(\Lambda, \Omega)$ does not depend on Ω

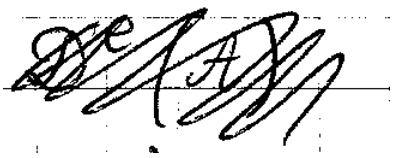
($\mathcal{D}(\Lambda, \Omega_1)$ and $\mathcal{D}(\Lambda, \Omega_2)$ are equivalent)

for different Ω_1 and Ω_2

Proof: application of symmetry.

Problem: find such [crossed out: $\mathcal{D}^{\ell}(\mathcal{A})$?] subcategories $\mathcal{A} \subset \mathcal{D}(\Lambda, \Omega)$, that

$$\mathcal{D}^{\ell}(\mathcal{A}) = \mathcal{D}(\Lambda, \Omega)$$



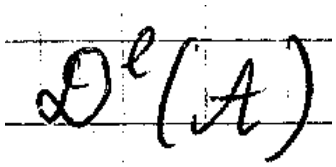
Original source details: GELFAND-P001444

Uncertain reading. The crossed-out formula tentatively reads $\mathcal{D}^{\ell}(\mathcal{A})$. The superscript also resembles e.



Original source details: GELFAND-P001444

Uncertain reading. The sign $\subset$ is retraced. An angular stroke is visible inside it.



Original source details: GELFAND-P001444

Uncertain reading. The sign above $\mathcal{D}$ in the manuscript resembles ℓ or e. Parthasarathy's paper uses $\mathcal{D}^b$ for the bounded derived category.

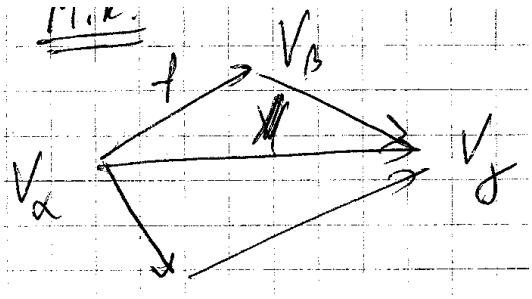
All of them have been found and are characterized by Dynkin graphs.

Reflections and gluings are used (and the philosophy of perverse sheaves)

Remark by I. M. G. The classification of vector bundles on CP^n can only be done in the language of derived categories.

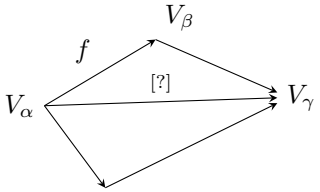
Grothendieck: through hard work he obtained the concepts leading to the derived category (like Michelangelo).

M. K.



Original source details: GELFAND-P001444

Diagram and typed description.



Above the middle arrow is a crossed-out mark, possibly g or φ. The lower vertex has no label.

Reflection a) $s_\alpha\Omega$ — reverse the arrows starting at α.

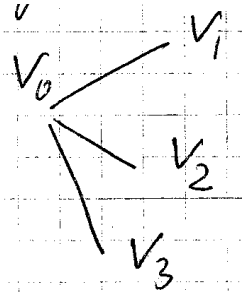
b)

$$(s_\alpha V)_{\beta\neq\alpha} = V_\beta$$

$$(s_\alpha V)_\alpha = \bigoplus_{\alpha\to\beta} V_\beta/(\text{the image of } V_\alpha)$$

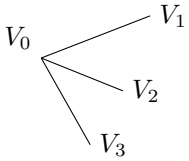
with the natural maps

There may be vectors in V_α that are mapped to 0 everywhere



Original source details: GELFAND-P001444

Diagram and typed description.

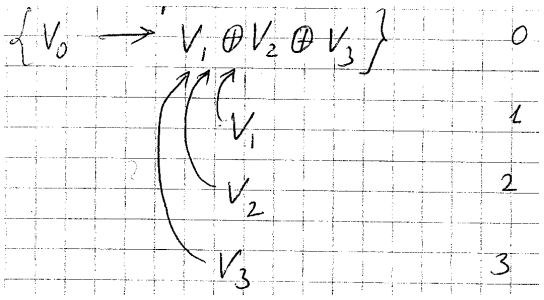


$$V_0 \longrightarrow V_1 \oplus V_2 \oplus V_3$$

GELFAND-P001445

[source number: 30]

Let us construct a complex from these data



Original source details: GELFAND-P001445

Diagram and typed description.

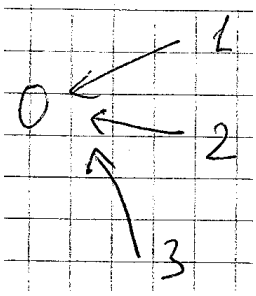
$\{V_0 \longrightarrow V_1 \oplus V_2 \oplus V_3\}$ 0

V_1 1

V_2 2

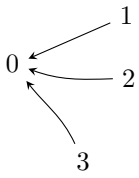
V_3 3

This is an object of the derived category (of quiver representations)



Original source details: GELFAND-P001445

Diagram and typed description.



To factorization in the ordinary category one must associate the map itself (the complex)

Parthasarathy poses and solves the question of when the derived category coincides with this one (for 3, but not for 4).



Original source details: GELFAND-P001445

Uncertain reading. The surname reads «Партасарати» (Parthasarathy), as on the preceding frame.

GELFAND-P001446

23. III. 88 Gelfand Seminar.

Zelevinsky, A. Matroids and Hall's theorem

I. M. G.

Matroid. [Above the next word, crossed out: illegible.] Example: linearly independent systems of vectors.



Original source details: GELFAND-P001446

Undeciphered passage. The writing above "Example" is crossed out and illegible.

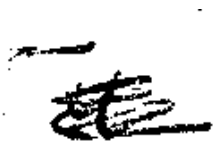
$$M_1, M_2 \text{ — sets}, \qquad \varphi \subset M_1 \times M_2$$

(relation)

Suppose M_1 has the structure of a matroid.

How do we introduce a matroid structure on M_2?

$$A \subset M_2, \qquad \pi_1\big(\pi_2^{-1}(A) \cap \varphi\big) \text{ — a set.}$$



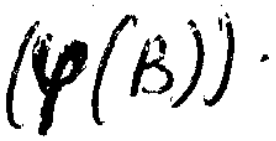
Undeciphered passage. The writing under the dash before "a set" is crossed out and illegible.

Original source details: GELFAND-P001446

The rank of the set A is determined

1°. Rado's theorem

$$r_{M_2}(A) = \min_{B \subset A}\big(r_{M_1}(\varphi(B)) + |A - B|\big), \qquad A \subset M_2.$$



Uncertain reading. The φ in $\varphi(B)$ is retraced in the rank formula.

Original source details: GELFAND-P001446

2°. Hall's theorem ≡ [Inserted above the surname: P.?]



Uncertain reading. The insertion before "Hall" tentatively reads «Ф.» (P.).

Original source details: GELFAND-P001446

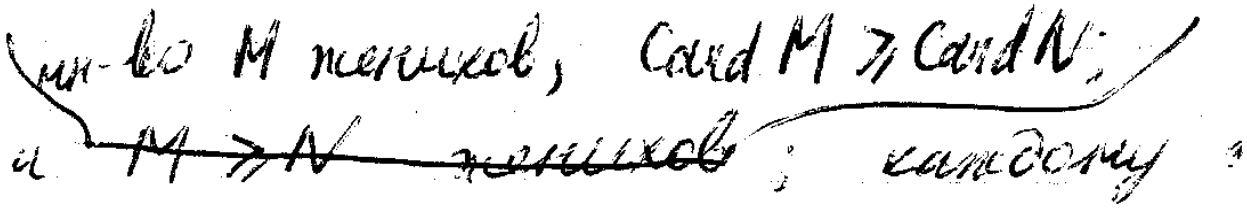
[Crossed out at the left: illegible.] Problem of choosing brides.



Undeciphered passage. The writing to the left of "Problem" is crossed out and illegible.

Original source details: GELFAND-P001446

If there is [inserted above N: a set] N of brides and [crossed out: $M \geq N$ grooms] [inserted above: a set M of grooms, Card $M \geq$ Card N], each groom $m \in M$ likes the brides $A(m) \subset N$;



Original source details: GELFAND-P001446

Uncertain reading. The insertion above the crossed-out "$M \geq N$ grooms" uses Card $M \geq$ Card N.

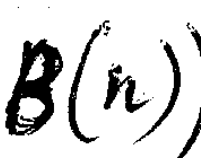
and [crossed out: $\forall A \subset N$] for every subset $K \subset N$,

$$\text{Card}\left(\bigcup_{n\in K} B(n)\right) \geq \text{Card}\, K.$$



Original source details: GELFAND-P001446

Undeciphered passage. The mark above Card is crossed out and illegible.



Original source details: GELFAND-P001446

Uncertain reading. The letter B is retraced in the last formula and reads $B(n)$.

GELFAND-P001447

[source number: 37]

(Here $B(n) \subset M$; [Below $B(n)$, joined by an equals sign:] the set of grooms who like n, i.e.

$$B(n) = \{m :\ n \in A(m)\}$$

[After $=$, crossed out: illegible.])



Original source details: GELFAND-P001447

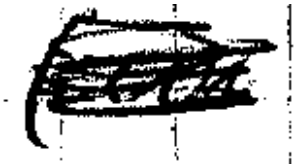
Undeciphered passage. The mark between $=$ and the opening brace is crossed out and illegible.

Then $\exists$ a one-to-one choice [crossed out: illegible]: a map

$$f : N \longrightarrow M,$$

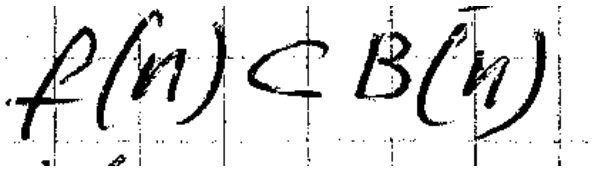
for which

$$f(n) \subset B(n) \qquad \forall n.$$



Original source details: GELFAND-P001447

Undeciphered passage. The writing after "choice" is crossed out and illegible.



Original source details: GELFAND-P001447

Uncertain reading. The manuscript uses $\subset$ between $f(n)$ and $B(n)$.

[Crossed-out line: Hall's theorem φ is a relation.]

[Crossed out: Theorem.] There exists a proof containing an algorithm.

Reduction to Rado's theorem: in N one must declare all sets independent.



Original source details: GELFAND-P001447

Uncertain reading. The letter after "in" is retraced heavily and reads N.

$2^\circ \Rightarrow 3^\circ$. Birkhoff's theorem.

It follows from Hall's theorem that if there is a subgroup $H \subset G$ [Below G: finite group], then $\exists$ a set of common representatives of left and right cosets.

I. M. G. There are several famous problems whose pointlessness is obvious to everyone.

Example. 1) Parallel axiom

(why this particular one?)

[Below the line:] they worked on it?

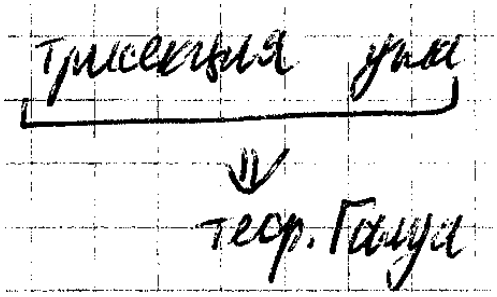


Original source details: GELFAND-P001447

Uncertain reading. Below the question the words tentatively read «занимались нею» ("they worked on it").

[To the right of the first two lines, beyond a vertical rule:] $\Rightarrow$ non-Euclidean geometry

2) Trisection of the angle and squaring the circle.



Original source details: GELFAND-P001447

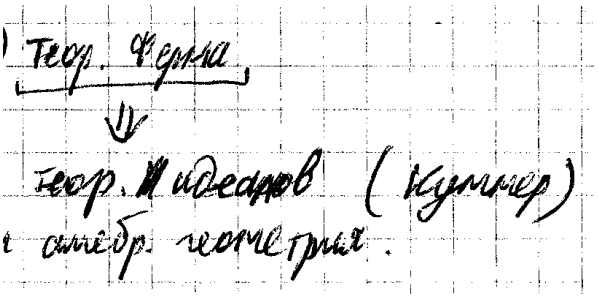
Diagram and typed description.

Trisection of the angle

$\Downarrow$

Galois theory

3) Fermat's theorem



Original source details: GELFAND-P001447

Diagram and typed description.

Fermat's theorem

$\Downarrow$

theory of [crossed out: illegible] ideals (Kummer) and algebraic geometry.

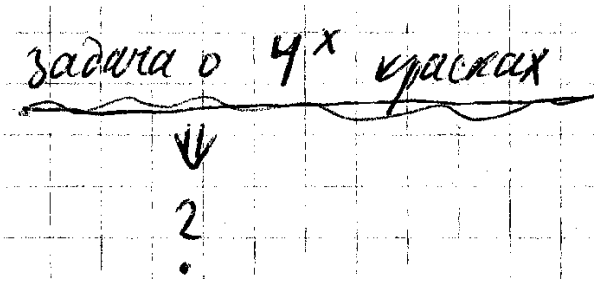


Original source details: GELFAND-P001447

Undeciphered passage. An illegible crossed-out mark stands before “ideals.”

[On the right, beyond a vertical rule:] work of Faltings.

4) The four-colour problem



Original source details: GELFAND-P001447

Diagram and typed description.
The four-colour problem
⇓
?

These works are worth going through.

5) Riemann hypothesis

GELFAND-P001448

A. Rudakov. Vector bundles, braid groups and Diophantine equations.

1°. On $\mathbb{P}^2$ $(= \mathbb{CP}^2)$

consider vector bundles (or coherent sheaves)



Original source details: GELFAND-P001448

Uncertain reading. The ending of the word for “sheaves” is heavily retraced.

$$E_1,\ E_2,\ E_3 \tag{1}$$



Original source details: GELFAND-P001448

Undeciphered passage. Illegible marks are crossed out after E_1.

$$\begin{aligned} &\operatorname{Ext}^k(E_j, E_i) = 0, \quad j > i, \quad k = 0, 1, 2, \\ &\operatorname{Ext}^k(E_i, E_i) = \begin{cases} 0, & k > 0, \\ \mathbb{C}, & k = 0. \end{cases} \end{aligned} \tag{2}$$



Original source details: GELFAND-P001448

Undeciphered passage. An illegible mark is crossed out before $k = 0, 1, 2$.

Such bundles exist, and if dim $E_j = x_j$, then

$$x_1^2 + x_2^2 + x_3^2 = 3x_1x_2x_3 \qquad \text{(Markov equation)} \tag{3}$$

(and all [inserted above: integer] solutions are realized this way!)

The braid group acts on the collections [inserted above: (1) (there are countably many of them)], and all the collections (1) form 1 orbit!

2°.

$$Q = \mathbb{P}^1 \times \mathbb{P}^1.$$



Undeciphered passage. An illegible mark is crossed out after $Q =$.

Original source details: GELFAND-P001448

Then take 4 bundles E_1, E_2, E_3, E_4 with the same conditions (2). The braid group acts here too.

In place of (3) we get a system of Diophantine equations (but the correspondence goes one way. It has not been proved that all solutions are obtained). It is something like a root system, with a braid group in place of the Weyl group.

3°. On $\mathbb{P}^n$ one can consider bundles

$$E_1, E_2, \ldots, E_{n+1} \qquad \text{with conditions (2)}$$

and again an action of the braid group arises.

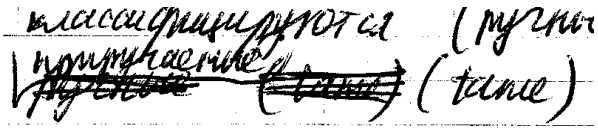
M. Kapranov: on the action of the braid group.

I. M. G.

Solvable and unsolvable classification problems.

No parameters $\Rightarrow$ can be classified (tame)

1 parameter $\Rightarrow$ [crossed out: «ручные» (“tame”) and illegible writing in parentheses$^?$; inserted above the line: semi-tame] (tame)

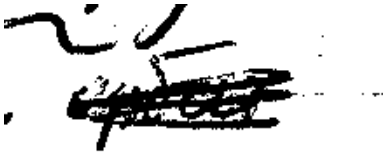


Uncertain reading. «ручные» (“tame”) and illegible writing in parentheses

Original source details: GELFAND-P001448

≥ 2 parameters $\Rightarrow$ unsolvable (e.g., pairs of matrices)

and classification up to



Undeciphered passage. Illegible writing is crossed out before the map.

Original source details: GELFAND-P001448

$$(A, B) \mapsto (C^{-1}AC, C^{-1}BC).$$

GELFAND-P001449

[source number: 38]

In unsolvable problems there is always a solvable part (an open orbit!)



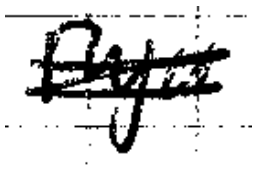
Original source details: GELFAND-P001449

Uncertain reading. The writing above the initials is crossed out. It resembles «Еф» or the start of Ext.

M. K.

$$\mathrm{Ext}^1(E,E)=0 \;\Rightarrow\; \text{rigidity}$$

(and such bundles have no parameters)



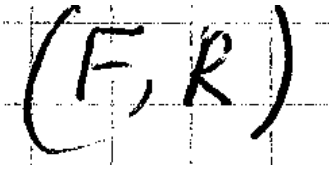
Original source details: GELFAND-P001449

Uncertain reading. The word crossed out above the beginning of the sequence tentatively reads «Пару» (“a pair”).

$$0 \longrightarrow R \longrightarrow E \otimes \mathrm{Hom}(E,F) \longrightarrow F$$

Call a pair (E,F) exceptional if $\mathrm{Ext}^j(F,E)=0$.

Then the pair (F,R) is also exceptional.



Original source details: GELFAND-P001449

Uncertain reading. The manuscript gives the order (F,R).

We obtain an action of $\mathbb{Z}$ on such pairs. The relations giving braids hold.

I. M. G. Work of Bernstein–Gelfand–Ponomarev:

in representations of graphs the Coxeter group acts.

What is needed from one's general education to work on, e.g., spectral theory.

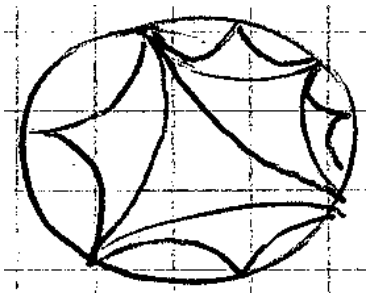
1) Representations of graphs (a quadruple of subspaces, etc.)

2) Derived category.

Beilinson, A. Perverse sheaves.

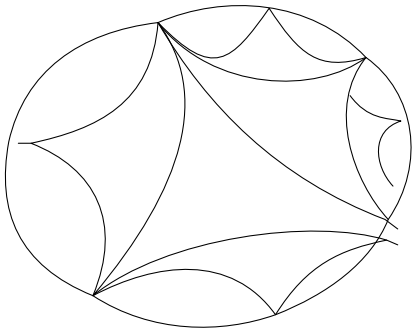
I. M. G.

(on MacPherson)



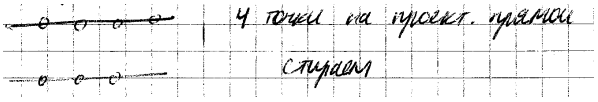
Original source details: GELFAND-P001449

Diagram and typed description.



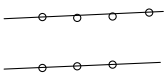
The strokes overlap at the lower-right junction. Their exact connections are unclear.

How are these regions to be parametrized?



Original source details: GELFAND-P001449

Diagram and typed description.



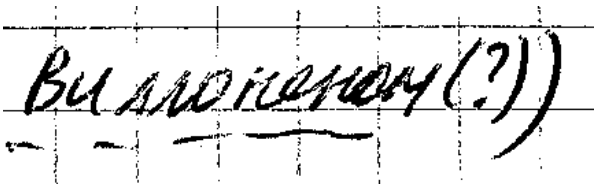
4 points on a projective line

erase

Siegel domains $\Rightarrow$ configurations of lines in $\mathbb{RP}^2$.

B.

On the work of MacPherson (on his work with Gelfand and Vilonen(?))



Original source details: GELFAND-P001449

Uncertain reading. The surname tentatively reads «Виллоненом» (Vilonen). The parenthesized question mark is in the manuscript.

Perverse sheaves?

A perverse sheaf corresponds to problems in linear algebra that describe it completely, but it is unclear how to single out the perverse sheaves.

GELFAND-P001450

X is a complex manifold (smooth)

Constructible sheaf.

A stratification is given $\{X_i\}$ [above X_i, with a leader: open], a sheaf $\mathcal{F}$ on X is called constructible [inserted below: with respect to it], if $\mathcal{F}|_{X_i}$ is a locally constant sheaf of finite-dimensional vector spaces.

(locally with respect to the usual topology)

Constructible complexes of sheaves

$$\ldots \longrightarrow \mathcal{F}^{-1} \longrightarrow \mathcal{F}^0 \longrightarrow \mathcal{F}^1 \longrightarrow \mathcal{F}^2 \longrightarrow \ldots \tag{1}$$

a complex of sheaves is called constructible if all the sheaves $H^i(\mathcal{F}^\bullet)$ are constructible.

(there are no conditions on $\mathcal{F}^i$!)

Singular support of a sheaf or complex.

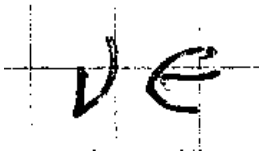
$\mathcal{F}^\bullet$ is a constructible complex.

$$SS(\mathcal{F}) \subset T^*X \tag{2}$$

[Below $SS(\mathcal{F})$: subset]

is defined as follows.

Let $\nu \in T^*_x X,\ \nu \neq 0$.



Original source details: GELFAND-P001450

Uncertain reading. The covector letter reads as ν or Latin v.

(3) Def.

$$\nu \notin SS(\mathcal{F}) \iff$$

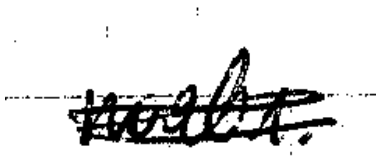
[Inserted above: if] one chooses a small neighbourhood $U \ni x$, a holomorphic function f on U, with $df_x = \nu$, $f(x) = 0$, then $\forall$ points $\varepsilon \in \mathbb{C}$ near 0

$$H^*\big(U, \underset{\cap U}{f^{-1}(\varepsilon)}; \mathcal{F}\big) = 0. \tag{4}$$

(relative cohomology groups mod the fibre)

(more precisely, one must pass to the inductive limit)

([crossed out: appearance] $\neq 0 \iff$ appearance of vanishing cycles)



Original source details: GELFAND-P001450

Uncertain reading. The crossed-out writing tentatively reads «появл.» (“appearance”).

(4) is independent of the choice of f

Zero section?

Add the zero section if $\mathcal{F}^\bullet$ is a nonzero complex.

$\mathcal{F}^\bullet$ is zero $\Rightarrow SS(\mathcal{F}^\bullet) \overset{\text{def}}{=} \varnothing$.

GELFAND-P001451

[source number: 39]

(5) Theorem. $SS(\mathcal{F})$ is a conic Lagrangian subset

(an analytic subset whose nonsingular points are Lagrangian in the usual sense [mark crossed out]; an equivalent condition is involutivity of the ideal)



Uncertain reading. The crossed-out mark before the semicolon resembles a closing parenthesis.

Original source details: GELFAND-P001451

Structure: a union of irreducible components (finitely many)

(6) Lemma. If $Y \subset T^*X$ is an irreducible Lagrangian conic subset, then

$$Y = N^*Z \qquad \text{for some irreducible } Z \subset X. \tag{7}$$

(conormal bundle)

(then $Z = \pi(Y)$, $\pi : T^*X \longrightarrow X$)

Here N^*Z is understood [crossed out: illegible] as the Zariski closure of N^*(smooth part of Z).



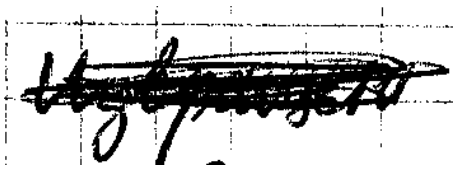
Undeciphered passage. The writing crossed out between “understood” and “as” is illegible.

Original source details: GELFAND-P001451



Undeciphered passage. An illegible mark is crossed out between the opening parenthesis and “smooth part.”

Original source details: GELFAND-P001451



Uncertain reading. The crossed-out writing above the heading tentatively begins «Извращ...» (“Perverse...”).

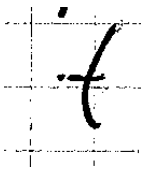
Original source details: GELFAND-P001451

(8) Definition. Let a conic Lagrangian subset be given $Y \subset T^*X$,

f is a holomorphic function on X.

f is called Morse with respect to Y, if it has the following property:

$df(X)$ (a Lagrangian subset (nonconic)) is transverse to Y,



Uncertain reading. The opening parenthesis after $df(X)$ has a short cross-stroke.

Original source details: GELFAND-P001451

that is, $df(X)$ intersects Y only at smooth points, and intersects transversely.

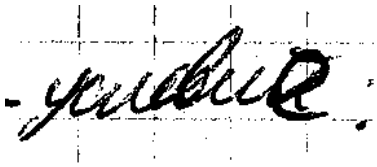
Examples.

1. $Y = X$ is the zero section.

then the Morse property of f with respect to $Y =$ the Morse property in the usual sense.

2. $Y =$ the conormal to Z, $Y = N^*Z$, Z is a smooth manifold.

Then the Morse property means satisfaction of [crossed out: two] the condition.



Original source details: GELFAND-P001451

Uncertain reading. The ending is overwritten. It tentatively reads «условие» (“condition”), possibly «условий» (“of conditions”).

[The label “a)” is crossed out.] $g = f|_Z$ is a Morse function (in the usual sense)

[The following two lines are boxed and crossed out with two diagonals.]

$$\text{b)} \quad dg(z) = 0 \;\Rightarrow\; df(z) \neq 0. \qquad (?)$$

(If Y contains the zero section, then this holds)

GELFAND-P001452

(9) Definition. Let $\mathcal{F}^\bullet$ be a constructible complex, and consider $SS(\mathcal{F}^\bullet)$, let f be Morse with respect to $SS(\mathcal{F}^\bullet)$ and let x be a critical point with respect to $SS(\mathcal{F}^\bullet)$ $\left(\overset{\text{def}}{\Longleftrightarrow} df(x) \in SS(\mathcal{F}^\bullet)\right)$.



Original source details: GELFAND-P001452

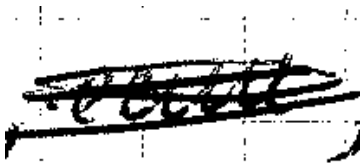
Uncertain reading. The equivalence arrow is heavily retraced, with “def” written above it.

Then

$$H^*\left(U, f^{-1}(\varepsilon); \mathcal{F}\right) \neq 0.$$

The complex $\mathcal{F}$ is called a perverse sheaf, if all such groups occur in only one degree (e.g., 0) [crossed out: if], that is, if

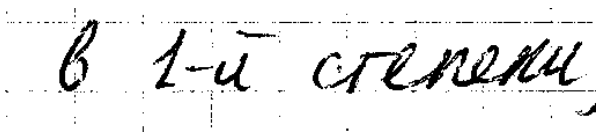
$$H^j\left(U, f^{-1}(\varepsilon), \mathcal{F}\right) = 0 \quad \text{for } j \neq 0. \qquad (10)$$



Original source details: GELFAND-P001452

Uncertain reading. The word crossed out after “e.g., 0” tentatively reads “if.”

(vanishing cycles occur only in the 1st degree)



Original source details: GELFAND-P001452

Uncertain reading. The note reads «в 1-й степени». The condition in (10) is written for $j \neq 0$.

Examples. a. $\mathbb{C}_{x_0}$ is a perverse sheaf. [Under the symbol: concentrated at a point.] (stalk $\mathbb{C}$ over x_0, 0 on the complement)

b. $\mathbb{C}_X[n]$ is the constant sheaf over X (shifted in grading) is a perverse sheaf.

$$n = \dim_{\mathbb{C}} X.$$

c. $Z \subset X$ [with a leader: smooth subset]

$$\mathbb{C}_Z[\dim Z]$$

[Under the symbol:] the constant sheaf on Z, extended by zero outside Z.

(or a shift of the Goresky–MacPherson complex)

Remark. [Under the heading: (approximate).] The strata with respect to which the sheaf $\mathcal{F}$ is constructible are the projections to the base of the irreducible components of its $SS(\mathcal{F})$.

Construction 1. Let $\mathcal{F}$ is a perverse sheaf. We construct a local system [inserted below: $\Phi(\mathcal{F})$] at the smooth points of its $SS(\mathcal{F})$.

Let [crossed out: illegible] z be a smooth point of $SS(\mathcal{F})$.

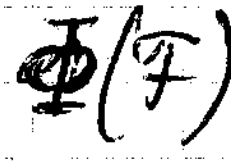


Undeciphered passage. Illegible writing is crossed out before z.

Original source details: GELFAND-P001452

Then

$$\Phi(\mathcal{F}) = H^0\big(U, f^{-1}(\varepsilon), \mathcal{F}\big), \qquad df(\pi(z)) = z. \tag{11}$$



Uncertain reading. The Φ in (11) is heavily retraced. The insertion above shows it clearly.

Original source details: GELFAND-P001452

(the choice of f makes no difference)

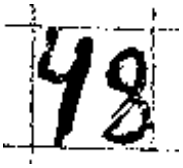
Next time in 2 weeks.

GELFAND-P001453

[source number: 48]

11/IV–88 Gelfand Seminar

D. B. Fuchs The four-colour problem. (Appel, Haken)



Uncertain reading. The number reads “48.” The lower loop of the second digit is thinner than the upper loop.

Original source details: GELFAND-P001453

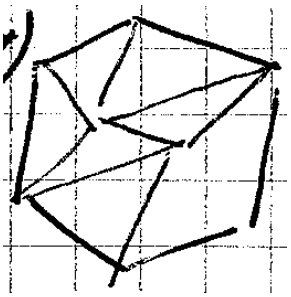
Def. A configuration is a triangulation of a disc.

[Crossed out on the right: a subdivision of a sphere (disc) (triangulation).]



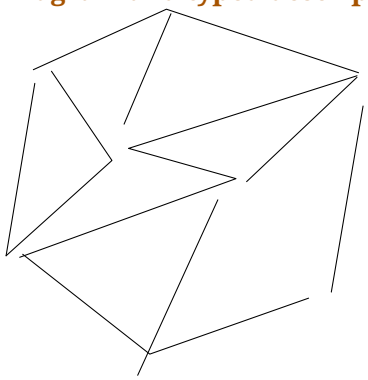
Original source details: GELFAND-P001453

Uncertain reading. The crossed-out word in the first parentheses tentatively reads “disc.”



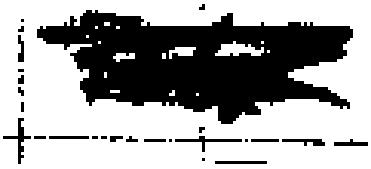
Original source details: GELFAND-P001453

Diagram and typed description.



The redraw preserves the visible gaps in the outline and interior strokes.

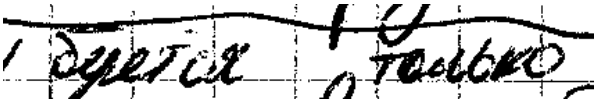
(1) We must prove that the vertices can be coloured with 4 colours [crossed out: illegible], so that adjacent vertices have different colours.



Original source details: GELFAND-P001453

Undeciphered passage. Illegible writing is crossed out after “4 colours.”

[Small annotation] (this is the dual problem to face colouring, so the other only



Original source details: GELFAND-P001453

Uncertain reading. The end of the note tentatively reads «другая только» (“the other only”). No closing parenthesis is visible.

(2) A set of configurations is called unavoidable, if one of them occurs in every triangulation.

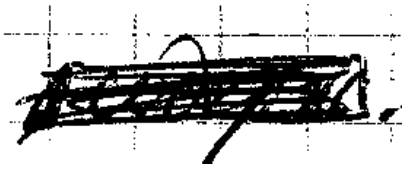
(3) A configuration is called reducible, if it cannot occur in a triangulation of the sphere that cannot be coloured with 4 colours and has the fewest vertices among such triangulations.

Reducible triangulation K: [crossed out: if; inserted above: suppose] there is a triangulation $T \supset K$ and it is known that $\forall$ configuration [crossed out: illegible] with fewer vertices (than T) can be coloured with 4 colours. Then T can also be coloured.



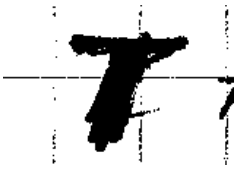
Original source details: GELFAND-P001453

Uncertain reading. The word crossed out before “there is” tentatively reads “if.”



Original source details: GELFAND-P001453

Undeciphered passage. The writing after “configuration” is crossed out and illegible.



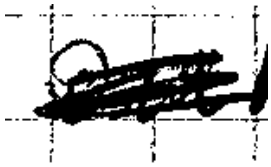
Original source details: GELFAND-P001453

Uncertain reading. The symbol before “also” is heavily retraced and reads T.

Induction on the number of vertices N.

Suppose it has already been proved for $N-1$ vertices.

configuration [crossed out: this is] K is called reducible, if [“from” inserted above] its occurrence in T with $|T| = N$ implies that T can be coloured.



Original source details: GELFAND-P001453

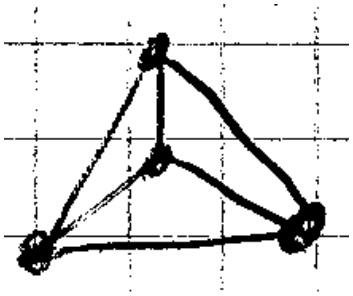
Uncertain reading. The word before K is crossed out, possibly «это» (“this is”).

[Crossed out: “Example.”]

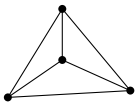
The main idea is to combine definitions (2) and (3).

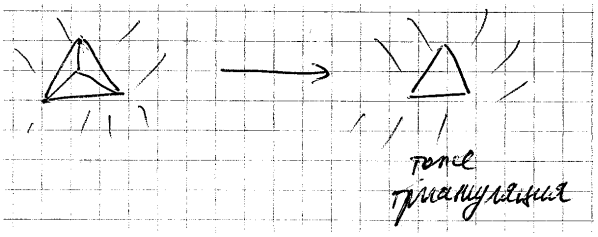
[Crossed out: “We need”; inserted above: “It suffices”] to construct an unavoidable set of reducible configurations.

(4) Example.



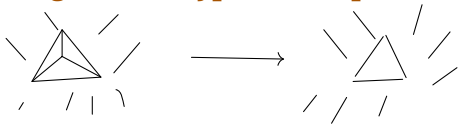
Original source details: GELFAND-P001453

Diagram and typed description. 
— a reducible configuration.



Original source details: GELFAND-P001453

Diagram and typed description.



also a triangulation

GELFAND-P001454

I. M. G. One should invent everything using the simplest examples. (I. Bernstein learned this from I. M. G.)

I. M. G. devised the whole theory of spherical functions using the formula

$$\cos(\alpha+\beta)+\cos(\alpha-\beta)=2\cos\alpha\cdot\cos\beta.$$

On $\mathbb{R}$ one should consider the group generated by translations and reflections.



Uncertain reading. The writing above the heading is crossed out and may read «Утв.» (“Proposition”).

Original source details: GELFAND-P001454

(5) Theorem

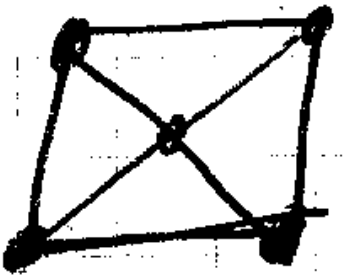
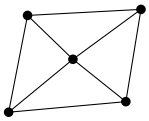


Diagram and typed description. 

reducible

Original source details: GELFAND-P001454

(Kempe, ~ 100 years ago)

Replace it with

(6)

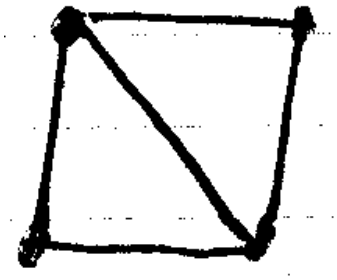
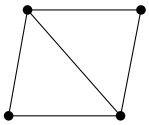


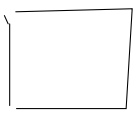
Diagram and typed description. 

Original source details: GELFAND-P001454

and colour the resulting triangulation.

(*) If any of the 4 have the same colour, colour the centre with the unused one.

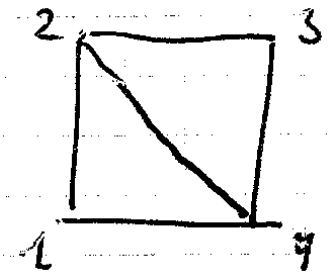


Diagram and typed description. 

Original source details: GELFAND-P001454

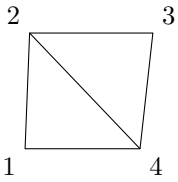
it remains to consider the case

(7)

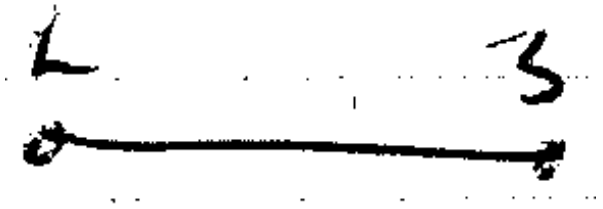


Original source details: GELFAND-P001454

Diagram and typed description.



Consider the subset K, consisting of vertices coloured 1 and 3 and edges



Original source details: GELFAND-P001454

Diagram and typed description.

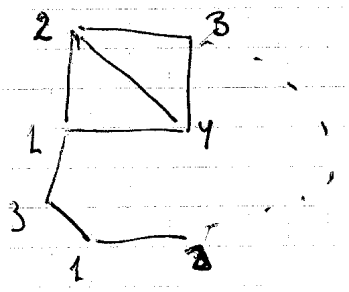


(8) 1 may a) be connected to 3 in K

or b) be disconnected from it.

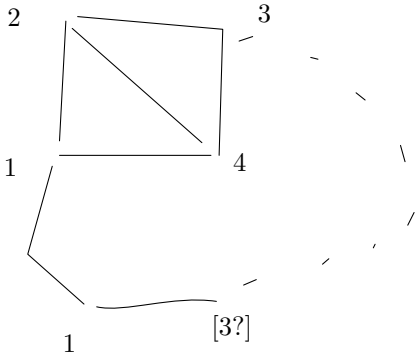
Case b). In the connected component of vertex 1 interchange 1 and 3 and reduce to the previous case (*).

(9)



Original source details: GELFAND-P001454

Diagram and typed description.



The lower-right digit is overwritten and may be 3. The continuation on the right is drawn as separate strokes.

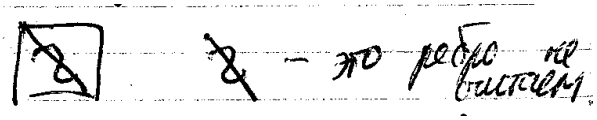
[To the right of the figure] Suppose we are in case 8a).

Then there is no path from 2 to 4 with [crossed out: illegible] 2 and 4.



Original source details: GELFAND-P001454

Undeciphered passage. The marks crossed out after “with” are illegible. The first resembles a 2.



Original source details: GELFAND-P001454

Diagram and typed description.



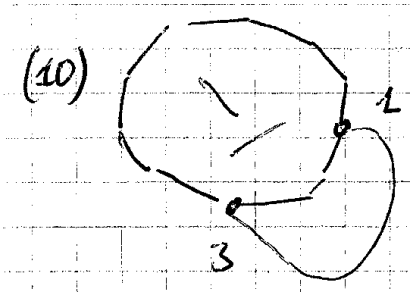
— do not count this edge.

(since it cannot cross the path of 1s and 3s)

and then reduce again to 8b).

GELFAND-P001455

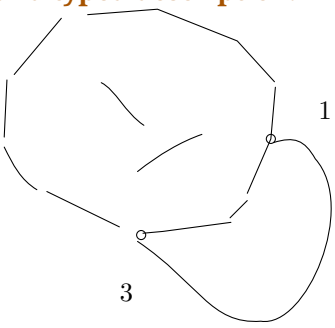
[source number: 49]



Original source details: GELFAND-P001455

Diagram and typed description.

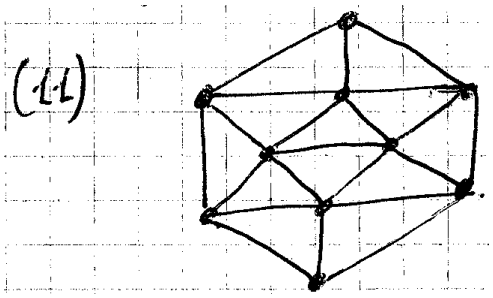
(10)



Consider a more complicated configuration and all possible colourings of its boundary. (With 4 colours)

Some extend to the interior and some do not.

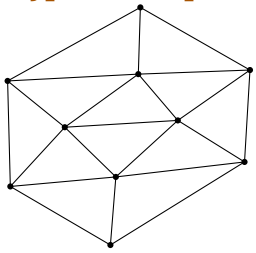
One can try recolouring as in (9) (using chains). The computer uses this method to look for reducible configurations.



Original source details: GELFAND-P001455

Diagram and typed description.

(11)



A reducible configuration (but Fuchs [crossed out: illegible mark] could not prove it)



Original source details: GELFAND-P001455

Uncertain reading. The crossed-out mark after “Fuchs” resembles «у» or «д».

I. M. G. Problem 6.



Original source details: GELFAND-P001455

Uncertain reading. The mark after “Problem” is heavily retraced. It may be the digit 6 or the letter «в».

Write a program for determining reducibility on a parallel computer.

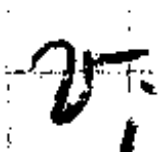
One of Appel and Haken's configurations (out of ~ 1700) was checked by hand (Allaire), since the computer could not handle it.

Unavoidable set.

Their construction is based on Euler's theorem

$$\mathrm{B} - \mathrm{P} + \Gamma = 2. \tag{12}$$

Kempe: let ν_i be the number of vertices joined to exactly i other vertices (with i edges issuing from each)

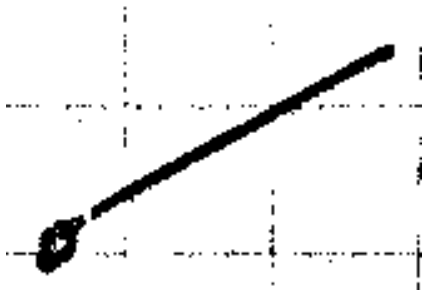


Original source details: GELFAND-P001455

Uncertain reading. The handwritten symbol can be read as ν_i or v_i.

$$\mathrm{B} = \nu_3 + \nu_4 + \nu_5 + \dots \tag{13}$$

$$\mathrm{P} = \frac{1}{2}(3\nu_3 + 4\nu_4 + 5\nu_5 + \dots) \tag{14}$$



Original source details: GELFAND-P001455

Diagram and typed description.

$$2\mathrm{P} = \text{the number of such pairs} = 3\nu_3 + 4\nu_4 + \dots$$

(a vertex and an edge issuing from it)

$$3\Gamma = 2\mathrm{P} \qquad \text{(since it is a triangulation)} \tag{15}$$

hence

$$\Gamma = \frac{3\nu_3 + 4\nu_4 + 5\nu_5 + \dots}{3} \tag{15'}$$

From (12) we find:

$$3\nu_3 + 2\nu_4 + \nu_5 - \nu_7 - 2\nu_8 - 3\nu_9 - \dots = 12, \tag{16}$$

that is,



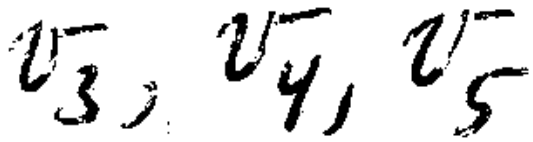
Original source details: GELFAND-P001455

Uncertain reading. A number resembling (17) or (16) is crossed out before (16').

$$\sum_{i=3}^{\infty}(6-i)\nu_i = 12. \tag{16'}$$

GELFAND-P001456

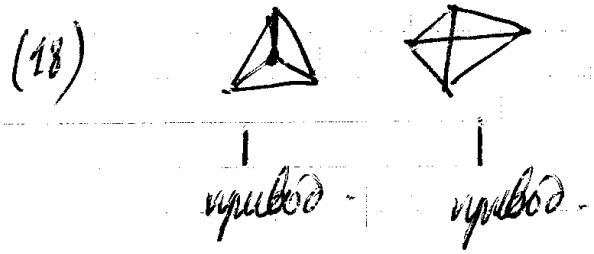
(17) Hence one of the numbers ν_3, ν_4, ν_5 is different from 0.



Original source details: GELFAND-P001456

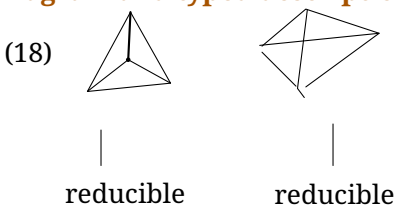
Uncertain reading. The letter in the three labels can be read as ν or v.

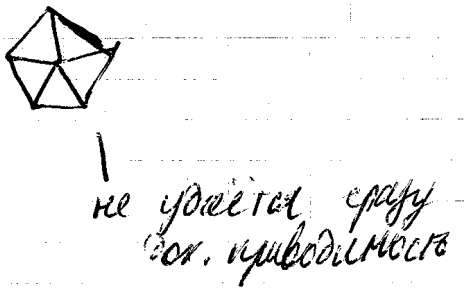
We obtain an unavoidable set



Original source details: GELFAND-P001456

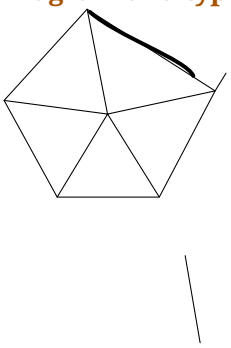
Diagram and typed description.





Original source details: GELFAND-P001456

Diagram and typed description.



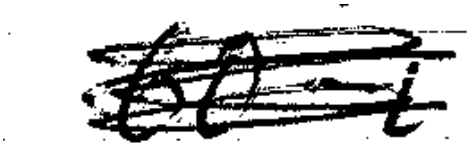
The upper-right side is heavily retraced. A short stroke of unclear purpose extends from its end.

Now one need only consider triangulations with

$$\nu_3 = \nu_4 = 0 \tag{19}$$

Discharging [Russian gloss: «разрядка»]

(20) Initial charge: to a vertex from which i edges issue, assign a charge [crossed out: $60 - i$] $60(6 - i)$.



Original source details: GELFAND-P001456

Uncertain reading. The crossed-out expression tentatively reads $60 - i$, possibly $6 - i$.

Redistribution procedure: construct another function on the vertices.

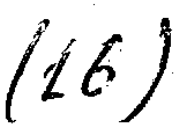
$$(16') \iff \text{sum of all charges} = 720.$$

(21) We want [inserted above: to make it so] that all charges are ≤ 0.
(which is impossible [crossed out: illegible] in view of equation (16))



Original source details: GELFAND-P001456

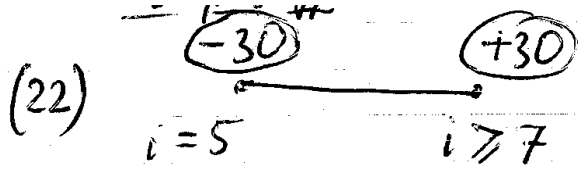
Undeciphered passage. Illegible writing is crossed out after “impossible.”



Original source details: GELFAND-P001456

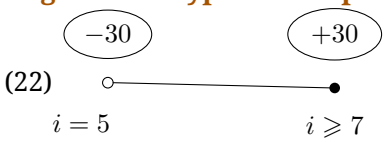
Uncertain reading. The equation number after “equation” is written without a prime.

Procedure 1.

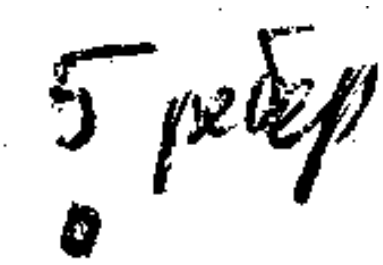


Original source details: GELFAND-P001456

Diagram and typed description.



(decrease by 30 at the 1st vertex, increase by 30 at the 2nd)
Some situations in which a positive charge remains somewhere.
(23) 1°.



Original source details: GELFAND-P001456

Diagram and typed description.



is joined to 0 or 1 [crossed out: edg...] vertices with number of edges $\geqslant 7$.



Original source details: GELFAND-P001456

Uncertain reading. The crossed-out word resembles «рёбр» or «рёбер» (“edges”).

(⇒ a vertex with positive charge remains



Uncertain reading. A separate mark, possibly 2, is crossed out before (24).

Original source details: GELFAND-P001456

(24) 2°.



Diagram and typed description. deg. 7

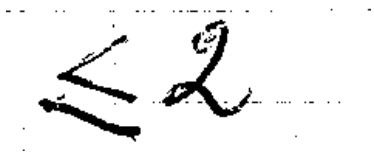
Original source details: GELFAND-P001456

is joined [crossed out: edge with] to $\leqslant 2$ vertices of degree 5



Uncertain reading. The crossed-out writing tentatively reads «ребро с» (“edge with”), possibly «ребра с» (“edges with”).

Original source details: GELFAND-P001456



Uncertain reading. The manuscript has $\leqslant$ before 2.

Original source details: GELFAND-P001456

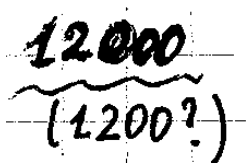
We end up with a finite set of configurations, which is unavoidable, since in the absence of all of them we can make every charge ≤ 0.

GELFAND-P001457

[source number: 50]

Now they must be checked for reducibility.

In 1977 the authors (Appel and Haken) thank the administration for 12 000 hours of computer time. [Written below the number: (1200?).]



Uncertain reading. The line gives “12000,” with “(1200?)” written below.

Original source details: GELFAND-P001457

Then the configurations are reworked into more complicated ones, and the transfer rules are reworked too.

A. Beilinson (continued)

(on work by K. Vilonen, S. Gelfand and R. D. MacPherson)

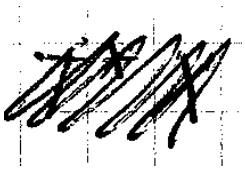
[Below "D.", with a leader: "Duncan."]

X is a smooth [inserted above: complex] algebraic variety.

$\mathcal{F}$ is a finite-dimensional constructible [crossed out: sheaf; inserted above: complex of sheaves] on X.

From these data one constructs

$SS(\mathcal{F})$ a Lagrangian conic set in [crossed out: T^*X] $\Omega X = T^*X$.



Original source details: GELFAND-P001457

Uncertain reading. The crossed-out notation before ΩX tentatively reads T^*X.

$$SS(\mathcal{F}) = \text{zero section} \iff H^i(\mathcal{F}) \text{ is smooth, i.e. locally constant.}$$

Def. $SS\mathcal{F}$

$$H^\bullet(\mathcal{F}) = 0 \quad \Rightarrow \quad SS(\mathcal{F}) = \varnothing \tag{1}$$

$$H^\bullet(\mathcal{F}) \neq 0 \quad \Rightarrow \quad SS(\mathcal{F}) \supset \text{zero section}, \quad \text{and}$$

$SS(\mathcal{F}) \setminus (\text{zero section})$ is described as follows

$$\nu \in \Omega^1_x, \qquad \text{choose } f\colon \quad df_x = \nu, \quad f(x) = 0$$

[Below ν, with a leader: "vector."]

Let U be a small neighbourhood of the point x

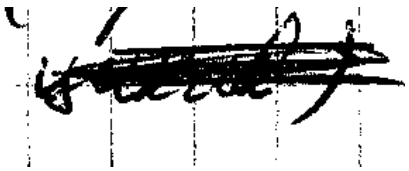
$$H^\bullet\big(U, U \cap f^{-1}(\varepsilon); \mathcal{F}\big) = \Phi^\bullet_f(\mathcal{F})_x \tag{2}$$



Original source details: GELFAND-P001457

Uncertain reading. Before the semicolon in (2), a symbol resembling $\mathcal{F}$ is crossed out.

(space of vanishing [crossed out: cocycles...] cycles)

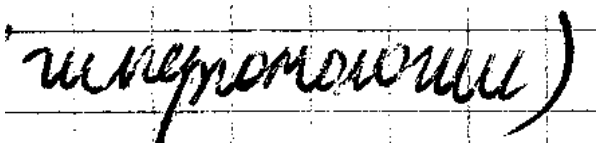


Original source details: GELFAND-P001457

Uncertain reading. The crossed-out word may read «коциклов» ("cocycles") or «когомол.» ("cohomology").

$\varepsilon \neq 0$ is small

(hyperhomology)



Uncertain reading. The manuscript tentatively reads “hyperhomology,” with “hypercohomology” another possible reading.

Original source details: GELFAND-P001457

(relative cohomology is defined via the cone) of the map, i.e. write down the bicomplex and totalize it

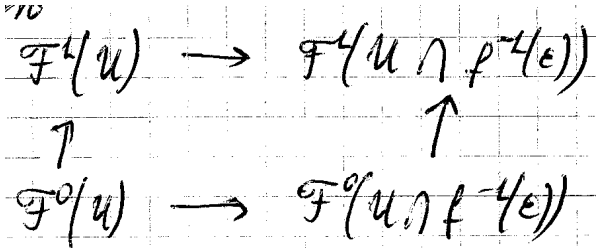


Diagram and typed description.

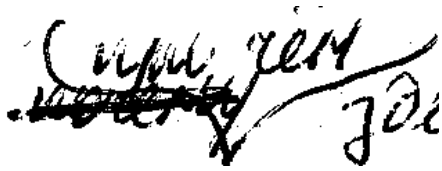
$$\begin{array}{ccc} \mathcal{F}^1(U) & \longrightarrow & \mathcal{F}^1(U \cap f^{-1}(\varepsilon)) \\ \uparrow & & \uparrow \\ \mathcal{F}^0(U) & \longrightarrow & \mathcal{F}^0(U \cap f^{-1}(\varepsilon)) \end{array}$$

Original source details: GELFAND-P001457

(if $\mathcal{F}$ is sufficiently acyclic)

GELFAND-P001458

an explanation of [crossed out: why; inserted above: what role] vanishing cycles play here.



Uncertain reading. The crossed-out word tentatively reads «почему» (“why”), possibly «зачем» (“what for”).

Original source details: GELFAND-P001458

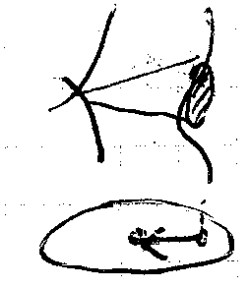
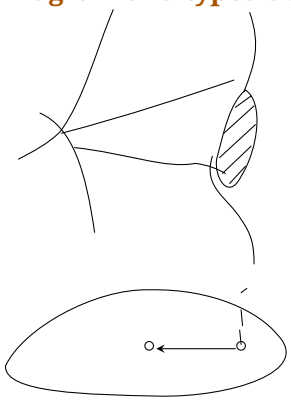


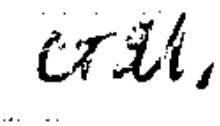
Diagram and typed description.



The lower arrow points left. Additional strokes inside the oval overlap it.

Original source details: GELFAND-P001458

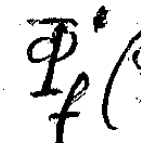
[On the right] the cycle contracts to a point



Uncertain reading. The abbreviation after “cycle” tentatively reads «стян.», possibly «стяг.».

Original source details: GELFAND-P001458

Df. $\nu \notin SS(\mathcal{F}) \iff$ vanishing cycles at ν are 0, i.e. $\Phi^{\bullet}_f(\mathcal{F})_x = 0$.



Original source details: GELFAND-P001458

Uncertain reading. The superscript on Φ_f resembles a heavy dot. It could also be i or $*$.

Consider the direct image of $\mathcal{F}|_U$ under the map $U \xrightarrow{f}$ disc.

Vanishing cycles measure how much the fibre at 0 differs from the neighbouring fibres.

f is called a Morse function on X, if [inserted above: f is holomorphic on X and] the graph $df(X) \subset$ [crossed out: illegible] Ω^1 intersects transversely $SS(\mathcal{F})$.



Original source details: GELFAND-P001458

Undeciphered passage. An illegible mark is crossed out before Ω^1.

A complex is called a perverse sheaf, if $\forall$ Morse with respect to $\mathcal{F}$ functions have vanishing cycles only in dimension 0.

(only $\Phi^0_f(\mathcal{F}) \neq 0$)

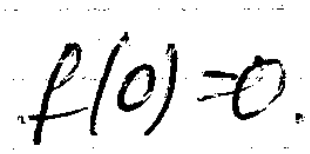
$\mathcal{F}$ is a perverse sheaf.

It gives a local system on the regular points of the singular support.

Its stalk over the point ν:

$$\Phi(\mathcal{F})_\nu = \Phi^0_f(\mathcal{F})_x$$

for a Morse function f with $df(x) = \nu$, $f(0) = 0$.



Original source details: GELFAND-P001458

Uncertain reading. The manuscript gives $f(0) = 0$.

What happens at singular points?

Problem. Describe perverse sheaves microlocally, i.e. in terms of vanishing cycles.

The work solves half of this problem.

Point. Where do perverse sheaves come from?

1°. $\mathcal{M}$ is an overdetermined system

[Under "system": (holonomic)]

[On the right] $\sum f_i \partial^i$

[Under f_i, with a vertical leader: holomorphic.]

$$\mathcal{F}^{0,0} \xrightarrow{\bar\partial} \mathcal{F}^{0,1} \longrightarrow \dots$$



Uncertain reading. Above the start of the line is a detached mark resembling 4 or an insertion sign.

Original source details: GELFAND-P001458

if $u \in \mathcal{F}^{0,0}$ is a solution of $\mathcal{M}$, then $\bar\partial u$ is too

The system is overdetermined, $\Rightarrow$ the solution complex is a perverse sheaf.

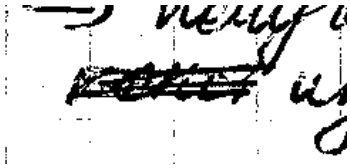
GELFAND-P001459

[source number: 51]

2°. Combinatorial method

(Goresky–MacPherson)

There is a [inserted above: general] method for extending sheaves to singular points $\Rightarrow$ this gives [word crossed out] perverse sheaves.

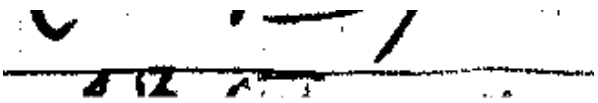


Uncertain reading. The cancellation obscures the word.

Original source details: GELFAND-P001459

The paper is devoted to the third method.

Next time $18/\mathrm{IV}$ a talk by Reshetikhin on quantum groups.



Uncertain reading. Clipped strokes are visible below the line. The shape at left may be «M» and is too incomplete to read with confidence.

Original source details: GELFAND-P001459

GELFAND-P001460

18.4.1988 Gelfand Seminar.

Titles of Bourbaki Seminar talks (87–88)

Lerner. “Uncertainty principle and microlocalization” (after Fefferman and Phong)

“Geometry of polynomials and Newton’s method” (after Smale and Shub)

“Nonlinear mixed problems”

“Noncommutative stochastic calculus”

“Metaplectic representations and the Howe conjecture”

“Schottky problem and Novikov conjecture”

Douady. “Siegel disks and Herman ring”

“Deformations of plane curves”

“Large deviations”

“Classification of unitary representations of real groups”

“Classification theory”

“Stable sheaves and [crossed-out fragment]



Uncertain reading. The crossed-out fragment is illegible. No closing quotation mark can be made out.

Original source details: GELFAND-P001460

“Moonshine”

Verdier. “Quantum groups [crossed-out superscript mark] (after Drinfeld)”



Uncertain reading. The mark after “groups” may be crossed-out closing quotation marks or an asterisk.

Original source details: GELFAND-P001460

Malgrange. “Geometric Fourier transform”

“A new invariant of homology spheres of dimension 3”

“Elliptic cohomology”

“Recent advances in Fermat’s theorem”.

GELFAND-P001461

[source number: 63]

Beilinson on work by Atiyah and somebody else.

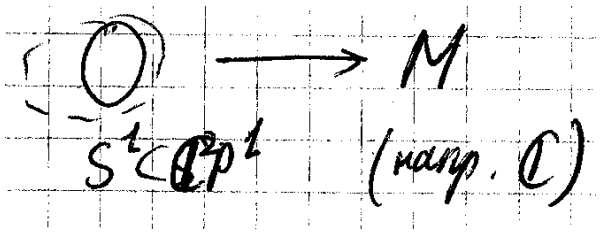
Semi-infinite Morse theory.

In the infinite-dimensional case there are 2 versions of Morse theory (for computing homology).

1) The geodesic problem with the energy functional resembles the finite-dimensional one. The negative index of inertia is finite and equals the number of conjugate points along the geodesic.

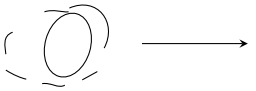
2) Infinite-dimensional manifold, a functional of energy type with isolated critical points but infinitely many $+$ and infinitely many $-$. But it turns out that at two distinct critical points the number of $-$ differs by a finite number. Homology is constructed by counting from $\infty/2$ (the topological version of Feigin and Tsygan)

Sensation: these invariants are related to Donaldson's invariants of 4-manifolds.



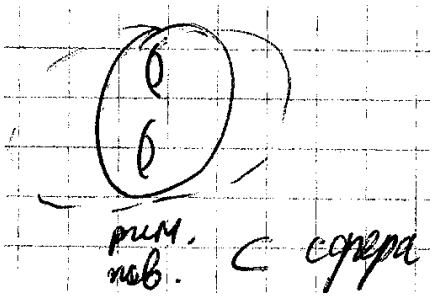
Original source details: GELFAND-P001461

Diagram and typed description.



$S^1 \subset \mathbb{C}P^1$ (e.g. $\mathbb{C}$)

For $S^1 \to \mathbb{C}$, distinguish 2 halves (extension inward and outward). They intersect in a finite-dimensional manifold.



Original source details: GELFAND-P001461

Diagram and typed description.

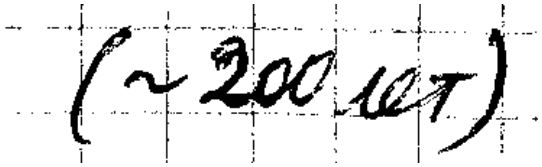


The exact junction of the upper inner stroke and the outline is unclear. The notes do not specify the meaning of the inner loops.

[Below the figure] Riemann surface $\subset$ sphere

[To the right of the figure] gauge fields; Y.–M. functional. critical points are flat connections.

I.M.G. The decomposition of $C^\infty(S^1)$ was used by many (~ 200 years)



Original source details: GELFAND-P001461

Uncertain reading. The apparent reading is "~ 200 years". Two rounded loops follow the 2.

(including τ-functions of Hirota–Sato, $\frac{1}{2}$-cohomology)

Remarkably, S^1 can be replaced by other manifolds. This will be for next time.

Reshetikhin Kolya.

~~Quantum groups.~~

$U_q(SU(2))$ and link invariants and representations of the braid group.

$$A = U_q(SU(2)); \qquad 1, H, X^{\pm} \quad \text{— generators}$$
assoc. alg.

$$[X^{\pm}, H] = \mp 2X^{\pm}; \tag{1}$$

$$[X^+, X^-] = \frac{\operatorname{sh}\left(\frac{h}{2}H\right)}{\operatorname{sh}(h/2)}; \qquad q = e^h$$

GELFAND-P001462

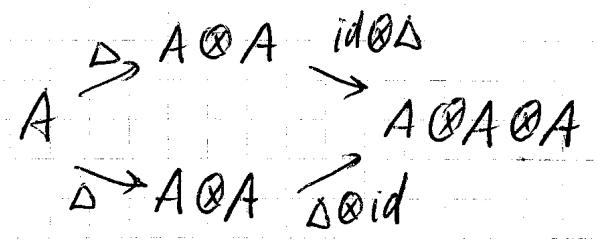
$$\left(q^{H/4}\right)^{\ell} H^n (X^+)^m (X^-)^k \quad \text{— basis as a vector space} \tag{2}$$

$$\Delta : A \to A \otimes A$$

$$\Delta(H) = H \otimes 1 + 1 \otimes H \tag{3}$$

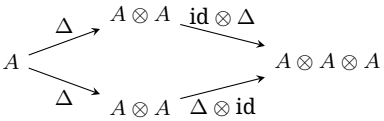
$$\Delta(X^{\pm}) = X^{\pm} \otimes q^{H/4} + q^{-H/4} \otimes X^{\pm}$$

1. Δ — an algebra homomorphism. $\Delta(ab) = \Delta(a)\Delta(b)$.
2. Δ — coassociative, i.e.



Original source details: GELFAND-P001462

Diagram and typed description.



[At right] (Hopf alg.)

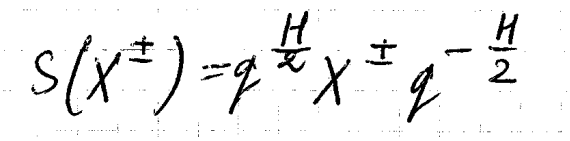
3. [crossed-out sign] $S : A \to A$ an antihomomorphism (analogous to passage to the inverse element in the algebra of functions on a group)



Original source details: GELFAND-P001462

Uncertain reading. The cancellation obscures the sign. Possibly Δ or «и».

$$S(X^{\pm}) = q^{H/2} X^{\pm} q^{-H/2} \tag{4}$$

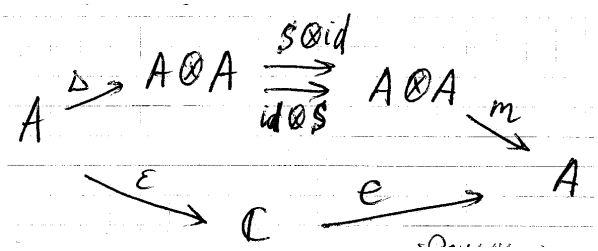


Original source details: GELFAND-P001462

Uncertain reading. In (4), $q^{H/2}$ immediately follows the equals sign. The formula follows the manuscript.

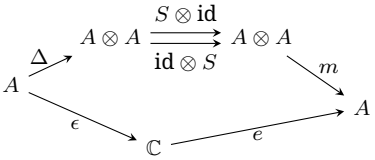
$$S(H) = -H$$

$$\epsilon(H) = \epsilon(X^{\pm}) = 0$$



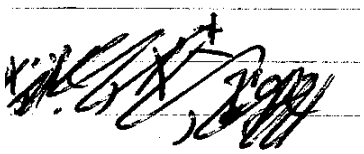
Original source details: GELFAND-P001462

Diagram and typed description.



[At right] (either arrow may be taken)

Turns into a quadratic algebra in [inserted above: other] generators [crossed-out fragment]



Original source details: GELFAND-P001462

Uncertain reading. Possibly K, then X^+ or $X^{\pm}$. The following strokes are illegible.

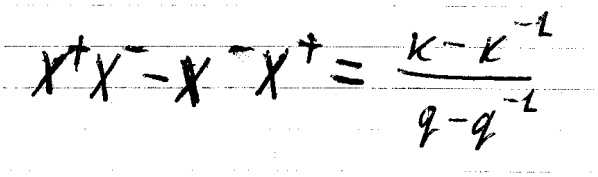
Consider.

$$K = q^{H/2}, \qquad K, X^{\pm}$$

[Crossed out at left: $K, X^{\pm}$.]

$$KX^{\pm} = q^{\pm 1} X^{\pm} K$$

$$X^+X^- - X^-X^+ = \frac{K - K^{-1}}{q - q^{-1}}$$



Original source details: GELFAND-P001462

Uncertain reading. The denominator $q - q^{-1}$ follows the manuscript.

GELFAND-P001463

[source number: 64]

(5) involution

$$(X^+)^* = X^-, \qquad H^* = H$$

finite-dimensional representations

$$\pi_j(X^\pm)e^j_m = \sqrt{\frac{\operatorname{sh}\left(\frac{h}{2}(j \mp m)\right)\operatorname{sh}\left(\frac{h}{2}(j \pm m + 1)\right)}{\operatorname{sh}^2\left(\frac{h}{2}\right)}}\, e_{m\pm1}, \tag{6}$$

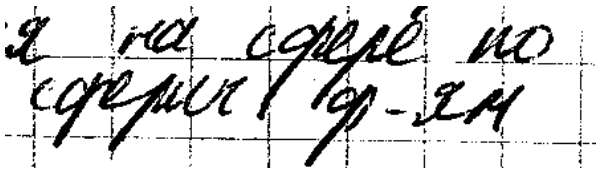
$$j \in \mathbb{Z}/2, \qquad -j \le m \le j.$$

$$\dim V^j = 2j + 1 \quad \text{(as for } \mathrm{SL}_2\text{)}$$

Sklyanin wrote in quadratic algebras.

Exercise (I. M. G.) $q = 1 \Rightarrow$ formulas from the book by Gelfand–Minlos–Shapiro

[At right, marked off by a bracket] specific problem: how to expand vector fields on a sphere in spherical harmonics



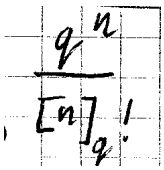
Original source details: GELFAND-P001463

Uncertain reading. The Russian abbreviation for “harmonics” appears to be «гар-ям», possibly «гар-м».

start with $\mathrm{SU}(2)$

(7) Thm. Set

$$R = \exp\left(\frac{h}{4}H \otimes H\right) \sum_{n\ge0} \frac{q^n}{[n]_q!}\, e^n \otimes f^n \tag{8}$$



Original source details: GELFAND-P001463

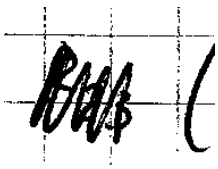
Uncertain reading. The numerator of the coefficient in (8) is q^n.

$$R \in U_q(\mathrm{SU}(2))^{\otimes 2},$$

where

$$e = q^{H/4}X^+, \qquad f = q^{-H/4}X^-, \tag{9}$$

$$1. \quad (\Delta \otimes \mathrm{id})R = R_{23}R_{13}; \qquad \left(\in U_q(\mathrm{SU}(2))^{\otimes 3}\right) \tag{10}$$



Original source details: GELFAND-P001463

Uncertain reading. An illegible fragment is crossed out before the parentheses in the first line of (10).

$$(\mathrm{id} \otimes \Delta)R = R_{12}R_{13};$$

$$2. \quad \sigma \circ \Delta(a) = R\,\Delta(a)R^{-1}; \qquad \text{(see (12))}$$

$$3.\quad R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}.$$

[At right] (Yang–Baxter equation with values in the tensor product of 3 copies of A)

In formula (8)

$$[n]_q! = [n]_q \dots [1]_q \qquad (q\text{-factorial}) \tag{11}$$

I. M. G. Quantum mechanics is the right way to introduce q.

Ex. what is Γ_q?

$$\sigma\colon A^{\otimes 2} \longrightarrow A^{\otimes 2}, \qquad \sigma(f \otimes g) = g \otimes f \tag{12}$$

In particular, it follows from $(10)_2$ that $\sigma \circ \Delta$ is also a comultiplication.

GELFAND-P001464

Define

$$R^{j_1 j_2}\colon V^{j_1} \otimes V^{j_2} \longrightarrow V^{j_2} \otimes V^{j_1}$$



Uncertain reading. The sign after $R^{j_1 j_2}$ reads as a colon over earlier strokes. The underlying sign is unclear.

Original source details: GELFAND-P001464

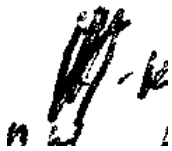
V^j — space of a finite-dimensional representation.

$$R^{j_1 j_2} = P^{j_1 j_2}\, \pi^{j_1} \otimes \pi^{j_2}(R) \tag{13}$$

[Beneath $P^{j_1 j_2}$, joined by a line] operator permuting the spaces

$$\begin{aligned}(1 \otimes R^{j_1 j_2})(R^{j_1 j_3} \otimes 1)(1 \otimes R^{j_2 j_3}) = \\ = (R^{j_2 j_3} \otimes 1)(1 \otimes R^{j_1 j_3})(R^{j_1 j_2} \otimes 1)\end{aligned} \tag{14}$$

q-analogues of the [crossed-out fragment] Clebsch–Gordan coefficients.

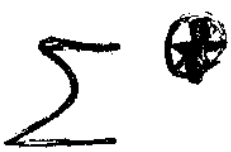


Uncertain reading. The crossed-out strokes before “coefficients” are illegible.

Original source details: GELFAND-P001464

all finite-dimensional representations are completely reducible.

$$V^{j_1} \otimes V^{j_2} = \sum_{|j_1 - j_2| \le j_3 \le j_1 + j_2}^{\oplus} V^{j_3} \tag{15}$$



Uncertain reading. The small sign above the sum reads $\oplus$, possibly over $\otimes$.

Original source details: GELFAND-P001464

V_j were found by Kulish and Reshetikhin (1981)
In the basis:

$$e^{j_1}_{m_1} \otimes e^{j_2}_{m_2} = \sum_{|j_1-j_2|\le j_3\le j_1+j_2} \begin{bmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{bmatrix} e^{j_3}_{m_3} \tag{16}$$

[A line joins the square bracket to the label] Clebsch–Gordan coefficients
where

$$m_3 = m_1 + m_2$$

Graphical notation:

$$A\colon V^{j_1} \otimes \ldots \otimes V^{j_N} \longrightarrow V^{j'_1} \otimes \ldots \otimes V^{j'_L}$$

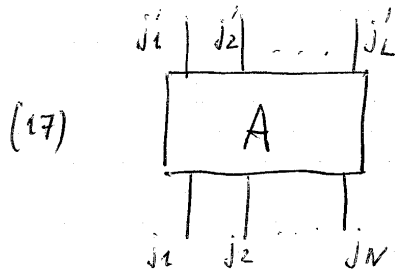
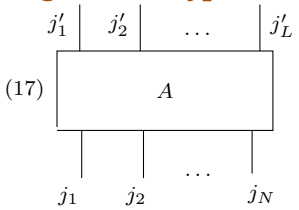
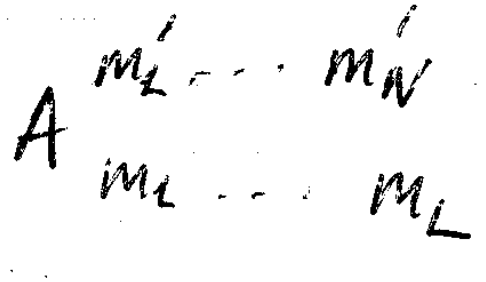


Diagram and typed description.



Original source details: GELFAND-P001464

$$A^{m'_1 \ldots m'_N}_{m_1 \ldots m_L}$$



Uncertain reading. In the manuscript coefficient the upper indices end at m'_N, the lower at m_L.

Original source details: GELFAND-P001464

$$-j_i \le m_i \le j_i;$$

[At lower right]

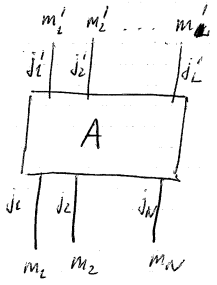
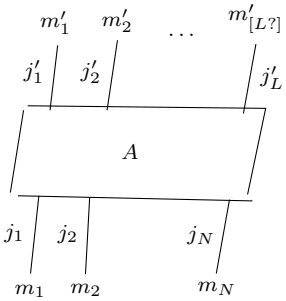


Diagram and typed description.



Original source details: GELFAND-P001464

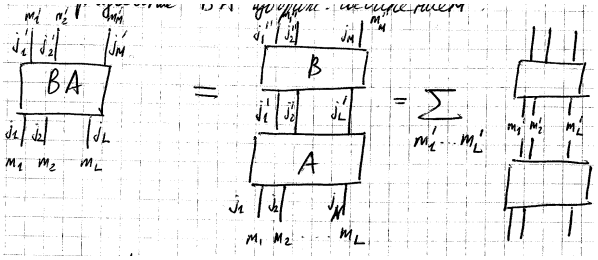
The upper-right m' subscript is overwritten. Its visible final stroke resembles L. The underlying form remains uncertain.

GELFAND-P001465

[source number: 65]

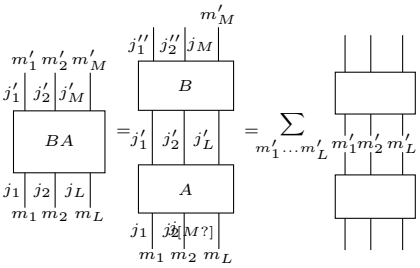
The product BA is shown by joining:

(18)



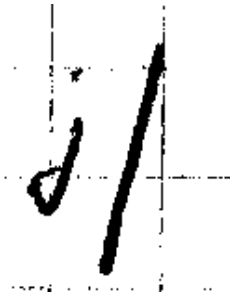
Original source details: GELFAND-P001465

Diagram and typed description.



In the middle drawing the first two upper j-labels at B have double primes. At the upper right j_M is legible and a prime is uncertain. The last lower j-index at A resembles M, though m_L stands below it.

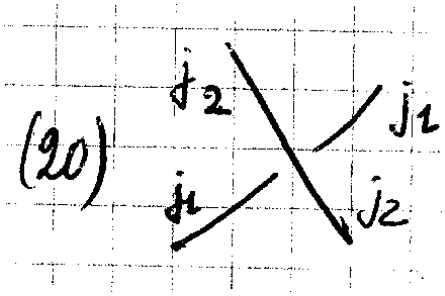
$V^j \xrightarrow{\text{id}} V^j$ is shown by a strand (19)



Original source details: GELFAND-P001465

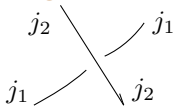
Diagram and typed description. j

(20)



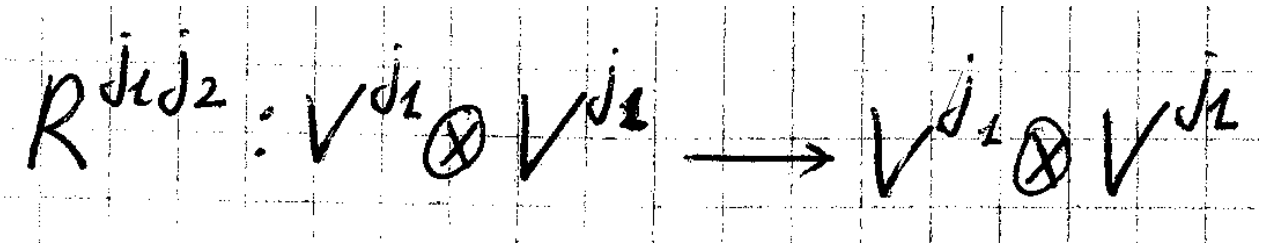
Original source details: GELFAND-P001465

Diagram and typed description.



A short side stroke touches the lower-right tip. A complete arrowhead cannot be made out.

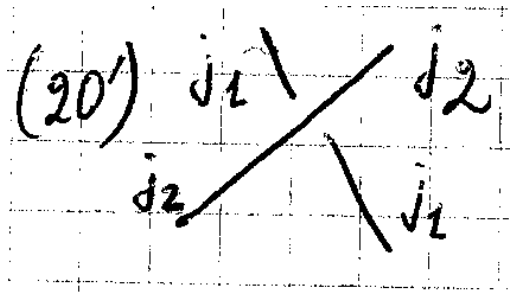
$$R^{j_1 j_2} : V^{j_1} \otimes V^{j_1} \longrightarrow V^{j_1} \otimes V^{j_1}$$



Original source details: GELFAND-P001465

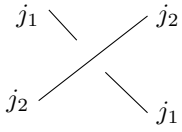
Uncertain reading. All four V factors read j_1, following the manuscript.

(20') $(R^{j_2 j_1})^{-1}$ in the same space.

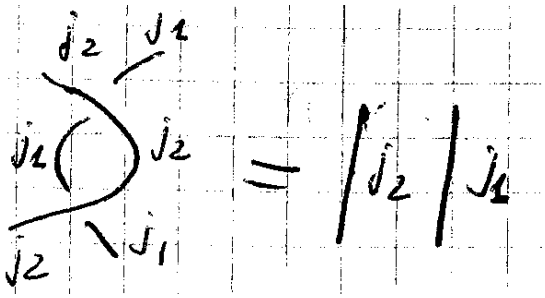


Original source details: GELFAND-P001465

Diagram and typed description.

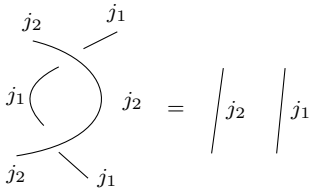


$$RR^{-1} = I$$



Original source details: GELFAND-P001465

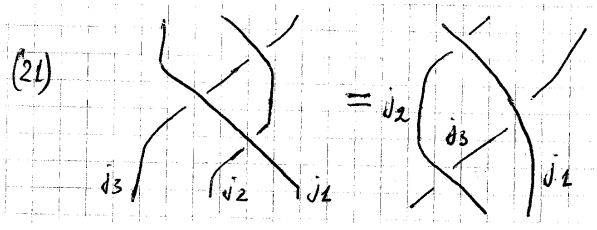
Diagram and typed description.



j_1, j_2 — colours of the strands

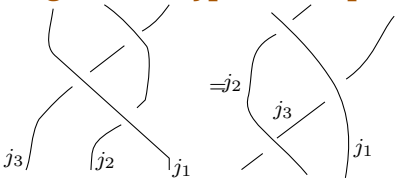
Yang–Baxter relation

(21)



Original source details: GELFAND-P001465

Diagram and typed description.



Additional vertices

(22)

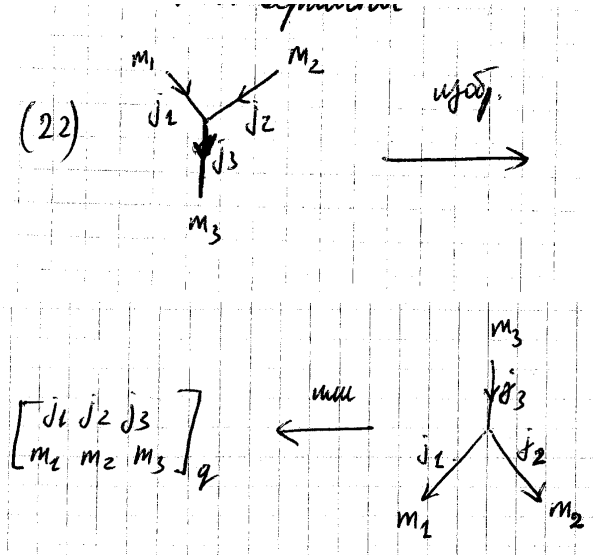
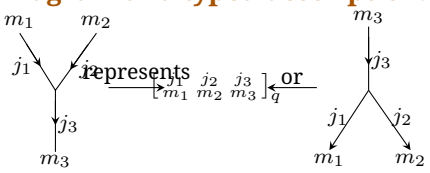


Diagram and typed description.



Original source details: GELFAND-P001465

GELFAND-P001466

(23)

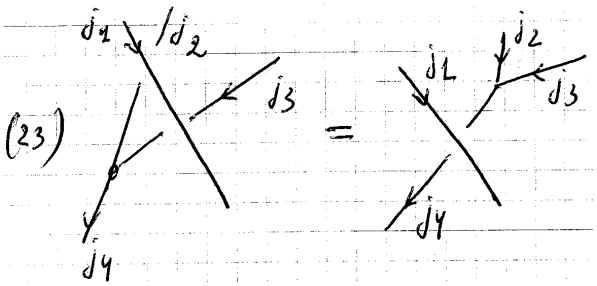
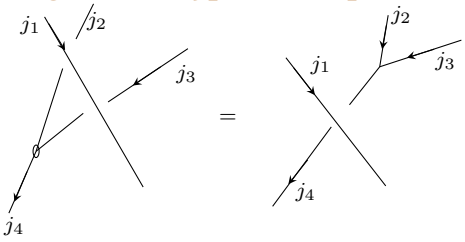


Diagram and typed description.



Original source details: GELFAND-P001466

(24)

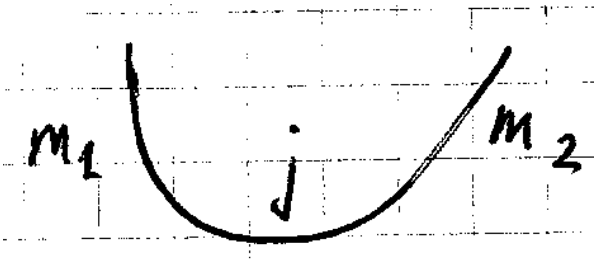


Diagram and typed description.

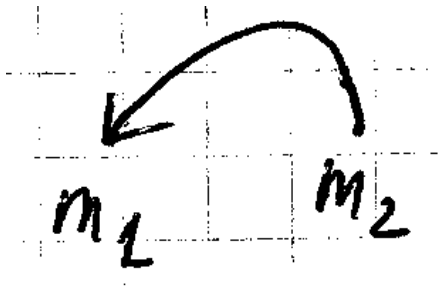


Original source details: GELFAND-P001466

$$\longrightarrow (-1)^{m_1} q^{m_1} \delta_{m_1,-m_2}$$

$-j \le m_i \le j, \quad i = 1, 2.$

(25)

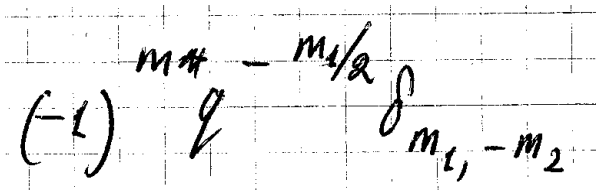


Original source details: GELFAND-P001466

Diagram and typed description.



$$\longrightarrow (-1)^{m[\text{crossed out}]}\, q^{-m_1/2} \delta_{m_1,-m_2}$$

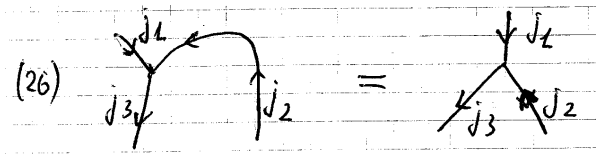


Original source details: GELFAND-P001466

Uncertain reading. A heavy cancellation follows m in the exponent of (-1). The exponent of q reads $-m_1/2$.

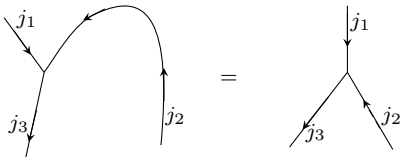
The following equalities hold:

(26)



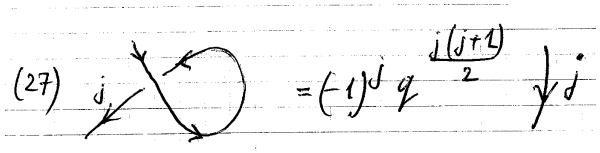
Original source details: GELFAND-P001466

Diagram and typed description.



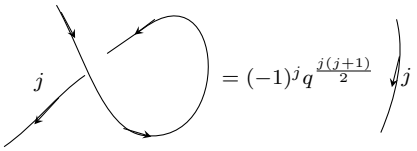
The arrow on the right j_2 branch is overwritten. The dotted arrow shows the possible direction towards the vertex.

(27)

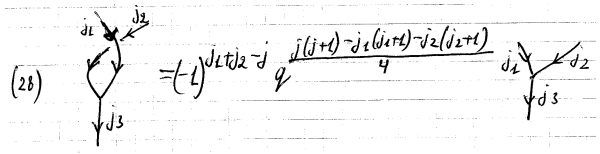


Original source details: GELFAND-P001466

Diagram and typed description.

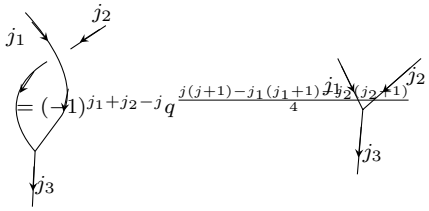


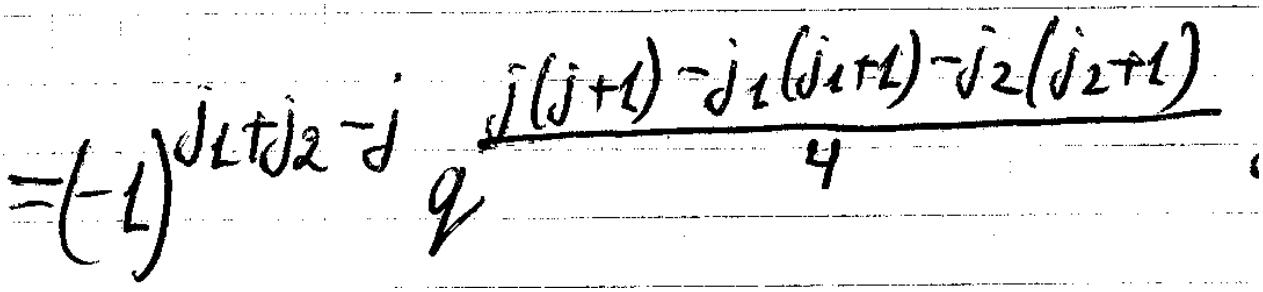
(28)



Original source details: GELFAND-P001466

Diagram and typed description.





Original source details: GELFAND-P001466

Uncertain reading. The coefficient uses j without a subscript. Both lower strands are labelled j_3.

This makes it possible to construct invariants of links, braids and knotted ribbons.

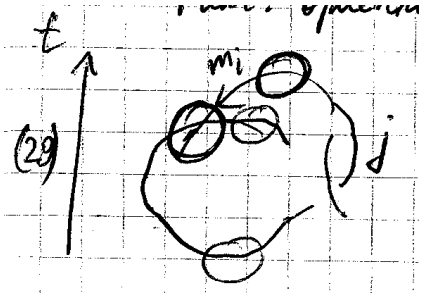
GELFAND-P001467

[source number: 66]

Link invariants.

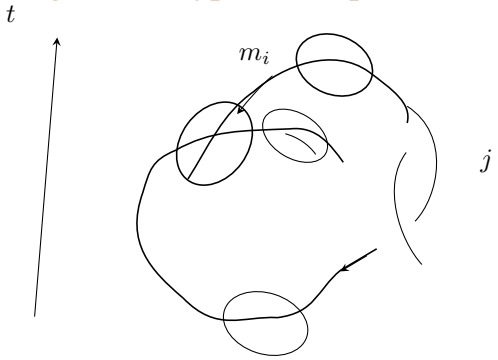
Fix an orientation and a direction t.

(29)



Original source details: GELFAND-P001467

Diagram and typed description.



The paths and individual arrow directions among the overlapping strokes are uncertain. The gaps at right are preserved.

Elementary pieces:

1) mark the points where the normal is $\|$ to the axis t (the chosen direction)

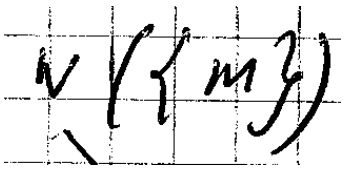
2) crossing points.

Place the numbers m_i at the joins between elementary pieces, so that

$$-j \le m_i \le j \qquad (j \text{ is chosen arbitrarily})$$

$$\sum_{\{m\}} \prod_{\substack{\text{over all} \\ \text{elementary} \\ \text{pieces}}} v(\{m\}) = \varphi_{j_1 \dots j_k}(D_L, q) \tag{30}$$

[From v a leader to the label] matrix entries computed above

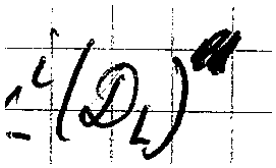


Original source details: GELFAND-P001467

Uncertain reading. The letter before $(\{m\})$ may be v or ν.

(31) Theorem

$$I_{j_1 \dots j_k}(L, q) = q^{\sum_i N_+^i(D_L) - N_-^i(D_L)} \varphi_{\{j\}}(D_L, q)$$



Original source details: GELFAND-P001467

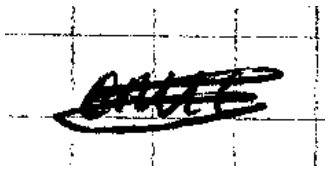
Uncertain reading. After $N_-^i(D_L)$ in the exponent, an illegible raised mark is crossed out.

— a link invariant (independent of the projection of the knot onto the plane).

$N_\pm^i$ — how many times the ith strand passes under and over

This contains the Jones invariant.

One can [crossed-out word] construct knot invariants from $\forall$ simple Lie algebras.



Original source details: GELFAND-P001467

Uncertain reading. The word crossed out between “can” and “construct” is illegible.

There is a conjecture that the SL_2-invariants suffice

There are also $6j$-symbols.

GELFAND-P001468

25.4.88. Gelfand Seminar

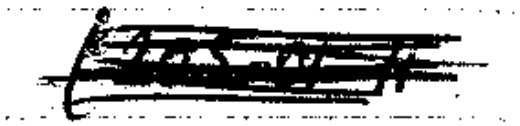
Ushveridze [crossed out] Alexander Grigoryevich. [crossed out]

The Riccati equation and quasi-exactly solvable problems.



Uncertain reading. A short entry between the surname and the given name is heavily crossed out.

Original source details: GELFAND-P001468



Uncertain reading. Several horizontal strokes cross out the writing after the patronymic. The text is illegible.

Original source details: GELFAND-P001468

$$y'(\lambda) + y^2(\lambda) + 2B(\lambda)y(\lambda) + C(\lambda) = 0, \tag{1}$$

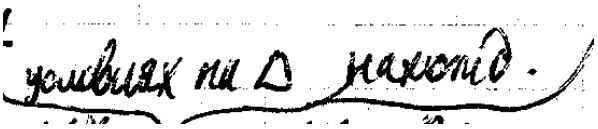
$$B(\lambda) = \sum_{\alpha=1}^{N} b_\alpha \Delta(\lambda - a_\alpha), \tag{2}$$

$$C(\lambda) = \sum_{\alpha=1}^{N} c_\alpha \Delta(\lambda - a_\alpha). \tag{3}$$

We seek a solution to (1) of the form

$$y(\lambda) = \sum_{i=1}^{M} \Delta(\lambda - \lambda_i) \tag{4}$$

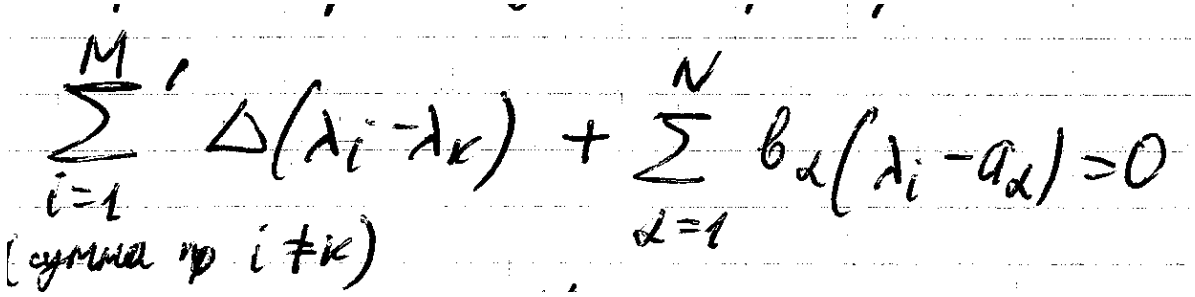
Then, under certain [inserted above: conditions on Δ, finding the] parameter values reduces to algebraic equations:



Uncertain reading. The final Russian abbreviation in the insertion appears to read «нахожд.», for “finding”.

Original source details: GELFAND-P001468

$$\begin{cases} \sum_{i=1}^{M}{}'\Delta(\lambda_i - \lambda_k) + \sum_{\alpha=1}^{N} b_\alpha(\lambda_i - a_\alpha) = 0 \\ \text{(sum over } i \neq k\text{)} \\ c_\alpha = 2b_\alpha \sum_{i=1}^{M} \Delta(\lambda_i - a_\alpha) \end{cases} \tag{5}$$



Original source details: GELFAND-P001468

Uncertain reading. In the first line of (5), $(\lambda_i - a_\alpha)$ immediately follows b_α. The first argument is written $\lambda_i - \lambda_k$.

(6) Properties of Δ and Ω (generalization of the addition theorem for elliptic functions)

$$1) \quad \Delta(\lambda_1)\Delta(\lambda_2) + \Delta(\lambda_2)\Delta(\lambda_3) + \Delta(\lambda_3)\Delta(\lambda_1) + \Omega(\lambda_1) + \Omega(\lambda_2) + \Omega(\lambda_3) = 0$$

when $\lambda_1 + \lambda_2 + \lambda_3 = 0$

$$2) \quad \Delta(\lambda) + \Delta(-\lambda) = 0$$

3) [crossed out] $\Omega(\lambda) - \Omega(-\lambda) = 0$



Original source details: GELFAND-P001468

Uncertain reading. In the crossed-out formula before $\Omega(\lambda)$, one sign resembles Δ. The remainder is illegible.

(7) Examples of solutions of system (6)

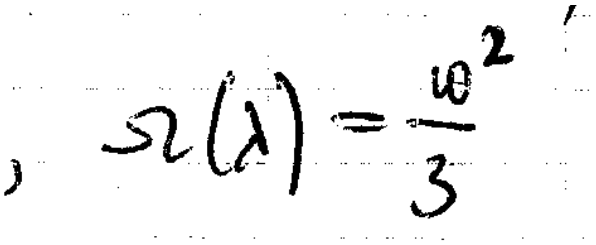
$$1) \quad \Delta(\lambda) = \frac{1}{\lambda}, \qquad \Omega(\lambda) = 0;$$



Original source details: GELFAND-P001468

Uncertain reading. At right of the first pair of formulas, the crossed-out fragment may begin with Δ.

2) $\Delta(\lambda) = \omega \operatorname{ctg} \omega\lambda, \qquad \Omega(\lambda) = \dfrac{\omega^2}{3}$

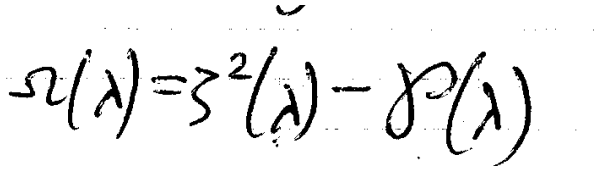


Uncertain reading. The manuscript gives $\omega^2/3$ with no minus sign.

Original source details: GELFAND-P001468

3) $\Delta(\lambda) = \zeta(\lambda), \qquad \Omega(\lambda) = \zeta^2(\lambda) - \wp(\lambda)$

[Below $\zeta(\lambda)$, with a short leader] Weierstrass function



Uncertain reading. The manuscript gives $\zeta^2(\lambda) - \wp(\lambda)$ without a numerical factor.

Original source details: GELFAND-P001468

(quasiperio



Uncertain reading. The word and parenthesis are unfinished on this leaf.

Original source details: GELFAND-P001468

GELFAND-P001469

[source number: 67]

Applications

A. $y = \dfrac{f'}{f} \quad \Rightarrow$ Schrödinger equation for f;

The solution of (6) with a singularity at 0 satisfies the equation

$$\Delta'(\lambda) + \Delta^2(\lambda) = 2\Omega(\lambda) + \Omega(0) \tag{8}$$

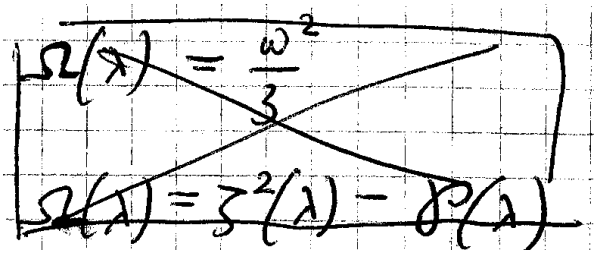


Diagram and typed description.

$\Omega(\lambda) = \dfrac{\omega^2}{3}$

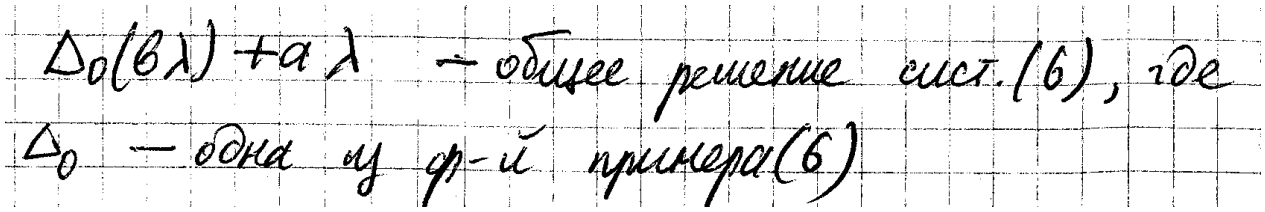
$\Omega(\lambda) = \zeta^2(\lambda) - \wp(\lambda)$

Both formulas are boxed and crossed out by two diagonals.

Original source details: GELFAND-P001469

(9) $\Delta_0(b\lambda) + a\lambda$ — the general solution of system (6), where

Δ_0 — one of the functions from example (6)

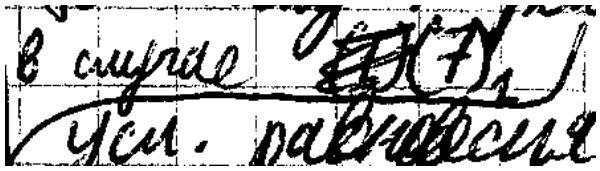


Original source details: GELFAND-P001469

Uncertain reading. There is no factor before $\Delta_0(b\lambda)$. The reference after "example" reads (6).

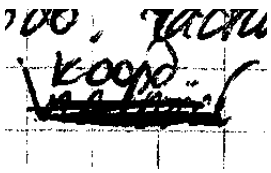
B. This also gives the Bethe ansatz solution for spin systems.

C. (5) $\Leftrightarrow$ [inserted above: in case [crossed out] $(7)_1$] equilibrium conditions for a classical system of charged particles of 2 kinds: fixed particles with coordinates a_α and charges b_α; mobile particles with [crossed out; inserted above: coordinates] λ_i and charge 1.



Original source details: GELFAND-P001469

Uncertain reading. An illegible entry is crossed out in the insertion "in case" before $(7)_1$.



Original source details: GELFAND-P001469

Uncertain reading. The abbreviation «коорд.» for "coordinates" is written above the crossed-out word.

[At right] Potential:

$$q_i q_k \ln|\lambda_i - \lambda_k|$$

In case $(7)_2$ the particles are on a circle with potential

$$q_i q_k \ln|\sin(\lambda_i - \lambda_k)|$$

In case $(7)_3$ the interaction potential is

$$q_i q_k \ln\theta(\lambda_i - \lambda_k)$$

(10) For the Schrödinger equation

λ_i — nodes of the wavefunction

In the Coulomb system

[crossed out] λ_i — coordinates of the charged particles



Original source details: GELFAND-P001469

Uncertain reading. A separate mark before λ_i is crossed out and illegible.

In the Bethe equation λ_i — rapidities

Stability of the quantum-mechanical system ⇔ stability of the Coulomb system

What to do with partial integrability? (with an incomplete system of integrals)

$$H = H_0 + L_1(H_1 - c_1) + L_2(H_2 - c_2), \qquad L_j \text{ — operators} \tag{11}$$

Let H_j commute for $j = 0, 1, 2$.

GELFAND-P001470

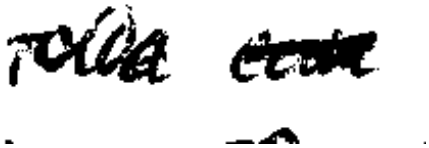
[Crossed out in the left margin]



Uncertain reading. The crossed-out mark is illegible.

Original source details: GELFAND-P001470

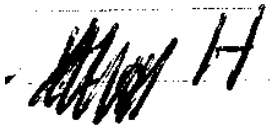
Consider a common eigenvector of H_0, H_1, H_2; then [crossed out]



Uncertain reading. The crossed-out word after “then” appears to be «если» (“if”).

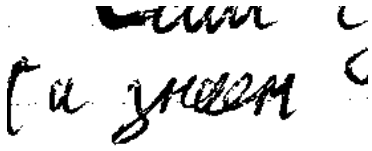
Original source details: GELFAND-P001470

If c_j — eigenvalues of H_j, they give an eigenvalue of [crossed out] H (and we know the eigenvectors — as many as the degeneracy)



Uncertain reading. The crossed-out strokes before H are illegible.

Original source details: GELFAND-P001470



Uncertain reading. The verb before “eigenvectors” tentatively reads «знаем» (“know”), possibly «умеем» (“can”).

Original source details: GELFAND-P001470

In this language

quantum nonlinear Schrödinger $\Longleftrightarrow$ harmonic oscillator

~~oper.~~ energy $\Longleftrightarrow$ particle-number operator

———— · ————

The spectrum of H_i is found from the Bethe ansatz equations

———— · ————

Reference:

Brief Reports on Physics, FIAN.

From the Riccati equation to the Schrödinger equation

This can be done in countless ways

$$y = \frac{f'}{f} \quad \Longrightarrow$$

$$f'' + 2Bf' + Cf = 0 \qquad \text{(the independent variable here is } \lambda\text{)} \tag{12}$$

Multiply [inserted above: (12)] by an arbitrary function A and then eliminate $f' \Longrightarrow$ [below the line (by changing the independent variable!)] $\Longrightarrow$ we obtain the Schrödinger equation

$$\left[-\frac{\partial^2}{\partial x^2} + V(x)\right] \Psi(x) = 0 \tag{13}$$

Each A gives its own potential V.

The function A can be chosen so that information about the numbers [crossed out] c_α, λ_i is concentrated in the x-independent part of the potential:

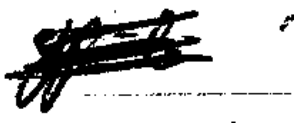


Uncertain reading. A small illegible entry is crossed out before c_α, λ_i.

Original source details: GELFAND-P001470

$$V(x) = U(x) - E(\lambda_i) \tag{14}$$

Then (13) [crossed out] $\sim$ the Schrödinger equation



Uncertain reading. An illegible entry is crossed out between (13) and the sign $\sim$.

Original source details: GELFAND-P001470

For this we need

$$A(\lambda) = \prod_{\alpha=1}^{N} \Xi(\lambda - a_\alpha), \qquad \frac{\Xi'(\lambda)}{\Xi(\lambda)} = \Delta(\lambda) \tag{15}$$

Change to x;

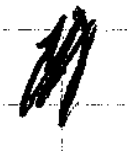
$$x = \int^{\lambda(x)} \frac{d\lambda}{\sqrt{A(\lambda)}} \tag{16}$$

GELFAND-P001471

[source number: 68]

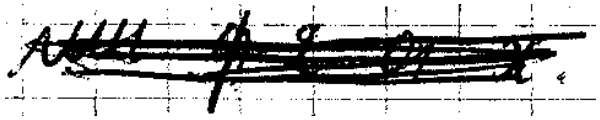
(17) [crossed out] In case $(7)_1$:

$$A(\lambda) = \prod_{\alpha=1}^{N} (\lambda - a_\alpha)$$



Original source details: GELFAND-P001471

Uncertain reading. The short crossed-out entry before “In case” is illegible.



Original source details: GELFAND-P001471

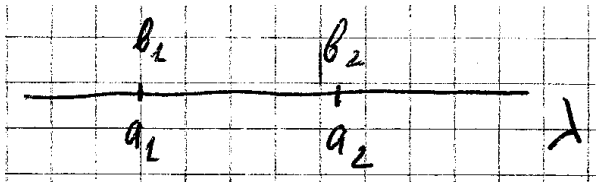
Uncertain reading. The crossed-out line reads “linear function of λ.”

($\boxed{\Xi}$ is a linear function of λ)

from here we get the Jacobi transformation (16)

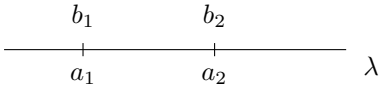
Use the Coulomb analogy to classify exactly and quasi-exactly solvable problems.

Let $N = 2$ in (2)–(3)



Original source details: GELFAND-P001471

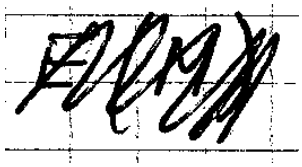
Diagram and typed description.



a_1, a_2 are the coordinates of the fixed particles (classical)

Mobile particles $\{\lambda_i\}$ — there are M of them
coord.

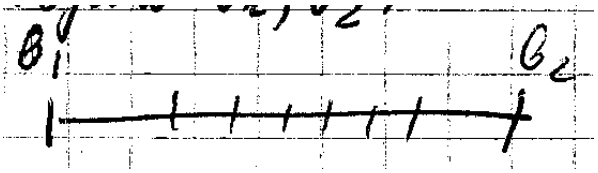
(18) [crossed out] The potential V does not contain M (!) $\Longrightarrow$ we have an exactly solvable model.



Original source details: GELFAND-P001471

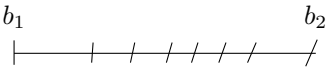
Uncertain reading. The crossed-out word before “The potential” tentatively reads «Если» (“If”).

(19) Let $b_1, b_2 > 0$



Original source details: GELFAND-P001471

Diagram and typed description.

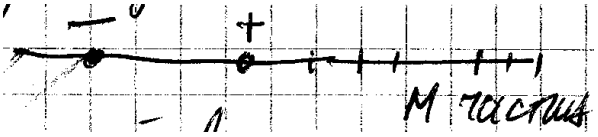


The particles inside cannot leave because of repulsion $\Longrightarrow$ $\exists$ min $\Longrightarrow$ $\exists$ a solution of the Schrödinger equation with M nodes $\Longrightarrow$ integrability!!!

(all the wave functions are found!!)

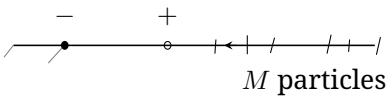
Another case

(20)

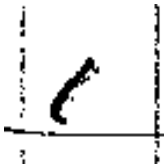


Original source details: GELFAND-P001471

Diagram and typed description.



The left end of the line is thickened. Faint strokes beneath it and the left point are shown in grey. An arrowhead at the left end cannot be made out with confidence.



Original source details: GELFAND-P001471

Uncertain reading. There is a short separate oblique stroke in the left margin below (20).

Stability condition for the system of M particles to the right of b_2:

$$b_1 + b_2 + M < 0 \tag{21}$$

(they do not approach b_2, and do not escape to ∞)

But this condition fails for large M! $\Longrightarrow$ only finitely many levels occur.

GELFAND-P001472

Choosing the intervals differently gives different problems.

Here we obtain exactly solvable models (magnets).

Beilinson A.

on the work of Atiyah



Original source details: GELFAND-P001472

Uncertain reading. The surname appears to read «Ати», rendered here as Atiyah. The final strokes also allow «Атии».

New invariants of 3- and 4-dimensional manifolds

(semi-infinite Morse theory)

$$\underset{\substack{\text{natural}\\ \infty\text{-dimensional manifolds}}}{X} \xrightarrow{f} \mathbb{R} \qquad \text{— Morse function.}$$

The critical points will contain ∞ many $+$ and $-$ squares, but different points differ by a finite number.

New homologies arise.

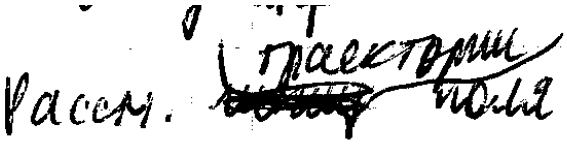
On the manifold X put a Riemannian metric

Suppose first that $\dim X < \infty$ and X is compact

g — metric,

$$\tau = \operatorname{grad} f$$

Consider [crossed out] [inserted above: trajectories] of the field τ.



Uncertain reading. The crossed-out word beneath "trajectories" is illegible.

Original source details: GELFAND-P001472

There are two types: points [below the line: (critical)] and trajectories connecting two critical points.

If the metric is in general position, consider the space $T(x_1, x_2)$ of all trajectories going from x_1 to x_2. Then

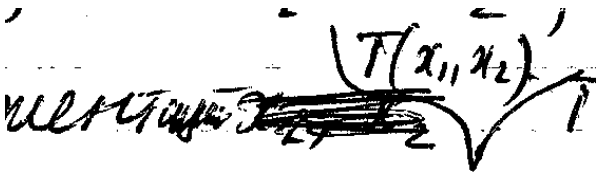
$$\dim T(x_1, x_2) = \operatorname{ind}_{x_1} f - \operatorname{ind}_{x_2} f - 1 \tag{1}$$

In particular, if the indices differ by 1, it is a discrete set.

Suppose

$$\operatorname{ind}_{x_1} f = \operatorname{ind}_{x_2} f + 1 \tag{2}$$

Then with each trajectory connecting x_1 and x_2 one associates a number ± 1, depending on the given orientations [crossed out] [inserted above: $T(x_1, x_2)$].



Uncertain reading. The crossed-out entry may contain x_1, x_2. The insertion above reads $T(x_1, x_2)$.

Original source details: GELFAND-P001472

Now construct the complex

[source number: 69]

$$\longrightarrow C_i \xrightarrow{d} C_{i-1} \longrightarrow \dots \tag{3}$$

C_i — vector space with basis $\{x_\alpha\}$, ind $x_\alpha = i$
d is given by a matrix;
index of an entry = pair of points of adjacent index x_1, x_2

$$d = \|d_{x_1 x_2}\|,$$

$$d_{x_1 x_2} = \sum_{\substack{\text{trajectories} \\ \text{connecting } x_1, x_2}} \pm 1 \tag{4}$$

———— · ————

Let M be a compact 3-dimensional manifold.
Consider the space of all [inserted above: irreducible] connections C [crossed out]

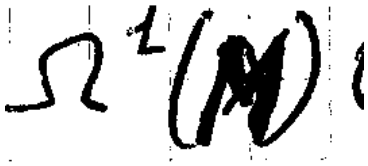


Original source details: GELFAND-P001473

Uncertain reading. The short entry to the right of C is crossed out and illegible.

[Beneath "connections": (Hermitian)]
on the trivial 2-dimensional bundle over M
Such a connection has the form $d + A$, where

$$A \in \Omega^1(M) \otimes \mathrm{su}_2$$



Original source details: GELFAND-P001473

Uncertain reading. The M is heavily retraced. The earlier strokes cannot be distinguished.

Consider them up to gauge equivalence

$$G \text{ — gauge group } SU_2^M, \qquad X = C/G \text{ — orbit space}$$

X — the main manifold.
On X define a Morse function, or more precisely its differential ν — a 1-form on C.
Tangent vector to C: matrix-valued 1-form
(C — affine space)

$$B \in \Omega^1(M) \otimes \mathrm{su}_2.$$

$$\langle \nu, B_\nabla \rangle = \int_M \mathrm{Tr}(F_\nabla \cdot B), \qquad B \in \Omega^1(M) \otimes \mathrm{su}_2 \tag{5}$$

[Beneath F_∇ with a leader: curvature of A]

[At right:] $F_\nabla \cdot B$ — exterior product, (3-form !!)

$$F_\nabla = dA + \tfrac{1}{2}[A, A] \tag{6}$$

GELFAND-P001474

(7) Properties of ν.

1. ν is pulled back from X
2. ν — closed 1-form
3. [inserted above: Primitive] ν — secondary characteristic class for c_2 (Chern class)

— Chern–Simons class for c_2:

$$\int_{M\times[0,1]} \mathrm{Tr}(F_{\widetilde{\nabla}})^2 \tag{8}$$



Uncertain reading. The first symbol with subscript 2 is heavily retraced and may be c_2 or e_2. The next line clearly has c_2.

Original source details: GELFAND-P001474

where

(9) [crossed out A_∇?] $\widetilde{\nabla}$ — connection on the trivial bundle over $M \times [0, 1]$ with potential at the point (x, t), equal to tA.

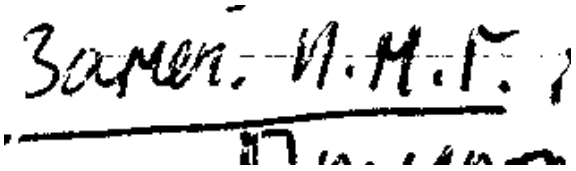


Uncertain reading. The crossed-out notation before $\widetilde{\nabla}$ resembles A_∇, possibly with a tilde in the subscript.

Original source details: GELFAND-P001474

This connection is on C, and not on X.

[small insertion between lines] Remark by I. M. G.: All of this is contained in the "simple" part of the work by Gabrielov–Gelfand–Losik

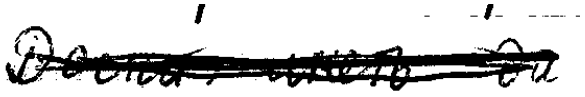


Uncertain reading. The middle initial resembles M. (M.), with H. (N.) also possible.

Original source details: GELFAND-P001474

Let us look at the critical points.

[crossed out]

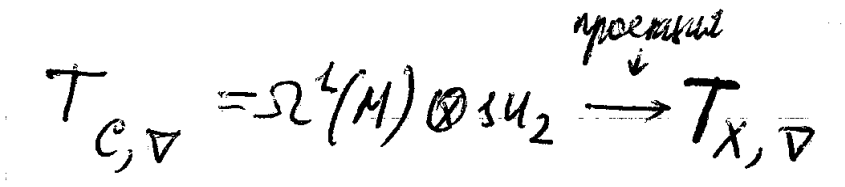


Uncertain reading. The short crossed-out line is illegible. Individual strokes remain above and below the cancellations.

Original source details: GELFAND-P001474

These are connections with curvature 0, i.e. integrable ones, i.e. representations of $\pi_1(M)$ Hessian.

$$\nabla \in C, \qquad T_{C,\nabla} = \Omega^1(M) \otimes \mathrm{su}_2 \xrightarrow{\text{projection}} T_{X,\nabla} \tag{10}$$



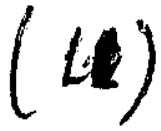
Original source details: GELFAND-P001474

Uncertain reading. Faint strokes above ∇ in the right subscript coincide with the ruling. A separate overbar cannot be established.

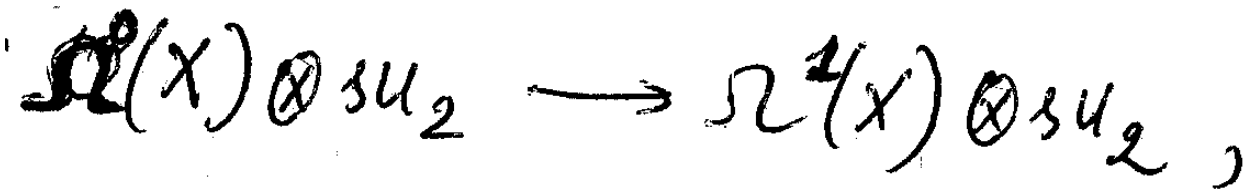
Introduce

$$\nabla_{\mathrm{ad}} \colon \Omega^0(X) \otimes \mathrm{su}_2 \longrightarrow \Omega^1(X) \otimes \mathrm{su}_2, \tag{11?}$$

Uncertain reading. The second digit is heavily overwritten. The sequence of formulas suggests (11).



Original source details: GELFAND-P001474



Original source details: GELFAND-P001474

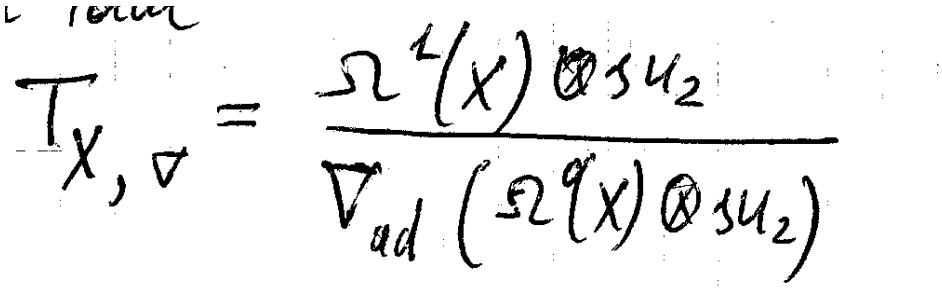
Uncertain reading. Both sides of the formula use X. The Ω^0 on the left is heavily retraced.

by the formula

$$\nabla_{\mathrm{ad}}(\varphi) = d\varphi + [A, \varphi] \tag{12}$$

and then

$$T_{X,\nabla} = \frac{\Omega^1(X) \otimes \mathrm{su}_2}{\nabla_{\mathrm{ad}}\left(\Omega^0(X) \otimes \mathrm{su}_2\right)} \tag{13}$$



Original source details: GELFAND-P001474

Uncertain reading. The numerator and denominator use X. Equation (10) uses M.

This holds for any connection.

GELFAND-P001475

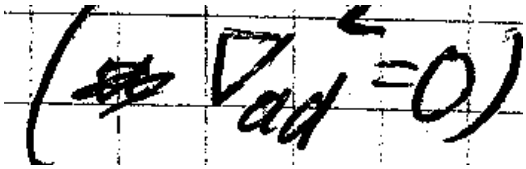
[source number: 70]

The Hessian is a quadratic form on the space (13), defined invariantly at a critical point.

$$H(\varphi) = \int_M \mathrm{Tr}(\nabla_{\mathrm{ad}}\varphi \cdot \varphi), \qquad \varphi \in T_{X,\nabla}. \tag{14}$$

$$\Omega^1(M) \otimes \mathrm{su}_2 \xrightarrow{\nabla_{\mathrm{ad}}} \Omega^2(M) \otimes \mathrm{su}_2. \tag{15}$$

If the connection was integrable, then ∇_{ad} is a complex ([crossed out] $\nabla^2_{\mathrm{ad}} = 0$)

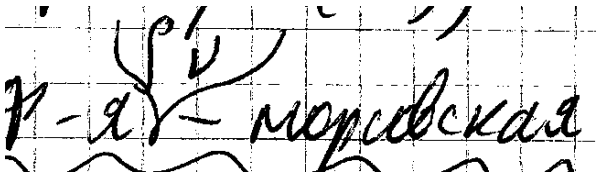


Original source details: GELFAND-P001475

Uncertain reading. The crossed-out sign before $\nabla^2_{\mathrm{ad}} = 0$ resembles φ or Φ.

well-defined $\nabla_{\mathrm{ad}}\varphi$ (independent of the representative in the quotient (13))
the function is Morse [inserted above: $\int \nu$?] $\iff$ ∇_{ad} has no degree-one cohomology, i.e.

$$H^1\left(M, \underset{\text{or } \forall \text{ local systems}}{\text{any representation of } \pi_1}\right) = 0 \tag{16}$$

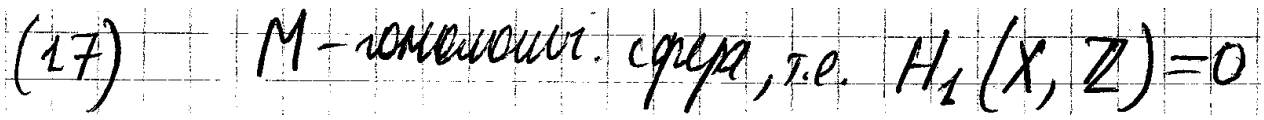


Original source details: GELFAND-P001475

Uncertain reading. The symbol after the integral sign may be ν or v.

This is equivalent to

$$M \text{ is a homology sphere, i.e. } H_1(X, \mathbb{Z}) = 0 \tag{17}$$



Original source details: GELFAND-P001475

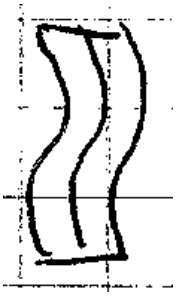
Uncertain reading. Equation (17) reads $H_1(X, \mathbb{Z}) = 0$, with M at the start of the line.

Following the standard scheme, consider gradient flows.

2 critical points $\longleftrightarrow$ 2 representations of π_1.

Join them by a trajectory of nonintegrable connections.

Choose a single coordinate system in a neighbourhood of the curve



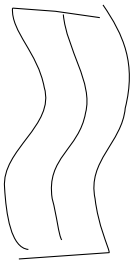


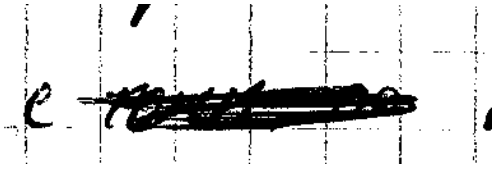
Diagram and typed description.

Original source details: GELFAND-P001475

and track the change in the matrix of second derivatives.

$$\begin{aligned}\text{Difference [inserted: of indices]} &= \text{number of points [inserted: on the path]} \\ &\quad \text{where a zero mode appears} \\ &= \text{spectral asymmetry.}\end{aligned} \tag{18}$$

This is done with [crossed out] the usual index theorem.



Uncertain reading. The crossed-out word after "with" is illegible.

Original source details: GELFAND-P001475

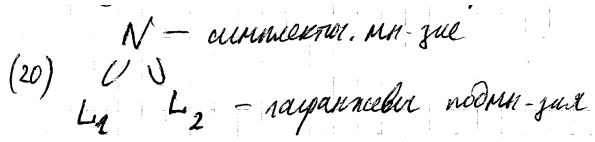
$$\text{The difference of indices } \underline{\mod 8} \text{ is well-defined!!} \tag{19}$$

GELFAND-P001476

Next example.

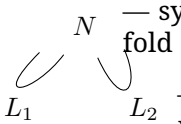
Computing the index from the finite-dimensional point of view. (Also for a homology sphere).

(20)



Original source details: GELFAND-P001476

Diagram and typed description.



N — symplectic manifold

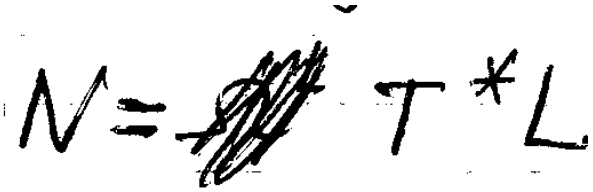
L_1 L_2 — Lagrangian submanifolds

Suppose

$$L_1 \cap L_2 \quad \text{is finite.} \tag{21}$$

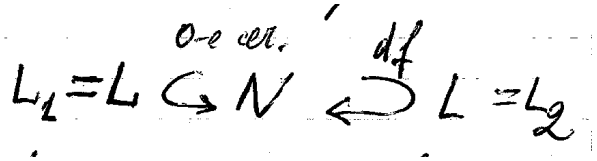
We want to compute the weighted number of intersection points.

(22) Example. $N =$ [crossed out] T^*L,



Original source details: GELFAND-P001476

Uncertain reading. The crossed-out sign between $N =$ and T^*L may be Ω.



Original source details: GELFAND-P001476

Diagram and typed description.

zero section $\qquad df$

$L_1 = L \hookrightarrow N \hookleftarrow L = L_2$

[Beneath $L_1 = L$ crossed out]



Original source details: GELFAND-P001476

Uncertain reading. The entry beneath $L_1 = L$ is crossed out. It may be H_2 or L_2.

f is Morse

$L_1 \cap L_2$ — critical points of f

they intersect transversely.

to an intersection point — an index

$$L_1 \cap L_2 \xrightarrow{\text{index}} \{0, 1, \dots, \dim L\}$$

$$\chi(N) = \sum_{x \in L_1 \cap L_2} (-1)^{\text{index}\, x} \tag{23}$$

General situation (20): define a relative index!

(24) X — paths joining L_1 and L_2

$$\begin{aligned} [0,1] &\longrightarrow N \\ 0 &\longrightarrow L_1 \\ 1 &\longrightarrow L_2 \end{aligned}$$

On such an X there is a multivalued Morse function

Critical points — points of intersection of the L_j

$\varphi \in X$, τ — tangent vector at φ, i.e. a vector field along φ

[From φ a line leads down to the label «path».]

GELFAND-P001477

[source number: 71]

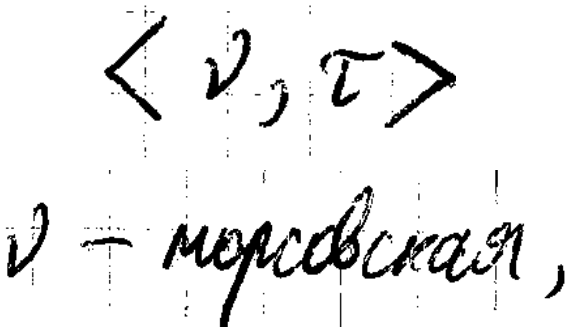
Then

$$\langle \nu, \tau \rangle := \int_0^1 \omega(\dot{\varphi}, \tau), \tag{25}$$

where

ω — symplectic form.

ν is Morse if L_1 and L_2 intersect transversely.



Uncertain reading. The two signs have the same shape and are rendered ν. The strokes also allow Latin v.

Original source details: GELFAND-P001477

The difference of indices is defined, but there is also an elementary construction (using the Maslov index)

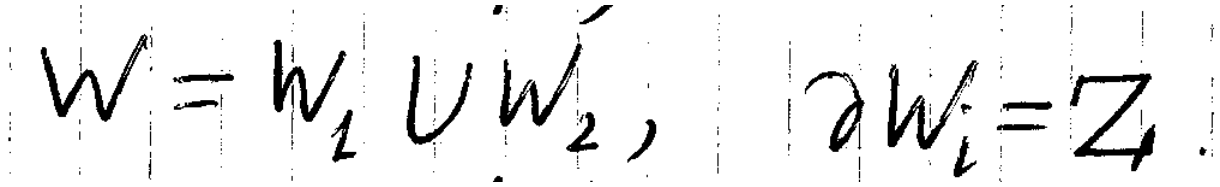
Consider a manifold M — a homology sphere. !

(26) Theorem

M can be cut along the Riemann surface Z,

(Heegaard splitting)

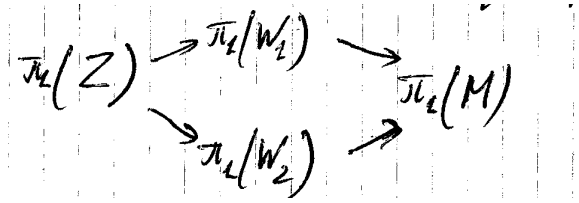
$$W = W_1 \cup W_2, \qquad \partial W_i = Z.$$



Original source details: GELFAND-P001477

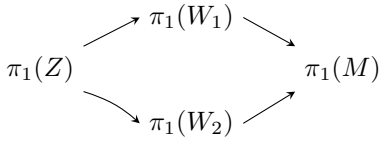
Uncertain reading. The formula has W. The preceding text has M.

(27)



Original source details: GELFAND-P001477

Diagram and typed description.



[at right] images determine $\pi_1(M)$

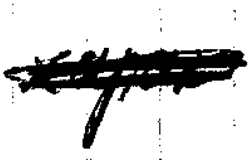


Original source details: GELFAND-P001477

Uncertain reading. The first word of the note at right tentatively reads «образы» (“images”).

$\pi_1(W_j)$ — free group on g generators

Representations of $\pi_1(M)$ [crossed out] — in it $\pi_1(W_j)$ are Lagrangian manifolds.

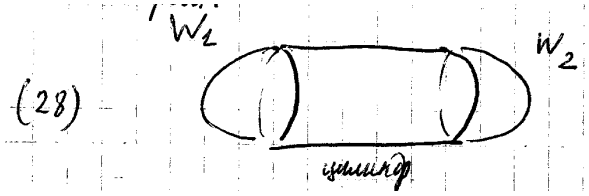


Original source details: GELFAND-P001477

Uncertain reading. The word after $\pi_1(M)$ is crossed out. Individual letters cannot be distinguished reliably.

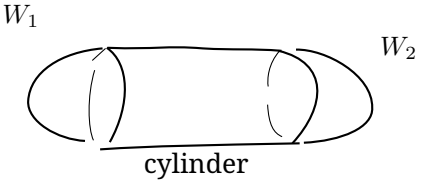
Let us construct

(28)



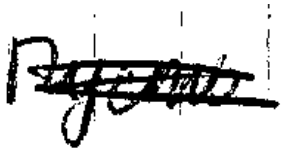
Original source details: GELFAND-P001477

Diagram and typed description.



Faint curves are shown as thin lines. The short slanting strokes at the upper joins may be part of the outline.

[crossed out] A path is a family of connections.



Original source details: GELFAND-P001477

Uncertain reading. The crossed-out word before “A path” starts with a letter resembling «P» (R). The remaining strokes cannot be read with confidence.

Second half of the work:

replace flat connections with instantons,

pass to M^4 and

Donaldson invariants

(M^4 is cut into 2 three-dimensional ones).



Original source details: GELFAND-P001477

Uncertain reading. The final words inside the parentheses are “2 three-dimensional ones,” followed by the closing parenthesis.

GELFAND-P001478

12.9.88 Gelfand Seminar.

1. «What is the smallest dimension of a space in which two two-dimensional planes can be skew?»

2. What is a determinant?

Two first-order equations can be replaced by [inserted above: one] second-order equation. How?

$$\begin{cases} \left(\frac{d}{dx} + q_{11}\right)\varphi_1 + q_{12}\varphi_2 = 0 \\ q_{21}\varphi_1 + \left(\frac{d}{dx} + q_{22}\right)\varphi_2 = 0 \end{cases} \tag{1}$$

a) First method: solve the first for φ_2 and substitute into the second

Second method: solve the second for φ_1 and substitute into the first

We get different equations!

but having solved one, we can solve the second.

This is known as the “Bäcklund transformation”.

We can take a linear combination of φ_1 and φ_2 and then a family of equations arises.



Uncertain reading. The first subscript is tentatively 1. It resembles 2, but its top stroke is closer to the 1 in (1).

Original source details: GELFAND-P001478

The answers are very beautiful

b) What about three first-order equations?

This is more complicated and the beauty disappears.

Gelfand and Retakh did it.

There is work by Sato and Kashiwara on noncommutative determinants.

c) Let $q_{ij} = \text{const}$

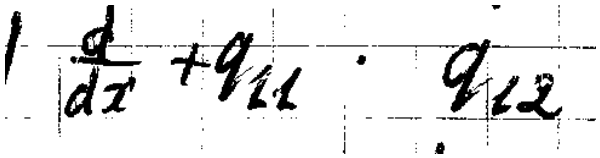
[crossed out at left] Then there is a determinant



Uncertain reading. The crossed-out label before the closing parenthesis may be «r)» or «1)».

Original source details: GELFAND-P001478

$$\begin{vmatrix} \frac{d}{dx} + q_{11} & \cdot & q_{12} \\ q_{21} & & \frac{d}{dx} + q_{22} \end{vmatrix} = \left(\frac{d}{dx}\right)^2 + (q_{11} + q_{22})\frac{d}{dx} + (q_{11}q_{22} - q_{12}q_{21}) \tag{2}$$



Uncertain reading. A separate dot appears between the top entries of the determinant. Its purpose is unclear.

Original source details: GELFAND-P001478

$\frac{d}{dx} \longleftrightarrow \lambda \quad \Longrightarrow$ we get the characteristic determinant of the matrix

in the general case we get [inserted above: (noncommutative analogue of)] the characteristic determinant.

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \quad \text{— ?}$$

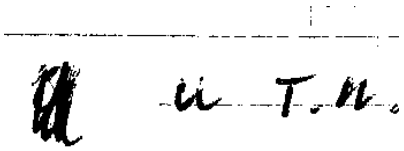
a_{ij} — a map $E_j \to E_i$,

so $a_{11}a_{22} - a_{12}a_{21}$ does not make sense.

There are 4 determinants in all:

$$a_{11} - a_{12}a_{22}^{-1}a_{21} \tag{3}$$

[crossed out] and so on.



Uncertain reading. Before “and so on” a heavily crossed-out sign resembles a or «и».

Original source details: GELFAND-P001478

GELFAND-P001479

[source number: 84]

Third order: 36 ways of writing it.

There arise noncommutative K-theory and noncommutative homologies.

c) 8 letters arranged in a cube $2 \times 2 \times 2$.



Uncertain reading. The overwritten label resembles «в)» (letter c), with «г)» (letter d) also possible.

Original source details: GELFAND-P001479

What is the determinant?

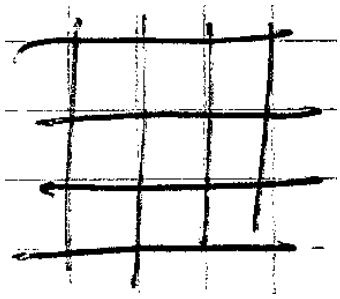
Cayley did this.

It is thought that after Hilbert the results of Gordan, Cayley, Clebsch and others lost their interest. They were declared old-fashioned nonsense. But that is untrue.

Moore. (GREGORY MOORE) (I.AS. Princeton)

Conformal field theory. (CFT)

Ising model



Original source details: GELFAND-P001479

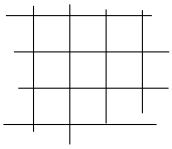
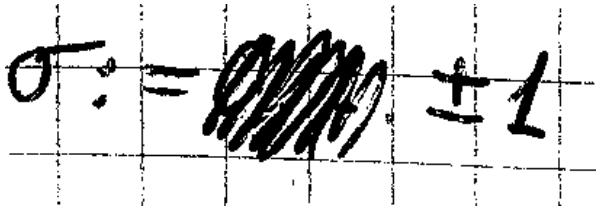


Diagram and typed description.

2-dimensional lattice [beneath “lattice”: (square)], at every point a variable $\sigma :=$ [crossed out] ± 1



Original source details: GELFAND-P001479

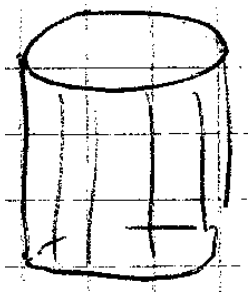
Uncertain reading. The entry between $\sigma :=$ and ± 1 is crossed out and illegible.

$$\text{Energy} \quad E = \sum_{\langle ij \rangle} \sigma_i \sigma_j \tag{1}$$

$\langle ij \rangle$ — neighbouring points

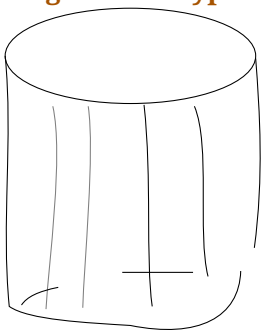
Partition function

$$Z = \sum e^{-\beta E} \tag{2}$$



Original source details: GELFAND-P001479

Diagram and typed description.

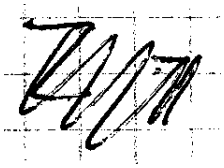


Two faint inner lines are shown in grey. The lower junctions and the purpose of the short strokes are unclear.

Consider it on a cylinder and refine the lattice

The asymptotics are expressed in terms of the functions

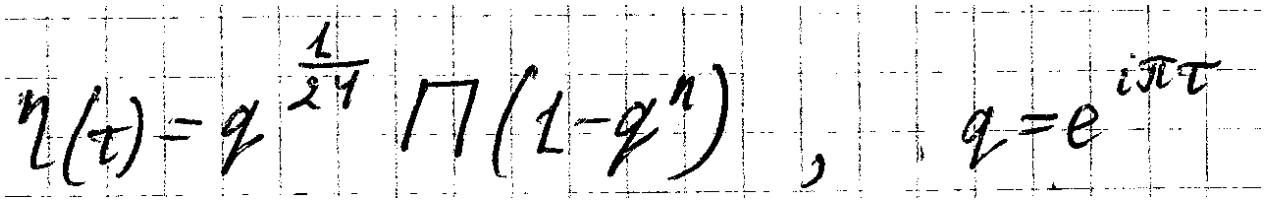
[crossed out at left]



Uncertain reading. The crossed-out entry begins with a sign resembling Z. The rest is illegible.

Original source details: GELFAND-P001479

$$\eta(\tau) = q^{1/24} \prod (1 - q^n), \qquad q = e^{i\pi\tau}$$



Original source details: GELFAND-P001479

Uncertain reading. The manuscript combines $q = e^{i\pi\tau}$ with $q^{1/24} \prod (1 - q^n)$. No product limits are written.

$$\theta_3(\tau) = \sum e^{i\pi\tau n^2}$$

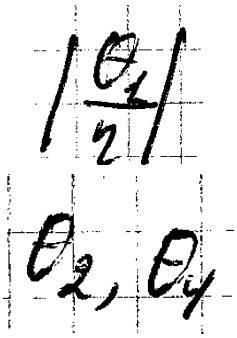
and similarly θ_2, θ_4

[above the line: $\tau = i\beta$]

Invariant under the substitution

$$\tau \longrightarrow -1/\tau$$

$$Z \longrightarrow \left|\frac{\theta_2}{\eta}\right| + \left|\frac{\theta_3}{\eta}\right| + \left|\frac{\theta_1}{\eta}\right| \qquad \text{(up to a factor)}$$



Uncertain reading. The final subscript resembles 1. In the upper excerpt it differs from the 4 of the earlier θ_4, shown below.

Original source details: GELFAND-P001479

$$Z(\beta) = Z\left(\frac{1}{\beta}\right)$$

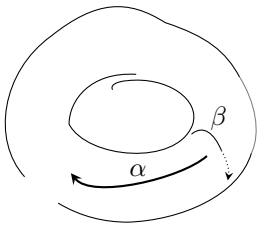
GELFAND-P001480

In CFT we associate real analytic functions to surfaces



Original source details: GELFAND-P001480

Diagram and typed description.



A short hooked stroke appears at the upper edge of the hole. An arrowhead there cannot be made out with confidence. The arc β has gaps.

$$= \sum_{i=1}^{N} f_i(\tau)\bar{g}_i(\tau), \qquad f_i := \left\{ \sqrt{\frac{\theta_2}{\eta}}, \sqrt{\frac{\theta_3}{\eta}}, \sqrt{\frac{\theta_4}{\eta}} \right\}$$

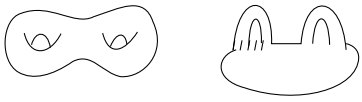
$$\tau \to -\frac{1}{\tau}, \qquad \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} \to \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix}$$

Other variants:

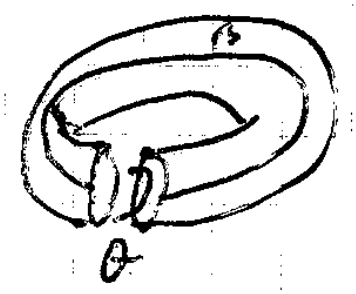


Original source details: GELFAND-P001480

Diagram and typed description.

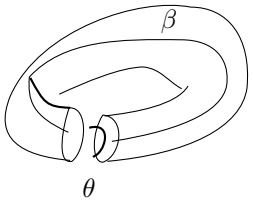


On every surface there is a corresponding model of CFT!!



Original source details: GELFAND-P001480

Diagram and typed description.



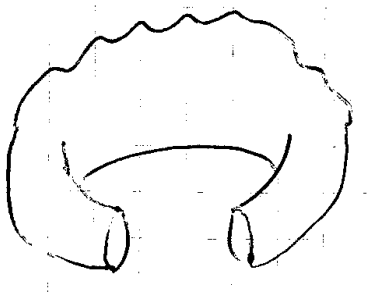
A heavy angular stroke appears at the left inner edge. Its purpose is unclear.

$$= f(\beta + i\theta, \beta - i\theta) \qquad \text{(real analytic functions)}$$

one can twist by θ

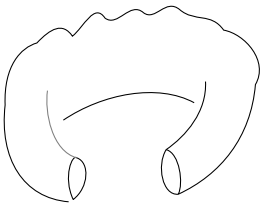
The torus parameters are 2 real numbers

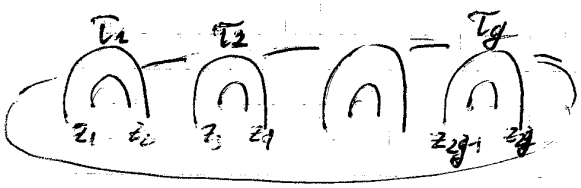
$$\tau = L + i\theta$$



Original source details: GELFAND-P001480

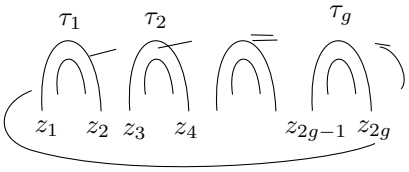
Diagram and typed description.





Original source details: GELFAND-P001480

Diagram and typed description.



Above the handles are τ_1, τ_2, τ_g. The rightmost subscript is heavily retraced and tentatively reads $2g$.

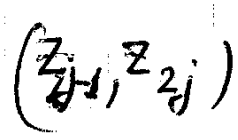
the lengths and twist rotations of the handles matter (handles)

$$\begin{array}{ccc} 0 & 1 & \infty \\ \uparrow & \uparrow & \uparrow \\ z_1 & z_2 & z_3 \end{array}$$

We obtain $3g-3$ complex parameters. $\qquad (g \geq 2)$

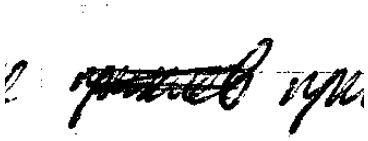
$$f(z_i, \tau_i; \bar{z}_i, \bar{\tau}_i)$$

(z_{2j-1}, z_{2j}) — points at which [crossed out] the handles are glued (thin).



Original source details: GELFAND-P001480

Uncertain reading. The first subscript tentatively reads $2j-1$. The 2 overlaps the base z. The second subscript is $2j$.



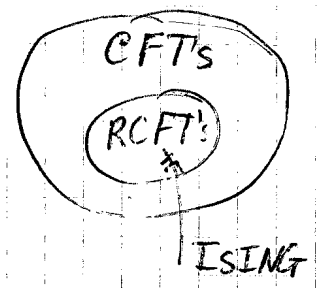
Original source details: GELFAND-P001480

Uncertain reading. The word before "glued" is crossed out. Loops and a long final stroke are visible.

τ_j — as in the torus.

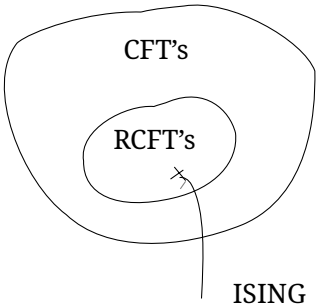
GELFAND-P001481

[source number: 85]



Original source details: GELFAND-P001481

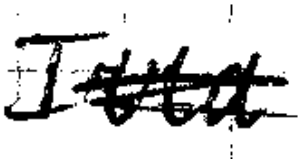
Diagram and typed description.



Inside the smaller oval, the line from ISING meets a mark resembling a cross or an arrowhead.

R — Rational

I [crossed out] RATIONAL. (the name is historical)



Original source details: GELFAND-P001481

Uncertain reading. The letters after the initial Latin I are crossed out and cannot be read with confidence.

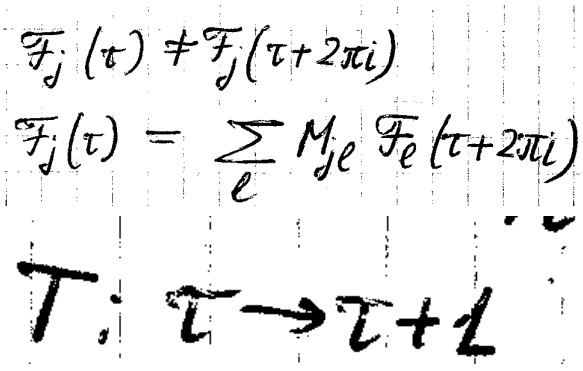
$$f(z_i,\tau_i,\bar{z}_i,\bar{\tau}_i)=\sum_{j=1}^{N<\infty}\mathcal{F}_j(z_i,\tau_i)\overline{\mathcal{F}}_j(\bar{z}_i,\bar{\tau}_i)$$

Torus:

$$\theta\to\theta+2\pi$$

$$\mathcal{F}_j(\tau)\neq\mathcal{F}_j(\tau+2\pi i)$$

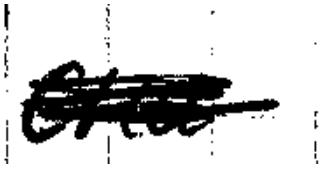
$$\mathcal{F}_j(\tau)=\sum_{\ell}M_{j\ell}\mathcal{F}_\ell(\tau+2\pi i)$$



Original source details: GELFAND-P001481

Uncertain reading. The two upper formulas have $\tau+2\pi i$. At the foot of the page, $T:\tau\to\tau+1$.

$M =$ monodromy matrix. [crossed out]



Original source details: GELFAND-P001481

Uncertain reading. The word after "monodromy matrix" is crossed out and illegible.

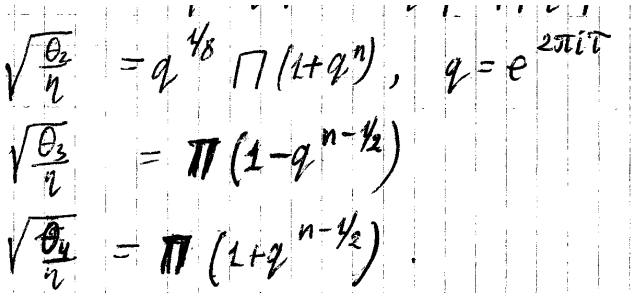
E.g. Ising

$$Z(\tau,\bar\tau) = \left|\sqrt{\frac{\theta_2}{\eta}}\right|^2 + \left|\sqrt{\frac{\theta_3}{\eta}}\right|^2 + \left|\sqrt{\frac{\theta_4}{\eta}}\right|^2$$

$$\sqrt{\frac{\theta_2}{\eta}} = q^{1/8}\prod(1+q^n), \qquad q = e^{2\pi i\tau}$$

$$\sqrt{\frac{\theta_3}{\eta}} = \prod(1-q^{n-1/2})$$

$$\sqrt{\frac{\theta_4}{\eta}} = \prod(1+q^{n-1/2}).$$



Original source details: GELFAND-P001481

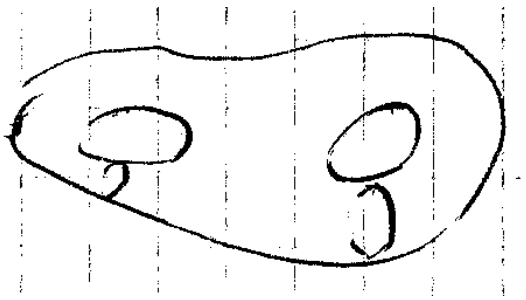
Uncertain reading. The manuscript has exponent $1/8$, minus in the θ_3 row and plus in the θ_4 row. The products have no written limits.

I. M. G. Elliptic functions are trigonometry with a parameter q.

$\mathcal{F}_j$ — are not invariant, but f is invariant

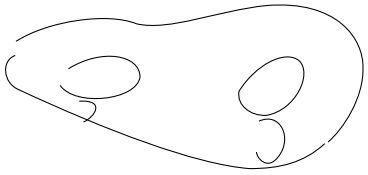
($\overline{\mathcal{F}}_j$ need not be complex conjugates of $\mathcal{F}_j$!)

The braid group appears.



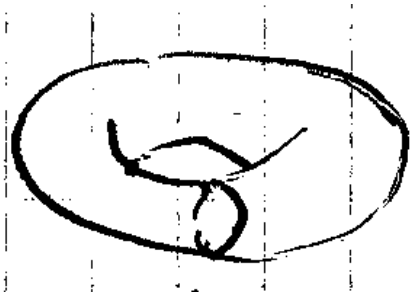
Original source details: GELFAND-P001481

Diagram and typed description.



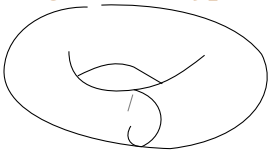
Generalize.

$g = 1$



Original source details: GELFAND-P001481

Diagram and typed description.



$$S : \tau \to -1/\tau \qquad T : \tau \to \tau + 1$$
$$S^2 = 1, \qquad (ST)^3 = I$$

GELFAND-P001482

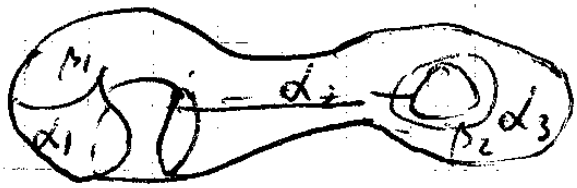
"DEHNTWIST"



Original source details: GELFAND-P001482

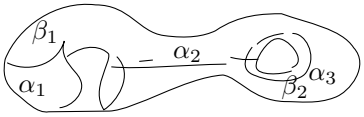
Uncertain reading. The H is overwritten. The earlier strokes cannot be read separately.

$g = 2$



Original source details: GELFAND-P001482

Diagram and typed description.



Where the strokes overlap near the holes, some of the curve junctions are indistinct.

$$z_i, \tau_i \to \tilde{z}_i, \tilde{\tau}_i$$

$$\alpha_1\beta_1\alpha_1 = \beta_1\alpha_1\beta_1$$
$$\alpha_2\beta_2\alpha_2 = \beta_2\alpha_2\beta_2$$
$$\alpha_2\beta_1\alpha_2 = \beta_1\alpha_2\beta_1 \qquad (\beta_1 \to \beta_2)$$
$$\alpha_1\alpha_2 = \alpha_2\alpha_1$$
$$\beta_1\beta_2 = \beta_2\beta_1 \qquad \text{etc.}$$
$$(\alpha_1\beta_1\alpha_2)^4 = \alpha_3^2$$
$$(\alpha_3\beta_2\alpha_2\beta_1\alpha_1^2\beta_1\alpha_2\beta_2\alpha_3)^2 = 1$$
$$\beta_1^{-1}\alpha_1\beta_1 = \alpha_1\beta_1\alpha_1^{-1}.$$

——— · ———

CFT.

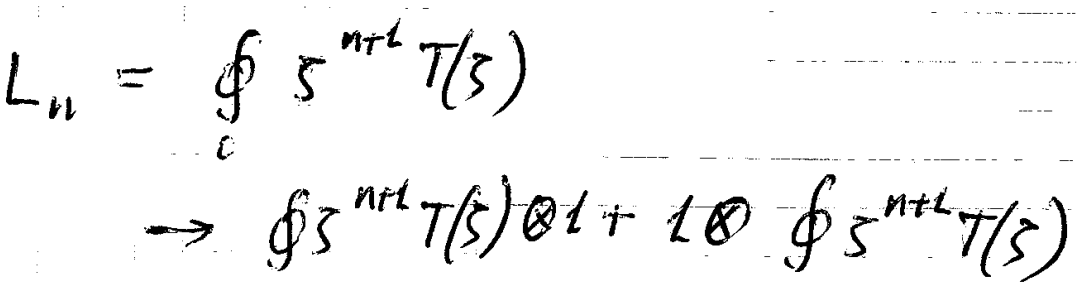
$$\mathcal{A} = \{(\sigma, \Delta) \text{ fields}\} = \text{operator algebra}$$

$$\mathcal{A} \qquad \{T(z), T'(z), \ldots\}$$

$$\mathcal{H} = \bigoplus_{i=1}^{N} \mathcal{H}_i \underset{\mathcal{A}}{\otimes} \mathcal{H}_{\bar{i}} \underset{\bar{\mathcal{A}}}{}$$

$$\Delta : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$$

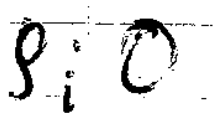
$$L_n = \oint_c \zeta^{n+1} T(\zeta)$$
$$\to \oint \zeta^{n+1} T(\zeta) \otimes 1 + 1 \otimes \oint \zeta^{n+1} T(\zeta)$$



Original source details: GELFAND-P001482

Uncertain reading. The variable resembles ζ or ξ, and the mark below the first integral resembles c or C. The differential is unwritten.

$\rho_i : \mathcal{A} \to \operatorname{End} \mathcal{H}_i, \qquad \rho_i\, C$



Original source details: GELFAND-P001482

Uncertain reading. The last mark resembles C or O. There is no visible continuation.

. . . . . .

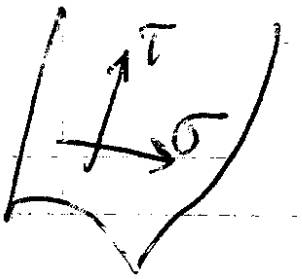
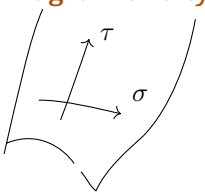
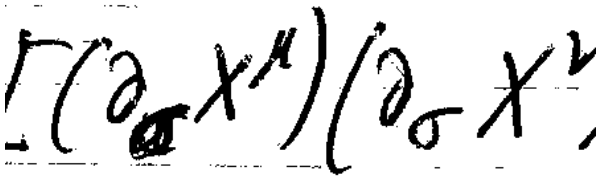


Diagram and typed description.



Original source details: GELFAND-P001482

$$S = \int g_{\mu\nu}(X)[(\partial_\sigma X^\mu)(\partial_\sigma X^\nu) - (\partial_\tau X^\mu)(\partial_\tau X^\nu)]$$



Uncertain reading. The first subscript σ is heavily retraced. The second subscript in the same product supports this reading.

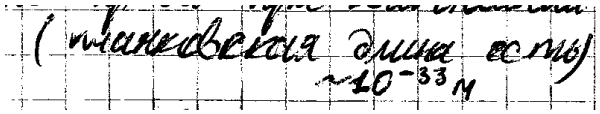
Original source details: GELFAND-P001482

Conformal Invariance $\rightarrow R_{\mu\nu} = 0$.

GELFAND-P001483

[source number: 86]

I. M. G. What should replace the notion of space in quantization?
(the Planck length is $\sim 10^{-33}$ m)



Uncertain reading. After 10^{-33}, the manuscript has a single sign read as “m”.

Original source details: GELFAND-P001483

1st variant: something discrete.

Riemann understood that this was important and gave thought to the notion of space.

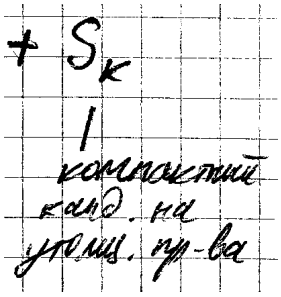
A thin film around a 4-dimensional surface.

A sensible exercise: naive geometry of 4-dimensional polyhedra.

—— · ——

$$S = \underbrace{\int_{R^4} g_{\mu\nu}(X)\left[(\partial_\sigma X^\mu)(\partial_\sigma X^\nu) - (\partial_\tau X^\mu)(\partial_\tau X^\nu)\right]}_{\substack{c_{R^4} + c_K = 26 \\ \| \\ 4}} + S_K$$

[Note joined by a stroke to S_K:] compact candidate for a thickening of space

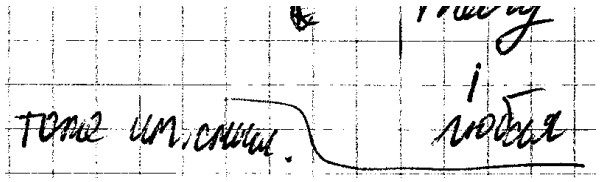


Original source details: GELFAND-P001483

Uncertain reading. Tentatively read as «канд. на утолщ. пр-ва». The first abbreviation may be «конд.», and the middle strokes of the abbreviation before «пр-ва» are indistinct.

can be replaced by ∀ other model with certain invariance properties

$$c\begin{pmatrix}\text{conformal Field}\\ \text{Theory}\end{pmatrix} — ?$$

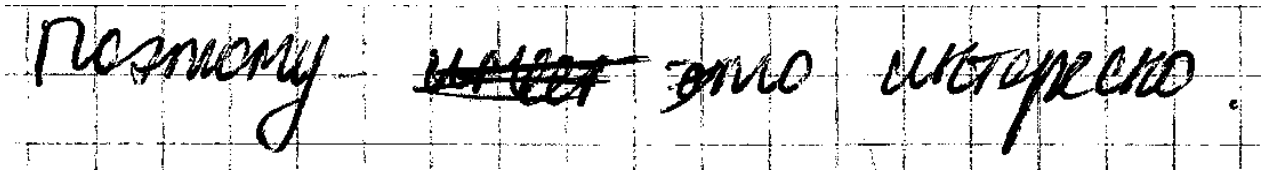


Original source details: GELFAND-P001483

Diagram and typed description.

Theory
/
also makes sense. any

Therefore [crossed out] this is interesting.



Original source details: GELFAND-P001483

Uncertain reading. The crossed-out word resembles «имеет» or «имеют», both forms of “has/have”.

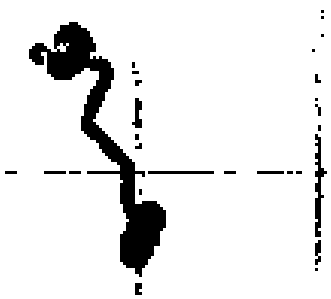
19.9.88 Gelfand Seminar.

Morozov A.

Conformal field theory (following Moore's talk).

String theory has been developing since 1984.

Starting picture

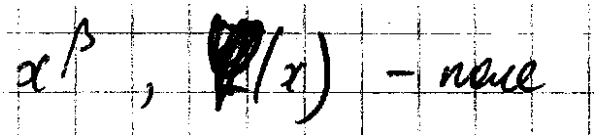


Original source details: GELFAND-P001483

Diagram and typed description.



x^β, $\varphi(x)$ — a field.



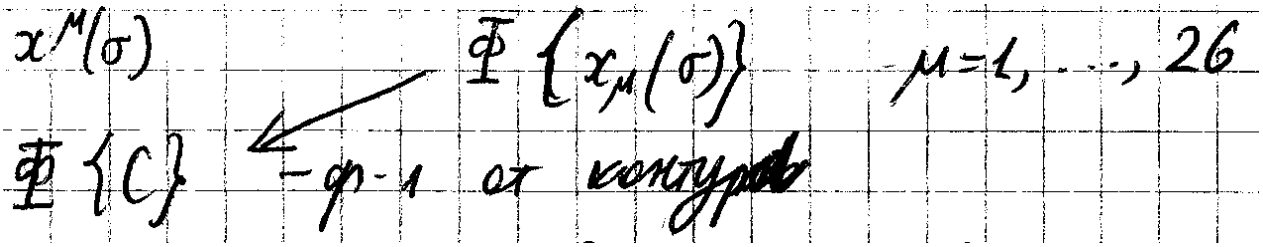
Original source details: GELFAND-P001483

Uncertain reading. The letter before (x) is overwritten and resembles φ or Φ.

$\Phi\{x_\mu(\sigma)\}$ $\mu = 1, \dots, 26$

$\Phi\{C\}$ ↙

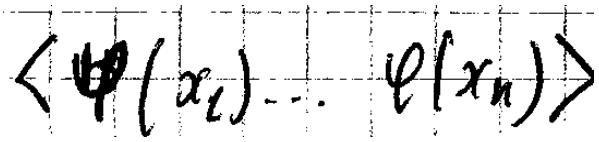
— functional of contours



Original source details: GELFAND-P001483

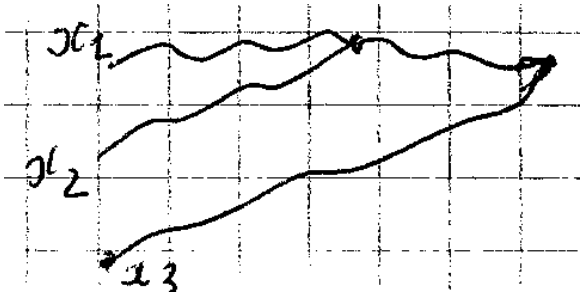
Uncertain reading. The ending is overwritten. "Of the contour" is also possible.

$$(1) \quad \langle \varphi(x_1) \dots \varphi(x_n) \rangle = \int D\varphi \, e^{\Sigma(\varphi)} \varphi(x_1) \dots \varphi(x_n)$$



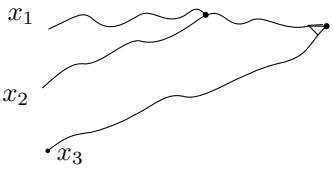
Original source details: GELFAND-P001483

Uncertain reading. The first field symbol in (1) is retraced. The second factor supports φ. Φ is also possible.



Original source details: GELFAND-P001483

Diagram and typed description.



The dot at the first junction is visible. At the right, short strokes form a dark tip or dot.

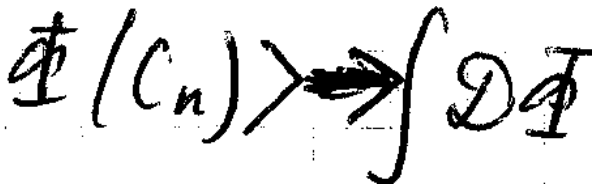
(There were 3 particles at the start, and nothing was left at the end)

$\Sigma(\varphi)$ is called the quantum action

GELFAND-P001484

We want the same in string theory

$$(2) \quad \langle \Phi(C_1) \dots \Phi(C_n) \rangle \longrightarrow \int D\Phi \, e^{\Sigma(\Phi)} \Phi(C_1) \dots \Phi(C_n);$$



Uncertain reading. The sign before the integral is overwritten. An arrowhead is visible at the right. The earlier strokes beneath it are indistinct.

Original source details: GELFAND-P001484

but there is no reasonable progress in this direction!

But the left-hand side of (2) can be described in other terms.

$$(3) \quad \Phi(C) = \sum \varphi_n(x) e^{in\sigma} \qquad \text{(closed string)}$$

φ_n are identified with the fields in (1)

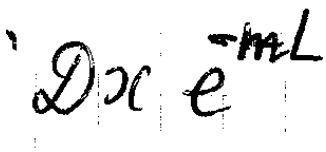
Consider

$$(4) \quad \langle \varphi_{N_1}(x_1) \dots \varphi_{N_n}(x_n) \rangle$$

and write another representation for it.

$$(5) \qquad \langle \varphi(x_1)\varphi(x_2) \rangle = \int Dx \, e^{-mL}, \qquad L = \int \sqrt{\dot{x}^2}\, dt$$

$$\| \qquad \int D\varphi \, e^{\Sigma_0(\varphi)} \varphi(x_1)\varphi(x_2)$$

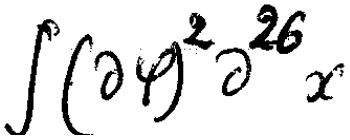


Uncertain reading. The letter after the minus is overwritten. The visible reading is $-mL$.

Original source details: GELFAND-P001484

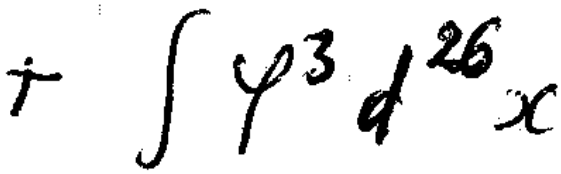
for the [inserted above: simplest] action Σ_0, describing a free particle.

$$(6) \quad \Sigma_0(\varphi) = \int (\partial\varphi)^2 \partial^{26}x + \int \varphi^3 d^{26}x$$



Original source details: GELFAND-P001484

Uncertain reading. In the first term the manuscript has ∂ before ^{26}x.



Original source details: GELFAND-P001484

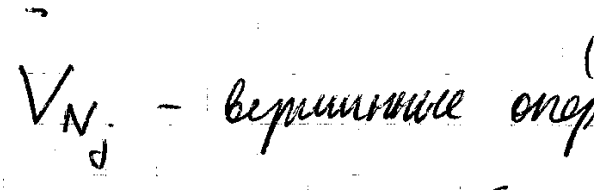
Uncertain reading. The second term is written $+\int \varphi^3 d^{26}x$.

Now

$$(7) \quad \langle \varphi_{N_1}(x_1) \dots \varphi_{N_n}(x_n) \rangle \sim \int Dx\, e^S\, V_{N_1}(x) \dots V_{N_n}(x),$$

[Below the integral] over worldsheets $x^\mu(\sigma, t)$ (nonsingular) and over all topologies

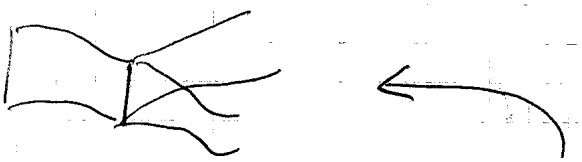
V_{N_j} — vertex operators



Original source details: GELFAND-P001484

Uncertain reading. The small index on N resembles j or d. There is a detached short stroke above V.

(8)



Original source details: GELFAND-P001484

Diagram and typed description.



The two inner curves cross. Their junctions with the middle stroke are partly indistinct. The arrow at the right points from the word “branching” toward the drawing.

String theory: [crossed out] we forbid branching



Original source details: GELFAND-P001484

Uncertain reading. The crossed-out fragment resembles «зап» or «запр», the start of “forbid”.

(9)

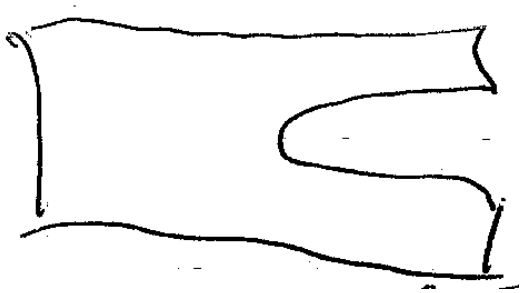
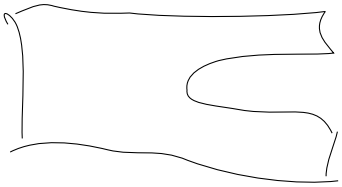


Diagram and typed description.



Original source details: GELFAND-P001484

(but such transformations remain!)
and we get a theory of free particles.

GELFAND-P001485

[source number: 87]

$$\int Dx\, e^{mL} \longrightarrow \int Dx\, D\lambda\, e^{\int \lambda^{-1}\dot{x}^2\, dt + \int m\lambda\, dt} \tag{10}$$

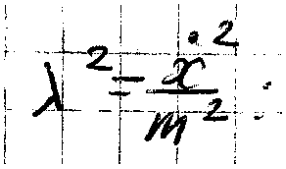


Uncertain reading. A separate sign before $m\lambda\, dt$ is crossed out. It may be m.

Original source details: GELFAND-P001485

L as in (5)
Integrating on the right by the saddle-point method (over λ), we obtain

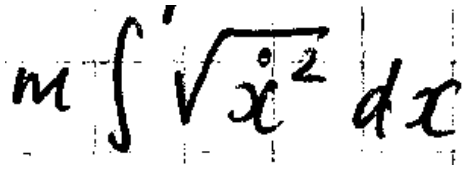
$$\lambda^2 = \frac{\dot{x}^2}{m^2}\ :$$



Uncertain reading. The denominator reads m^2. In (10), the writing after the cancellation reads $m\lambda$.

Original source details: GELFAND-P001485

$$\int Dx\, e^{m\int \sqrt{\dot{x}^2}\, dx} \tag{11}$$

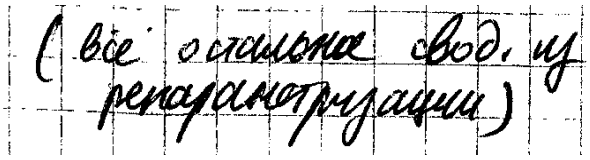


Uncertain reading. The last letter resembles x, as in Dx on the left. Equation (5) has dt.

Original source details: GELFAND-P001485

((11) ⇒ (10) gets rid of the square root). It turns out that the equality is exact!!

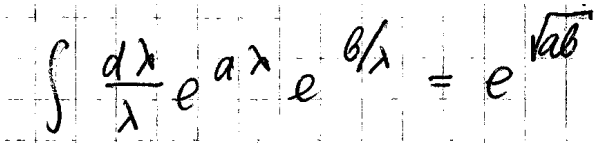
The number λ has the meaning of a one-dimensional metric (everything else reduces from reparametrization)



Original source details: GELFAND-P001485

Uncertain reading. The abbreviation after “else” resembles «свод.» (reduces). «своб.» (free/freedom) is also possible.

$$\int \frac{d\lambda}{\lambda} e^{a\lambda} e^{b/\lambda} = e^{\sqrt{ab}} \tag{12}$$



Original source details: GELFAND-P001485

Uncertain reading. The manuscript has $e^{\sqrt{ab}}$ on the right. No integration limits or additional factors are written.

Green and others did a similar thing in string theory, passing to

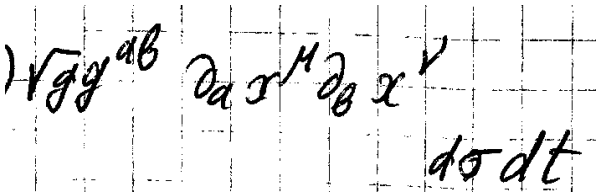
$$\sum_{\text{topol.}} \int Dx\, Dg\, e^{\int \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\mu \, d\sigma\, dt} \underbrace{e^{\int \sqrt{g}\, d\sigma\, dt}}_{\text{this can be removed}} \tag{13}$$

Generalization:

curved 26-dimensional space

$$\sum_{\text{topol.}} \int Dx\, Dg\, e^{\int G_{\mu\nu}(x) \sqrt{g}\, g^{ab} \partial_a x^\mu \partial_b x^\nu} \tag{14}$$

$$d\sigma\, dt$$



Original source details: GELFAND-P001485

Uncertain reading. The manuscript puts $d\sigma\, dt$ below the right end of the exponent. It is unclear whether it belongs within the exponent.

We want Einstein’s equations to hold

$$R_{\mu\nu}(x) = 0 \tag{15}$$

$(15) \Leftrightarrow$ dilatation invariance!

$\Downarrow$

Yang–Mills equations

All invariant equations for external fields are consequences of dilatation invariance (in the one-loop approximation)



Original source details: GELFAND-P001485

Uncertain reading. The word after “for” is tentatively read as «внешних» (“external”).

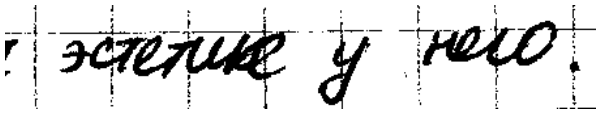
26/IX-88 Gelfand Seminar.

I. M. G. Read Feynman's book on QED and learn aesthetics from him.



Original source details: GELFAND-P001486

Uncertain reading. The verb tentatively reads «учите». The final strokes also allow «учить». The sense could be "to learn".



Original source details: GELFAND-P001486

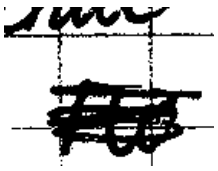
Uncertain reading. The last mark resembles «е», giving «эстетике». «эстетику» is also possible.

There QED is presented without formulas, with pictures alone.

Kazakov. Combinatorics and new work on strings

Developments have taken place (in combinatorics)

Polyakov and [crossed out] Zamolodchikov — Knizhnik — Polyakov



Original source details: GELFAND-P001486

Uncertain reading. The crossed-out strokes may begin with «по».

showed that the models given above really do describe two-dimensional quantum gravity.

(Knizhnik — Polyakov — Zamolodchikov)

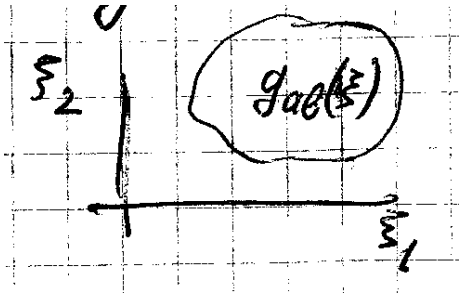
[Below the names] in English alphabetical order.

Z.–K.–P. considered a model of the bosonic string.

$$S = \int d^2\xi \, \sqrt{g} \, \partial_a x^\mu \partial_b x^\mu \, g^{ab} + \beta \int d^2\xi \, \sqrt{g} \tag{1}$$

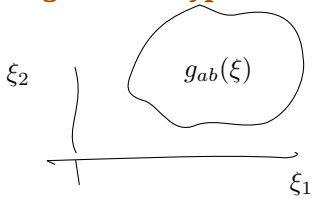
$$Z(\beta) = \int Dg_{ab} \int Dx^\mu \, e^{-S} \tag{2}$$

$\mu = 1, \ldots, D$.



Original source details: GELFAND-P001486

Diagram and typed description.

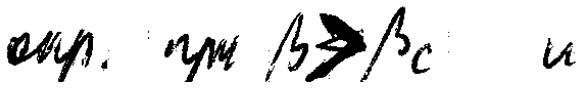


The right end of the horizontal line curls. Distinct arrowheads are not fully discernible at the ends of the axes.

GELFAND-P001487

Integral (2) is defined for $\beta > \beta_c$ and there behaves like

$$(\beta - \beta_c)^{2-\gamma_{\text{str}}(D)} \tag{3}$$



Original source details: GELFAND-P001487

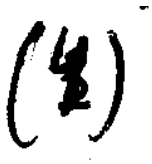
Uncertain reading. The inequality sign is retraced. The angle reads as $>$. A stroke of $\geq$ is also possible.

(at β_c the volume of the surface becomes infinite)

str = string

K.–P.–Z. proved that

$$\gamma_{\text{str}} = \frac{1}{12}\left(D - 1 - \sqrt{(D-1)(D-25)}\right). \tag{4}$$



Original source details: GELFAND-P001487

Uncertain reading. The digit is overwritten. Its upper strokes and the sequence of formulas support (4).

This makes sense for $D \leq 1$ and $D \geq 26$.



Original source details: GELFAND-P001487

Uncertain reading. The right part of the line reads $D \geq 26$. There is a separate slanting stroke below $>$.

This was obtained using Kac–Moody algebras arising from the study of the Lagrangian.

Kazakov and others predicted several values of γ_{str}:

for $D = -2, 0, \frac{1}{2}, 1$

[Mark in left margin] It all agreed!!



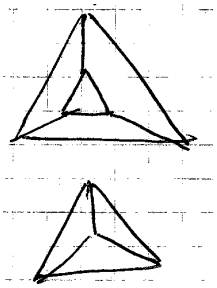
Original source details: GELFAND-P001487

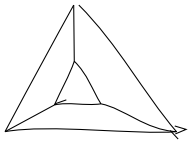
Uncertain reading. The intertwined strokes resemble the monogram «ИГ» or «ИМ».

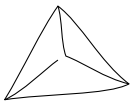
Define

$$Z(\beta) = \sum_{n=0}^{\infty} e^{-2n\beta} Z_{2n}, \tag{5}$$

where Z_n — the number of planar graphs [inserted above: with n vertices] with the topology of a sphere (3 edges must meet at a vertex)







Original source details: GELFAND-P001487

Diagram and typed description. Both drawings are in the right margin. Two thickened marks in the upper drawing resemble arrowheads. In the lower drawing the left segment stops short of the other two segments' junction.

Then

$$Z(\beta) \sim (\beta - \beta_c)^{2-\gamma_{\text{str}}(D)} \tag{6}$$

[Bracket above $\gamma_{\text{str}}(D)$: $= -\frac{1}{2}$.]

This corresponds to $D = 0$

[crossed out] Asymptotics



Original source details: GELFAND-P001487

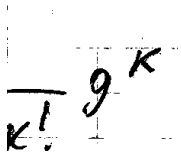
Uncertain reading. The crossed-out word before "Asymptotics" may read «число» ("number").

$$Z_{2n} \sim \left(\frac{256}{27}\right)^n n^{-3+\gamma_{\text{str}}} \tag{7}$$

γ_{str} — a universal number, independent of the way the planar graphs are defined (one can have 4 edges at 1 vertex, and so on)

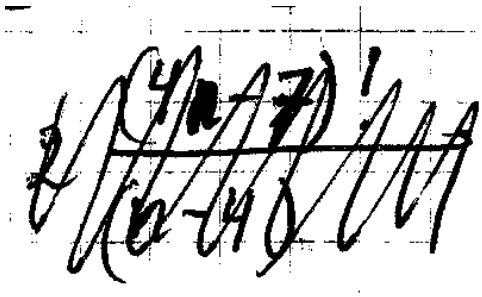
Formula for Z_n:

$$Z_{2k} = 2\frac{(4k-3)!}{(3k-1)!\,k!}\,9^k \tag{8}$$



Original source details: GELFAND-P001487

Uncertain reading. The last factor is tentatively read as 9^k. The base also resembles g.



Original source details: GELFAND-P001487

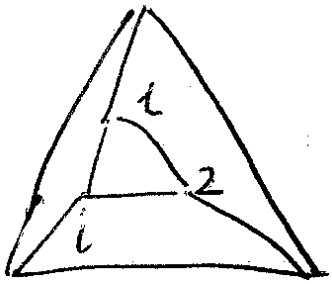
Uncertain reading. A fraction is crossed out after the equals sign. The numerator tentatively reads $(4n+7)!$, and $(n-4)$ is visible in the denominator. The indices and remaining strokes are indistinct.

Z_{2k} — the number of planar graphs inscribed in a triangle with the number of internal vertices $= 2k$.

GELFAND-P001488

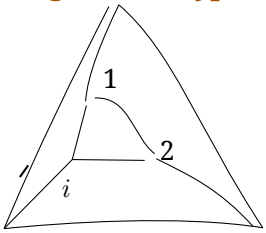
[source number: 91]

Tutte solved this problem by purely combinatorial means.



Original source details: GELFAND-P001488

Diagram and typed description.



A thickening on the left edge of the outer triangle resembles a downward arrowhead. It may be overlapping strokes.

Consider

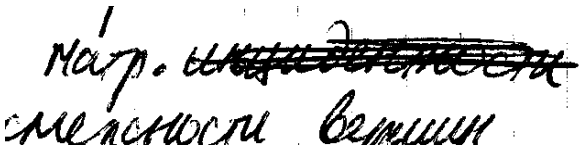
$$Z_n = \sum_{\substack{\text{graphs}\\ \text{with } n\\ \text{vertices}}} \int \prod_{i=1}^{n} d\vec{x}_i \, e^{-\sum_{\langle ij\rangle}(x_i^\mu - x_j^\mu)^2} \tag{9}$$

$\langle ij\rangle$ — nearest neighbours in the graph

internal vertices

(10) The integral in (9) equals $\det^{-D/2}(3I - G)$.

[Stroke leading from G:] Matrix of [crossed out] vertex adjacency

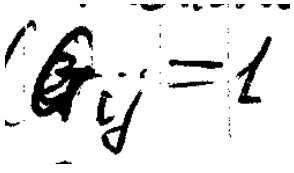


Original source details: GELFAND-P001488

Uncertain reading. The crossed-out word after “Matrix of” tentatively reads “incidence”.

($G_{ij} = 1 \Longleftrightarrow i, j$ are joined by an edge

$G_{ij} = 0$ [crossed out $\Longleftrightarrow$] otherwise).



Original source details: GELFAND-P001488

Uncertain reading. The first G is overwritten. Strokes beneath it resemble E.



Original source details: GELFAND-P001488

Uncertain reading. After $G_{ij} = 0$, a two-headed arrow is visible beneath dark strokes.

Kapranov Misha.

On the work by [crossed out B.–G.–G.] on $\mathcal{D}$-modules. [Inserted below the abbreviation: Bernst.–Gelf.–Gelf.]

V — a vector [crossed out] space, $P(V)$ — its projectivization.



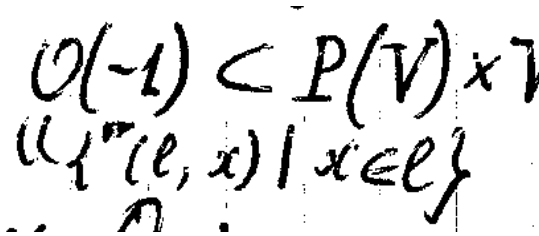
Original source details: GELFAND-P001488

Uncertain reading. A short fragment between "vector" and "space" is crossed out. It may be the start of «пространство» ("space").

$O(-1)$ — [inserted above: tautological] bundle on $P(V)$ (fundamental);

[At right]

$$O(-1) \subset P(V) \times V$$
$$\| \{(\ell, x) \mid x \in \ell\}$$



Original source details: GELFAND-P001488

Uncertain reading. Two curved strokes below $O(-1)$ may be a vertical equals sign. A separate short mark follows the opening brace.

The points of $P(V)$ — are the lines ℓ in V, passing through 0;

the line itself is the fibre over ℓ.

$$O(-k) = O(-1)^{\otimes k}, \qquad k > 0;$$

$O(+k)$ — the dual bundle of $O(-k)$

A way to construct new bundles:

$\Lambda(V)$ — the exterior algebra on V;

$\Lambda^k(V)$ — the homogeneous degree k part of $\Lambda(V)$.

We consider modules over $\Lambda(V)$ with a grading.

$$\{\ldots\ M_{-1};\ M_0;\ M_1;\ \ldots\}, \qquad \Lambda^k M_i \subset M_{i+k}$$

Exercise for next time: write the answers for [crossed out] $\dim V = 1$ and 2.



Original source details: GELFAND-P001488

Uncertain reading. The crossed-out sign before $\dim V$ resembles M or V.

GELFAND-P001489

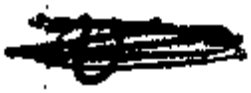
$V \ni v$, $v^2 = 0$ in $\Lambda(V)$.

Consider multiplication by v:

$$(1) \qquad \ldots \to M_{-1} \xrightarrow{v} M_0 \xrightarrow{v} M_1 \xrightarrow{v} M_2 \to \ldots$$

$v^2 = 0 \Rightarrow$ we obtain a complex.

$$\operatorname{Ker} v \supset \operatorname{Im} v, \qquad \text{[crossed out]}$$

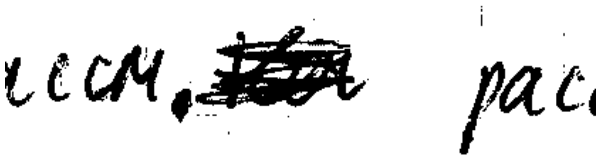


Undeciphered passage. The fragment to the right of Im v is crossed out with several horizontal lines and is illegible.

Original source details: GELFAND-P001489

Consider [crossed out] the bundle

$$v \longmapsto \frac{\operatorname{Ker}\{v : M_0 \to M_1\}}{\operatorname{Im}\{v : M_{-1} \to M_0\}}, \qquad v \neq 0.$$



Undeciphered passage. The crossed-out fragment between “Consider” and “the bundle” is illegible.

Original source details: GELFAND-P001489

This is a “bundle”, or more precisely a “sheaf”, over $\mathbb{P}(V)$.

Examples.
of modules

1). The exterior algebra itself $\Lambda(V)$

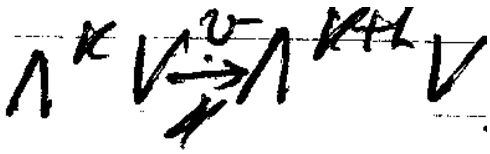
$$\underset{0}{\mathbb{C}} \xrightarrow{v} \underset{1}{V} \xrightarrow{v} \underset{2}{\Lambda^2 V} \xrightarrow{v} \ldots \xrightarrow{v} \underset{N}{\Lambda^N V}$$

The zero bundle corresponds to it; Ker $v =$ Im v

2) The truncated algebra Λ, i.e. $\Lambda^{\geq k}(V)$

(keep only the components of degrees $\geq k$).

$$\{0;\ \Lambda^k V \xrightarrow{v} \Lambda^{k+1} V;\ \ldots\}$$

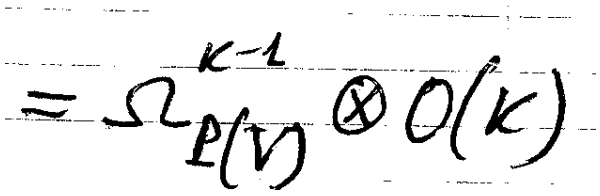


Uncertain reading. A short sign below the arrow is crossed out. It resembles v or k.

Original source details: GELFAND-P001489

Bundle:

$$\mathrm{Ker}\{v : \Lambda^k V \to \Lambda^{k+1} V\} = \Omega^{k-1}_{\mathbb{P}(V)} \otimes O(k)$$

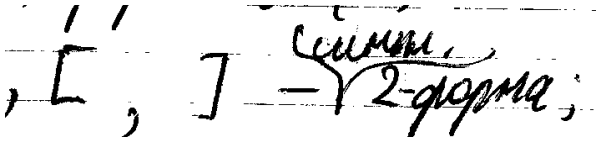


Uncertain reading. The end of the formula clearly reads $O(k)$.

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[Below $\Omega^{k-1}_{\mathbb{P}(V)}$:] $(k-1)$-forms on $\mathbb{P}(V)$

3) Let V — be a symplectic space, $[\ ,\ ]$ — [inserted above: symm.] 2-form;



Uncertain reading. The abbreviation «симм.» (“symm.”) is above “2-form” and joined to it by a bracket.

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consider the module

$$0 \to \underset{-1}{\mathbb{C}} \xrightarrow{v} \underset{0}{V} \xrightarrow{v} \underset{1}{\mathbb{C}} \to 0$$

$$v : w \longmapsto [v, w]$$

$$v \longmapsto v^{\perp}/\mathbb{C}v, \qquad \perp \text{ — orthogonality with respect to } [\cdot, \cdot]$$

Theorem. Every algebraic vector bundle over $\mathbb{P}(V)$ arises from some $\Lambda(V)$-module

$$E = \frac{\mathrm{Ker}\{v : M_0 \to M_1\}}{\mathrm{Im}\{v : M_{-1} \to M_0\}}, \quad \text{and the other factors } = 0.$$

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[source number: 92]

Proof see the appendix by S. Gelfand to the book by Okonek and others, “Bundles over Projective Space”.

Statement: note in FA in 1978.

The proof uses derived categories. Can this be avoided? S. Gelfand thinks it cannot.

Kapranov’s result: a description of bundles over a quadric in $\mathbb{P}(V)$.

$$V, \quad \langle\,\rangle, \qquad \mathbb{P}(V) \supset Q = \{v : \langle v, v\rangle = 0\}$$

[Below $\langle\,\rangle$: symmetric form.]

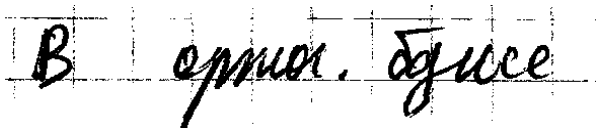
We construct a graded Clifford algebra [inserted above with a bracket: $A(V)$], generated by the generators

$$v, \quad \deg v = 1 \text{ and } h, \quad \deg h = 2$$

[Below h: fixed.]

$$vv' + v'v = \langle v, v'\rangle h$$

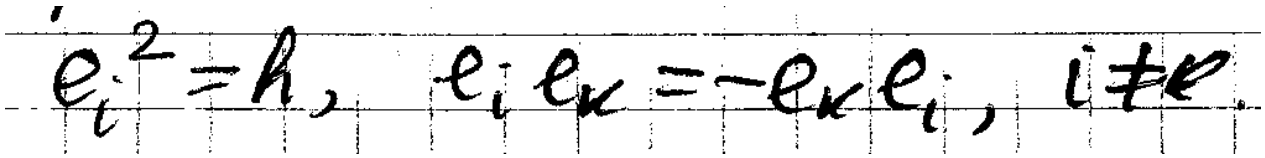
In an orthonormal basis $e_1, \dots, e_r$



Uncertain reading. The abbreviation tentatively reads «ортон.» (“orthonormal”). «ортог.» (“orthogonal”) is also possible.

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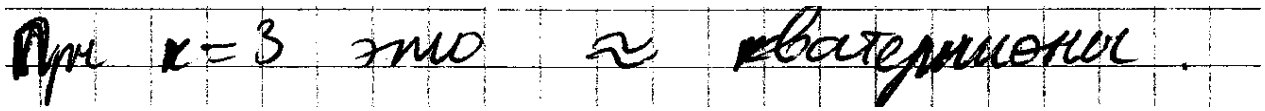
$$e_i^2 = h, \qquad e_i e_k = -e_k e_i, \quad i \neq k.$$



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Uncertain reading. The basis line reads $e_i^2 = h$. The preceding relation has $\langle v, v'\rangle h$ on the right.

For $r = 3$ this is $\approx$ the quaternions.



Original source details: GELFAND-P001490

Uncertain reading. The letter before $= 3$ is heavily retraced. The preceding notation e_r supports a tentative r.

$$\langle v, v\rangle = 0 \;\Rightarrow\; v^2 = 0$$

and we construct the same kind of bundle.

Then the same kind of theorem holds.